

Program overview

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Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Master AM 2023							
Common Core AM 2023							
Common Core AM 2023							
WI4800	Scientific Writing for Applied Mathematics	3					
WM1028AM	Ethics for Applied Mathematics	3					
Orientation Courses AM 2023							
WI4201	Scientific Computing	6					
WI4227-14	Discrete Optimisation	6					
WI4430	Martingales, Brownian Motion	6					
WI4635	Linear Algebra and Optimization for Machine Learning	6					
WI4645	Introduction to Quantum Information and Computing	6					
WI4655	Perturbation and Variational Methods for Partial Differential Equations	6					
WI4675	Introduction to Financial Mathematics	6					
Specialisation Computational Science and Engineering AM 2023							
Compulsory Courses (select 3 out of 4)				Compulsory Courses Computational Science and Engineering AM 2023			
WI4019-SP	Non-linear Differential Equations	6					
WI4204	Advanced Modeling	6					
WI4205	Applied Finite Elements	6					
WI4475	Data Assimilation	6					
Recommended Courses				Recommended Courses Computational Science and Engineering AM 2023			
IN4049TU	Introduction to High Performance Computing	6					
WI4011-17	Computational Fluid Dynamics	6					
WI4207	Continuous Optimization	6					
WI4209	Systems and Control	6					
WI4212	Advanced Numerical Methods	6					
WI4221	Control of Discrete-Time Stochastic Systems	6					
WI4260TU	Scientific Programming for Engineers	3					
WI4450	Special Topics in Computational Science and Engineering	6					
WI4520	Nonlinear Analysis and Partial Differential Equations	6					
WI4635	Linear Algebra and Optimization for Machine Learning	6					
WI4655	Perturbation and Variational Methods for Partial Differential Equations	6					
WI4680	Applications in Partial Differential Equations	6					
WI4771TU	Object Oriented Scientific Programming with C++	3					
Specialisation Financial Engineering AM 2023							
Compulsory Courses				Compulsory Courses for Financial Engineering AM 2023			
WI4079	Financial Mathematics	6					
WI4154	Computational Finance	6					
WI4201	Scientific Computing	6					
WI4430	Martingales, Brownian Motion	6					
Recommended Courses				Recommended Courses for Financial Engineering AM 2023			
WI4019-SP	Non-linear Differential Equations	6					
WI4052	Risk Analysis	6					
WI4156(TU)	Game theory	6					
WI4205	Applied Finite Elements	6					
WI4224	Special topics in Financial Engineering	6					
WI4230	Time Series & Extreme Value Theory	6					
WI4425	Financial Markets Theory	6					
WI4505	Quantitative Risk Management	6					
WI4614	Stochastic Simulation	6					
WI4615	Stochastic Calculus	6					
WI4630	Statistical Learning	6					
WI4635	Linear Algebra and Optimization for Machine Learning	6					
WI4771TU	Object Oriented Scientific Programming with C++	3					
Specialisation Discrete Mathematics and Optimization AM 2023							
Compulsory Courses				Compulsory Course for Discrete Mathematics and Optimization AM 2023			
Compulsory Courses (select 2				Compulsory Courses for Discrete Mathematics and			

out of 4)		Optimization(select 2 out of 4) AM 2023		
WI4410	Advanced Discrete Optimization	6		
WI4515	Relaxations and Heuristics	6		
WI4635	Linear Algebra and Optimization for Machine Learning	6		
WI4670	Extremal Combinatorics	6		
Recommended Courses		Recommended Courses for Discrete Mathematics and Optimization AM 2023		
ET4030	Error Correcting Codes	4		
WI4156(TU)	Game theory	6		
WI4207	Continuous Optimization	6		
WI4630	Statistical Learning	6		
Specialisation Stochastics AM 2023				
Compulsory Courses		Compulsory Course for Stochastics AM 2023		
WI4455	Statistical Inference	6		
Compulsory Courses (select 2 out of 6)		Compulsory Courses for Stochastics AM 2023 (select 2 out of 6)		
WI4052	Risk Analysis	6		
WI4230	Time Series & Extreme Value Theory	6		
WI4465	Advanced Topics in Probability	6		
WI4615	Stochastic Calculus	6		
WI4630	Statistical Learning	6		
WI4665	Advanced Topics in Statistics	6		
Recommended Courses		Recommended Courses for Stochastics AM 2023		
WI4050	Uncertainty and Sensitivity Analysis	6		
WI4138	Decision Theory/Expert Judgment	6		
WI4156(TU)	Game theory	6		
WI4505	Quantitative Risk Management	6		
WI4614	Stochastic Simulation	6		
WI4640	High-dimensional Probability	6		
Specialisation Mathematics of Data Science AM 2022		Specialisation courses for Mathematics of Data Science AM 2023		
Compulsory Courses		Compulsory Courses for Mathematics of Data Science AM 2023		
WI4630	Statistical Learning	6		
Compulsory Courses (select 2 out of 4)		Compulsory Courses for Mathematics of Data Science AM 2023 (select 2 out of 4)		
WI4203	Applied Functional Analysis	6		
WI4455	Statistical Inference	6		
WI4614	Stochastic Simulation	6		
WI4640	High-dimensional Probability	6		
Recommended Courses		Recommended Courses for Mathematics of Data Science AM 2023		
CS4230	Machine Learning 2	5		
EE4685	Machine Learning, a Bayesian Perspective	5		
WI4207	Continuous Optimization	6		
WI4230	Time Series & Extreme Value Theory	6		
WI4410	Advanced Discrete Optimization	6		
WI4665	Advanced Topics in Statistics	6		
Specialisation Partial Differential Equations AM 2023		Specialisation courses for Partial Differential Equations AM 2023		
Compulsory Courses (select 2 out of 6)		Compulsory Courses for Partial Differential Equations AM 2023 (select 2 out of 6)		
WI4019-SP	Non-linear Differential Equations	6		
WI4201	Scientific Computing	6		
WI4203	Applied Functional Analysis	6		
WI4430	Martingales, Brownian Motion	6		
WI4520	Nonlinear Analysis and Partial Differential Equations	6		
WI4660	Dynamical Systems and Chaos	6		
Recommended Courses		Recommended Courses for Partial Differential Equations AM 2023		
WI4006	Special Functions and Representation Theory	6		
WI4046	Spectral Theory of Linear Operators	6		
WI4204	Advanced Modeling	6		
WI4205	Applied Finite Elements	6		
WI4211	Advanced Topics in Analysis	6		
WI4480	Internet Seminar on Evolution Equations	9		
WI4485	Harmonic Analysis	6		
WI4615	Stochastic Calculus	6		
WI4680	Applications in Partial Differential Equations	6		
Specialisation Mathematics of Quantum Technology and Computation AM		Specialisation courses for Mathematics of Quantum Technology and Computation AM 2023		
Compulsory Courses (select 2 out of 4)		Compulsory Courses Mathematics of Quantum Technology and Computation AM 2023 (select 2 out of 4)		

4)			
WI4006	Special Functions and Representation Theory	6	
WI4203	Applied Functional Analysis	6	
WI4650	Applied Quantum Algorithms	6	
Recommended courses	Recommended courses Mathematics of Quantum Technology and Computation AM 2023		
AP3452	Quantum Error Correction	4	
CS4090	Quantum Communication and Cryptography	5	
QIST4300	Qubit Dynamics and Quantum Control	4	
QIST4320	Quantum Computation and Quantum Programming	4	
WI4046	Spectral Theory of Linear Operators	6	
WI4207	Continuous Optimization	6	
WI4211	Advanced Topics in Analysis	6	
WI4635	Linear Algebra and Optimization for Machine Learning	6	
Non-Mathematics Electives AM 2023			
IFEEMCS520200	Interdisciplinary Advanced Artificial Intelligence Project	15	
TPM301B	Spoken English for Academic Purposes Intermediate	2	
TPM302B	Spoken English for Academic Purposes Advanced	2	
TPM303A	Intermediate Writing in English for the University	2	
TPM304A	Advanced Writing in English for the University	2	
TPM305A	Writing a Masters Thesis in English	2	
TUD4040	Joint Interdisciplinary Project	15	
WI5215	Internship AM	15	
WM0375TU	Ethics of Technological Risks	3	
WM1135TU	Advanced English for the University	3	
WM1301TU	Ethics of Transportation	3	
WM1401TU	Ethics of Healthcare Technologies	3	
Thesis project 2023			
WI5010	Thesis Project	45	
WI5010EXT	Thesis Project (external)	45	

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Master AM 2023

MSc Coordinator	Dr.ir. J.H. Weber
Program Title	Applied Mathematics
In association with the University of	TU/e and UTwente
Introduction 1	The Master of Science programme in Applied Mathematics is intended for students who have obtained a Bachelor of Science degree in (Applied) Mathematics or in another area that provides a sufficiently advanced background in mathematics. Students may apply for admission through the admissions office of the Delft University of Technology. Any student who has completed a Bachelor of Science degree in (Applied) Mathematics at a Dutch research university is automatically qualified for admission to the M.Sc. programme. Also, a completed bridging programme in Applied Mathematics from TU Delft gives access. Individuals holding a foreign bachelor's degree will be assessed on an individual basis by the admission committee. The objective of the MSc programme in Applied Mathematics is to provide students with a thorough knowledge of applied mathematics and to develop their expertise in applying the methods and tools of mathematics to problems in science and engineering. It is therefore crucial for students to obtain broad experience in the simulation and mathematical modeling of such problems, the mathematical analysis of these models and the implementation of the results. Moreover, they should have sufficient insight into the underlying mathematical theory to be able to develop new mathematical methods and techniques, if needed. The M.Sc. programme in Applied Mathematics is a two-year full-time programme, to be concluded with a masters thesis. In total, the programme requires 120 EC.
Program Structure 1	The Master's Degree Programme in Applied Mathematics has seven specialisations: - Computational Science and Engineering, - Financial Engineering, - Discrete Mathematics and Optimization, - Stochastics, - Mathematics of Data Science, - Partial Differential Equations, - Mathematics of Quantum Technology and Computation. An Individual Exam Programme (IEP) consists of 120 EC, subdivided into the following parts: - Common core courses worth 6 EC, - Orientation course of 6 EC, - Specialisation courses worth 48 EC, of which a maximum of 12 EC may be spent on non-mathematical specialisation courses, - Non-mathematical electives worth 15 EC, - A thesis project of 45 EC. During their education, students are required to gain experience as a mathematician in interdisciplinary work. This requirement can be satisfied in four ways: - through an internship of 15 EC (WI5215), - through a Joint Interdisciplinary Project (TUD4040), - through an Interdisciplinary Advanced AI Project (IFEEMCS520200), - through a graduation project performed in an interdisciplinary setting outside the Delft Institute of Applied Mathematics. Students that have been admitted through a bridging programme (i.e. with a bachelor degree different from (Applied) Mathematics) or that are following the SEC specialisation are exempt from this requirement. The joint masters programme in Computer Simulations for Science and Engineering (COSSE) deviates. Students who have followed the minor Science Education and Communication (SEC) in their bachelors programme and wish to obtain the certificate granting 1st degree teacher (eerstegraads bevoegdheid) status in the Netherlands may follow the SEC specialisation (Verdiepingsdeel) at the faculty of Applied Science within their Applied Mathematics Masters programme.

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Master Applied Mathematics

Common Core AM 2023	
Introduction 1	The core consists of a. The common core courses and b. The mandatory orientation course for each specialisation

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Master Applied Mathematics

Common Core AM 2023	
Introduction 1	1. The common core consists of two courses, TPM005A Scientific Writing for Applied Mathematics and WM1028AM Ethics for Applied Mathematics

WI4800	Scientific Writing for Applied Mathematics	3
Responsible Instructor	Dr. K. van Nispen	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	In this course, you learn to report on the outcomes of your study in a scientific paper. More specifically, you will learn to write a literature review about a subject of your own choice. This is a necessary skill for anyone who wants to pursue an academic career after their graduation, but it can also be used immediately for all academic texts you will write during your Master program, such as your Master thesis. In eight weeks, we will go through all steps of the writing process, from formulating a good main question and finding relevant literature to the actual writing, re-writing and final editing. You will exercise with finding, reading and managing relevant academic literature, writing in an academic style, building a comprehensive argumentation, reviewing fellow students' articles and using other students and the instructor's comments to improve your own work.	
Study Goals	In this course you will learn to communicate scientific findings in a report/paper To reach this aim, you will work on the following subaims: 1) Students are able to identify the relevant author guidelines for their text genre 2) Students are able to formulate a main question (or aim/purpose statement) with subquestions 3) Students support writing sufficiently and correctly with literature 4) Students support writing with argumentation 5) Students can structure a text according to the scientific conventions 6) Students can implement knowledge about the specific elements of a scientific text and their generic requirements in their own text. 7) Students use visualisations, such as tables and figures, appropriately. 8) Students apply a correct formal scientific style	
Education Method	Writing a scientific paper can be an overwhelming task. In this course we divide the workload over several lectures and assignments. There are 6 mandatory sessions: 5 lectures and one teacher consultation - one missed session is allowed only. In addition, you will work on (bi)weekly assignments that compose parts of your writing task. Handing in all assignments in time is a requirement for passing this course. Students who receive a pass for this course are rewarded with 3 ects. This equals 84 hours of study. A total of 10,5 hours is spent on attending classes. The remaining 44 hours is for self-study, writing, and revising. This means that from preparing lecture 2 up to handing in your final article in week 8, you will have to spend at least 9 hours of self study on this course, every week. It is important that you make sure you have this time in your personal schedule. At the beginning of the course you will mainly be reading up about scientific writing and the subject of your text. As the course proceeds, you will be spending more of your time on writing, giving feedback and revising your own text.	
Assessment	The final grade of the course consists of the following components: - Assignment (100%) Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Assignment: resubmit deliverable You write a scientific article (word limit will be communicated on Brightspace). You have to hand in parts of your article every week. Your final grade is based on the final article. An evaluation form for the grading of the article is available throughout the course. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). The grade will be pass/fail	

WM1028AM	Ethics for Applied Mathematics	3
Module Manager	A.R. Gammon	
Responsible for assignments	Dr. S.N.R. Buijsman	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course is a study of broadly normative issues that arise in relation to applied mathematics. The following kinds of questions are examined: How can applied systems be built to promote, protect, preserve, and foster the pursuit of various values related to human wellbeing? What risks and uncertainties need to be accounted for in such systems? To what extent is it possible to empower individuals with applied systems while limiting the extent to which they can harm each other and themselves?	
Study Goals	1. Recognize and discuss normative and societal implications of mathematical applications and cases. 2. Explain key course concepts (e.g., bias, normativity, modeling assumptions) and their ethical dimensions. 3. Apply basic ethical frameworks and concepts to mathematical applications in society. 4. Think critically about the roles and responsibilities of mathematics professionals.	
Education Method	1 lecture and 1 tutorial per week	
Literature and Study Materials	Readings will be provided on Brightspace.	
Assessment	Final Assignment (Exam): 50% (individual) Tutorial Presentation: 25% (small groups) Ethical Assessment paper: 25% (small groups)	
Remarks	Contact the course instructor for disability accommodations: S.N.R.Buijsman@tudelft.nl	

Orientation Courses AM 2023	
Introduction 1	2. The second part of the core of your IEP consists of an orientation course, which is compulsory for your specialisation. WI4201 Scientific Computing, mandatory for Computational Science and Engineering WI4227-14 Discrete Optimisation, mandatory for Discrete Mathematics and Optimization WI4430 Martingales, Brownian Motion, mandatory for Stochastics WI4635 Linear Algebra and Optimization for Machine Learning, mandatory for Mathematics of Data Science WI4645 Introduction to Quantum Information and Computing, mandatory for Mathematics of Quantum Technology and Computation WI4655 Perturbation and Variational Methods for PDE, mandatory for Partial Differential Equations WI4675 Introduction to Financial Mathematics, mandatory for Financial Engineering It is allowed to take more than one orientation course, you can include the extra course(s) under your specialisation in your IEP.

WI4201	Scientific Computing	6
Responsible Instructor	Prof.dr.ir. C. Vuik	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	A basic knowledge on partial differential equations (PDEs), on numerical methods for solving ODEs/PDEs, and on linear algebra.	
Course Contents	During the course, the important steps towards the solution of real-life applications dealing with partial differential equations will be outlined. Based on a well-known basic partial differential equation, which is representative for different application areas, we treat and discuss direct and iterative solution methods from numerical linear algebra in great detail. The discretization of the equation will result in a large system of discrete equations, which can be represented by a sparse matrix. After a discussion of direct solution methods, the iterative solution of such systems of equations is an important step during numerical simulation. Emphasis is laid upon the so-called Krylov subspace methods, like the Conjugate Gradient Methods.	
Study Goals	A. General Study Objectives 1. After having completed this course, the student is able to use finite differences to approximate the solution of a partial differential equation and use an efficient and robust method to compute or approximate the solution of the resulting large system of linear equations. 2. The student will be able to assess the reliability in terms of the accuracy and stability of the numerical approximation of the solution. 3. The student will be able to use methods to approximate eigenvalues of large matrices. B. Specific Study Objectives 1. Finite Difference Method: Boundary Conditions, Grid, Numbering of the unknowns, Error Analysis - The student will be able to discretize 2-dimensional Poisson type problems. The student is able to implement the boundaries in various ways. - The student will be able to analyze the properties of the resulting matrix, as there are symmetry, positive definiteness, M-matrix etc. Also the sparseness of the matrix can be investigated and a notion of band and profile matrices should be known. 2. Direct Solution Methods Gaussian elimination, pivoting, iterative refinement, band and profile methods - The student will be able to understand and use direct solution methods based on Gaussian elimination and LU decomposition. - The student will be able to estimate the error due to rounding errors in the answer and to use pivoting to eliminate most of the errors. - The student will be able to estimate the amount of flops for direct solution methods and how to use more efficient variants as there are Cholesky method, band and profile methods. 3. Basic Iterative Methods - The student will be able to construct Basic Iterative Methods based on the splitting of the matrix. Furthermore an analysis of the convergence could be made. - A comparison of the various method should be possible together with the dependence on the gridsize. - The student will be able to choose suitable starting vectors and stopping criteria. 4. Krylov Subspace methods CG, SPD, preconditioning, general matrices, CGNR, Bi-CGSTAB and GMRES - The student will be able to use and understand the conjugate gradient method, together with a good understanding of the (super)linear convergence. - The student will be able to make a good choice for Krylov Subspace methods in the non-SPD case. - The student will be able to combine Krylov Subspace methods with preconditioners and know three classes of preconditioners: diagonal scaling, BIM and incomplete decompositions. 5. Eigenvalue methods Power method, Lanczos method, Arnoldi method - The student will be able to use the power method and is able to investigate its convergence. - The student will be able to choose suitable starting vectors and stopping criteria. - The student will be able to use the Lanczos and Arnoldi method.	
Education Method	Lectures/computer exercises	
Literature and Study Materials	Lecture notes, for further reading the book Matrix Computations, G.H. Golub and C.F. van Loan, the Johns Hopkins University, Baltimore, 2013, can be used.	
Assessment	The final grade of the course consists of the following components: - homework exercises deadline start of Q2 leads to grade G1, - take home exam deadline half of January grade G2 - written exam grade G3. The final grade is $(G1+G2+2*G3)/4$, provided that all grades are larger than or equal to 5. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - homework exercises: resubmit deliverable - take home exam: resubmit deliverable - written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Prof.dr.ir. M.B. van Gijzen	

WI4227-14	Discrete Optimisation	6
Responsible Instructor	Dr. F.M. de Oliveira Filho	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Specific prior knowledge: * Linear programming: simplex method, LP-duality * Basic knowledge of graphs/networks This is covered in the course TW2020 (Optimalisering) General knowledge: * Proficiency in reading and writing proofs * Sufficient mathematical maturity	
Course Contents	Discrete optimization is about the problem of finding a best solution among a (finite) set of feasible solutions. A well-known example is the traveling salesman problem, where we are asked to find a shortest tour in a graph visiting every node exactly once. This course is an introduction to the area, focusing on fundamental methods like linear programming and its theoretical framework (polyhedra, integrality, etc.). The following topics will be covered: - Convex sets and their structure. - Polyhedra, cones, and their structure. - Linear programming, the simplex method. - Systems of linear inequalities, Fourier-Motzkin elimination. - Farkas's lemma, linear programming duality. - Several applications and examples: flows in networks, matrix games, bipartite matching, etc.	
Study Goals	By the end of this course, students will be able to - Recognize convex sets and their applications; - Apply the theory of convexity to concrete examples; - State the main results/theorems concerning polyhedra; - Model problems as linear programming problems; - Recognize total unimodularity and use it to prove integrality of polyhedra; - Construct the dual of a linear program.	
Education Method	Lectures	
Literature and Study Materials	The course is based on lecture notes and no specific book will be followed.	
Books	No specific book will be followed in this course. The following books contain most of the course's material: * A. Barvinok, A Course in Convexity, American Mathematical Society, Providence, 2002. * V. Chvatal, Linear Programming, W.H. Freeman and Company, New York, 1983. * W.J. Cook, W.H. Cunningham, W.R. Pulleyblank, and A. Schrijver, Combinatorial Optimization, John Wiley & Sons, New York, 1998. * A. Schrijver, Theory of Linear and Integer Programming, John Wiley & Sons, New York, 1986.	
Prerequisites	* Linear programming: simplex method, LP-duality * Basic knowledge of graphs/networks This is covered in the course TW2020 (Optimalisering) General knowledge: Proficiency in reading and writing proofs Sufficient mathematical maturity	
Assessment	The final grade of the course consists of the following components: - Written exam (100%) Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	No materials are permitted during the test (except pencil, ruler, eraser etc). In particular, calculators are not allowed.	

WI4430	Martingales, Brownian Motion	6
Responsible Instructor	Prof.dr. F.H.J. Redig	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	1. Conditional expectation. 2. Martingales, stopping times, optional sampling. 3. Martingale convergence theorems, inequalities. 4. Elementary continuous-time stochastic processes: Poisson process, continuous-time Markov chains. 5. Brownian motion: definition, construction, basic properties. Martingales associated to Brownian motion, brownion motion with drift. 6. Elementary introduction to the stochastic integral, Ito isometry	
Study Goals	1. Understanding the concept of conditional expectation. Being able to compute conditional expectations in concrete examples, and being apply the tower property, and the take-out-what-is-known property. 2. Being able to recognize whether a process is a martingale, sub- or super martingale. 3. Being able to use martingales in computations such as gambler's ruin and distributions of hitting times. 4. Being able to apply the martingale convergence theorems and martingale inequalities in concrete contexts such as law of large numbers, branching process. 5. Being able to perform concrete computations with Brownian motion and its martingales (such as computation of hitting time distribution, recurrence transience). 6. Understanding the definition and construction of the stochastic integral and being able to perform elementary computations with it. 7. Being able to reproduce correctly definitions, statements and proofs and the main definitions and theorems as indicated in a list (this concerns the so-called theory question).	
Education Method	Lectures and problem sessions.	
Literature and Study Materials	1. Lecture notes based on chapter 10 of A. Gut ``A graduate course in probability" for the martingale part of the course. 2. Lecture notes for Brownian motion part of the course. The lecture notes are provided via Brightspace (course content).	
Reader	1. Lecture notes based on chapter 10 of A. Gut ``A graduate course in probability". 2. For the material on Brownian motion we will also sometimes use P. Morters and Y. Peres, Brownian Motion, (2010) freely available at http://people.bath.ac.uk/maspm/book.pdf . Also lecture notes will be available for that part. 3. Other recommended literature: Le Gall: Brownian motion, martingales and stochastic calculus, Springer 2016.	
Assessment	The final grade of the course consists of the following components: - Two graded homeworks - a written exam A homework grade h is computed as the average of the scores on the two graded homeworks. Course grade is the final exam grade f or $0.6f+0.4h$, whichever is larger, provided $f > 5$. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - homework assignments and written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). The homework grade does not count for the resit.	

WI4635	Linear Algebra and Optimization for Machine Learning	6
Responsible Instructor	Dr. A. Heinlein	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	The course material and literature are mostly self-contained, but prior knowledge equivalent to contents of the courses AM2010 Linear Algebra 2 and AM2020 Optimization from the Bachelor of Applied Mathematics programme is strongly recommended. Programming experience in Python or Matlab is expected.	
Course Contents	State-of-the-art machine learning algorithms combine techniques from many different areas of mathematics. In this course, we will discuss algorithmic foundations from the areas of linear algebra and optimization. This also includes aspects of their theory and efficient implementation. Among others, we will study + matrix decomposition and factorization techniques, + regression and classification problems and regularization, + iterative and continuous optimization methods, + hyper parameter optimization, + graph-based algorithms and clustering, + basic introduction to neural networks and algorithmic aspects neural networks (e.g. computational graphs and back-propagation)	
Study Goals	Upon completion of the course, the student should be able to: + Categorize typical data sets based on their structure + Derive suitable mathematical formulations for common supervised and unsupervised machine learning tasks + Choose suitable algorithmic building blocks for solving mathematical formulations of common machine learning tasks + Analyze machine learning algorithms with respect to their properties, such as efficiency and robustness + Implement machine learning algorithms in a computationally efficient way	
Education Method	The course comprises 14 lectures and 14 exercise classes.	
Literature and Study Materials	Lecture notes, slides, and the main literature: + Charu Aggarwal. Linear algebra and optimization for machine learning. Springer, 2020. + Gilbert Strang. Linear algebra and learning from data. Wellesley-Cambridge Press Cambridge, 2019. + Jeremy Watt, Reza Borhani, and Aggelos K Katsaggelos. Machine learning refined: Foundations, algorithms, and applications. Cambridge University Press, 2020.	
Assessment	The final grade of the course consists of the following components: - Two group projects (0.4) - Written exam (0.6) Final grade calculation: $(0.4 * \text{group projects} + 0.6 * \text{written exam})$ Note: For the written exam, a score of at least 5.0 must be obtained. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Group projects: resubmit deliverable - Written exam: There will be a resit for the written exam at a later point in the same academic year Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	D. de Laat	

WI4645	Introduction to Quantum Information and Computing	6
Responsible Instructor	Dr. B. Janssens	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic probability theory. Linear algebra.	
Course Contents	Quantum mechanics is not a prerequisite. We outline the mathematical framework of quantum mechanics, and give an introduction to quantum information theory, quantum computing and quantum algorithms. Since this is the orientation course for the specialization 'Mathematics of quantum technology and computation', we also pay special attention to the role in quantum theory by some of the other topics covered in this specialization: functional analysis, spectral theory, representation theory, semidefinite optimization and Lie theory.	
Study Goals	At the end of this course, you will be able to: - Explain the Bell inequalities in terms of games. - Explain the basic postulates of quantum mechanics for closed systems in terms of operators and spectral theory, and solve the Schrodinger equation for simple systems such as qubits and the harmonic oscillator. - Manipulate quantum circuits, and analyse simple quantum algorithms such as quantum FFT, order finding and the factoring algorithm. - Quantify the amount of entanglement on a composite system. - Explain the basic postulates of quantum mechanics for open systems, and analyse operations between different quantum systems in terms of completely positive maps (such as qubit gates and POVMs). Explain the Stinespring theorem and the Choi/Kraus decomposition, and derive the no-cloning theorem. - Explain the connection between symmetries, conserved quantities and Lie theory, and analyse spin and momentum in terms of rotation and translation symmetry.	
Education Method	-2 hours of lectures per week, 14 weeks in total. -Biweekly exercises.	
Literature and Study Materials	We will use the book by Nielsen and Chuang - Quantum computation and quantum information - 10th edition. In addition, we use lecture notes that will be provided via Brightspace.	
Books	Nielsen and Chuang - Quantum computation and quantum information (10e edition). Lecture notes (provided via Brightspace)	
Assessment	The final grade of the course consists of the following components: - Written exam (100%) Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI4655	Perturbation and Variational Methods for Partial Differential Equations	6
Responsible Instructor	Dr. H.M. Schuttelaars	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Course Contents	In the course two different approaches to analyse PDEs will be presented. The first method will focus on approximating solutions to PDEs. In the lectures we elaborate on asymptotic approximations, matched asymptotic expansions, multiple scales, WKB method, bifurcation and stability analysis. In the second approach, a modern introduction to the field of elliptic PDEs is given. In the lectures we consider weak solutions to elliptic equations (such as Laplace equation), Sobolev spaces. Modern PDEs uses tools from functional analysis and measure theory. The precise regularity and behavior of solutions of elliptic equations is one of the main features in the lectures. For non-linear PDEs such results are fundamental. The course might be useful for student in analysis, PDE and numerics, or anyone who wants to know more about Sobolev spaces and second order equations. (Key words: elliptic, weak derivatives, Sobolev spaces extension theorem, trace theorem, Sobolev inequalities, Hilbert techniques, compactness theorems, second order elliptic equations, Weak solution, energy estimates, regularity, eigenvalue techniques).	
Study Goals	Global objectives for the Approximation approach: After following this course the student will be able to apply fundamental and advanced analytical techniques and methods to study properties of solutions of ordinary and partial differential equations. The mathematical methods and techniques that the student is able to use are mentioned under "course contents". After studying the lecture notes and after making the exercises successfully the student can start to work on research problems in the field of nonlinear differential equations. Global objectives for the Variational approach: After the course the student - knows what a weak derivative is - understands the definition of L^p and Sobolev spaces. - can apply basic Sobolev space techniques (approximation, extension, trace, Sobolev embedding, compactness) - can give the definition of a weak solution of a second order elliptic PDE - can apply Sobolev space techniques to prove existence and uniqueness of weak solutions of second order elliptic PDEs - can derive properties (regularity/estimates) of solutions of second order elliptic PDEs Objective per topic: Sobolev spaces the student -knows the definition of Hölder spaces -knows the definition of the weak derivative -can calculate weak derivatives -knows the definition of Sobolev spaces -can approximate Sobolev functions by smooth functions -can explain the extension theorem -can explain the trace theorem -knows Sobolev inequalities -can apply Sobolev inequalities -can derive Sobolev inequalities -knows the compactness theorem -can apply the compactness theorem -understands what the dual space of a Sobolev space is Second order elliptic equations the student -knows the definition of a weak solution. -can give energy estimates -can apply the Lax-Milgram theorem to obtain existence of weak solutions -knows what regularity is -can state regularity results for second order equations (interior and boundary) -can apply eigenvalue techniques.	
Education Method	Perturbation approach: Introduction to Perturbation Methods (second edition) by M.H. Holmes Variational approach: lecture notes. Optional literature: Partial Differential Equations, by Lawrence C. Evans, AMS, 1998, 662 pp., Hardcover, ISBN 0-8218-0772-2. Assessment	
Assessment	The final grade of the course consists of the following components: - Take home assignments (two per quarter) - final oral exam Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Take home assignments: resubmit deliverable - Oral exam: oral resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr.ir. M.C. Veraar	

WI4675	Introduction to Financial Mathematics	6
Responsible Instructor	Dr. A. Papapantoleon	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	This is a compulsory course of the Financial Engineering Specialization. Students need to be familiar with probability theory and analysis. Knowledge of discrete time stochastic processes and measure theory is beneficial. Students from other faculties are welcome, no need to write in advance, but they should be aware that this is a highly formalized course, belonging to the Master of Applied Mathematics	
Course Contents	The aim of this course is to introduce students to financial derivatives and the mathematics of finance. The course starts with basic financial derivatives (futures, options, swaps) and then introduces the probabilistic modeling and study of financial markets. Topics covered include: no-arbitrage theory and the fundamental theorems of asset pricing, risk-neutral pricing of derivatives, discrete-time models, the binomial asset pricing model, American and exotic derivatives and the Black-Scholes model.	
Study Goals	At the end of the course, the students will know: - the basics of financial derivatives - the fundamentals of asset pricing in discrete time - modeling financial markets in discrete time - option pricing and hedging in discrete time - valuation of exotic and American derivatives - the Black-Scholes model.	
Education Method	Lectures and possibly case studies.	
Literature and Study Materials	Föllmer & Schied: Stochastic Finance: An Introduction in Discrete Time, de Gruyter, 2016	
Prerequisites	This course is compulsory for all the Applied Maths students in the "Financial Engineering" specialization. For all the others the course can be chosen as an elective, but it must be stressed that this is a rather mathematical course. The treatment of financial topics is at a high level of formalization.	
Assessment	The final grade of the course consists of the following components: - Written exam (70%) - Assignments (30%) Final grade calculation: $(0.7 * \text{written exam}) + (0.3 * \text{assignments})$ Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Written exam: written resit - Assignments: oral exam on the material of the assignments Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks		

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Specialisation Computational Science and Engineering AM 2023

Introduction 1

Specialisation Computational Science and Engineering

Compulsory orientation course:
WI4201 Scientific Computing

Compulsory specialisation courses (choose 3 out of 4 courses):
WI4204 Advanced Modeling in Science
WI4205 Applied Finite Elements
WI4475 Data Assimilation
WI4019 Nonlinear Differential Equations

All of the compulsory specialisation courses can also be chosen as recommended specialisation course.

Recommended specialisation courses:
WI4011-17 Computational Fluid Dynamics
WI4207 Continuous Optimization
WI4212 Advanced Numerical Methods
WI4450 Special Topics in Computational Science and Engineering
WI4209 Systems and Control
WI4221 Control of Discrete‐Time Stochastic Systems
MM Inverse Problems in Imaging
IN4049TU Introduction to High Performance Computing
WI4260TU Scientific Programming for Engineers
WI4655 Perturbation and Variational Methods for PDE
WI4635 Linear Algebra and Optimisation for Machine Learning
WI4771TU Object Oriented Scientific Programming with C++
WI4520 Nonlinear Analysis and Partial Differential Equations
WI4680 Applications in Partial Differential Equations

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Compulsory Courses (select 3 out of 4)

WI4019-SP	Non-linear Differential Equations	6
Responsible Instructor	Dr.ir. W.T. van Horssen	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Introductory course on Differential Equations e.g. Boyce and DiPrima Chapters 2-7, 9.	
Course Contents	Existence and uniqueness of solutions of initial value problems, Gronwall's lemma, Autonomous systems, Critical points, Periodic solutions, Stability theory, Linear systems and Floquet theory, Perturbation theory and asymptotic methods, Poincare-Lindstedt method, Averaging method, Multiple time-scales method, Elementary bifurcations.	
Study Goals	After following this course the student will be able to apply fundamental and advanced, analytical techniques and methods to study properties of solutions of nonlinear ordinary differential equations. The mathematical methods and techniques are mentioned under "course contents". After studying the lecture notes and after making the exercises successfully the student can start to work on research problems in the field of nonlinear differential equations.	
Education Method	Lectures and a MAXIMA practical work. In total 14 lectures of 2 hours each will be given. The MAXIMA practical work is about 4 hours work. Following and preparing the lectures and the MAXIMA practical work takes about 60 to 70 hours. Making the 30 take-home exercises takes about 60 hours. Preparing the 45 minutes oral exam takes about 30 hours.	
Literature and Study Materials	F. Verhulst, Nonlinear Differential Equations and Dynamical Systems, Springer-Verlag 2nd edition, ISBN 3-540-60934-2.	
Assessment	The final grade of the course consists of the following components: - Take home exam (50%) - oral exam (50%) Final grade calculation: $(0.5 * \text{take home exam}) + (0.5 * \text{oral exam})$ The take-home exam consists of 30 exercises. The oral exams will be held at the end of the 2nd quarter, and the student is only allowed to take part in this oral exam when at least 70% of the take-home exercises are made correctly and when the MAXIMA practical work is followed successfully. In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5 take home exam: resubmit deliverable oral exam: oral resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI4204	Advanced Modeling	6
Responsible Instructor	Dr. J.L.A. Dubbeldam	
Contact Hours / Week x/x/x/x	2e semester intensieve cursus	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The participants will benefit from a solid knowledge of advanced calculus, basic knowledge of ordinary and partial differential equations and programming experience with Matlab (or an equivalent language).	
Course Contents	This modelling course is introducing the principles and practice of mathematical modelling. An important part of the course is to recognize the essential mechanisms governing a phenomenon. These mechanisms have to be translated into mathematical equations and formulas, and included into a mathematical model. This activity requires both a good understanding of the system under consideration and good mathematical skills. Mathematical modelling is essential for understanding processes in real life, complementing expensive experiments that sometimes cannot even be carried out at all. This course will concentrate on applications in physics and life sciences and focus on models in terms of partial differential equations. The course focuses on working in a small group on a specific project. The topics for project work may differ from year to year. For the academic year 2020-2021 some emphasis will be on continuum mechanics	
Study Goals	General 1. Has knowledge of mathematical modelling tools that are used for model applications in physics and for mathematical models formulated in terms of partial differential equations. 2. The student is able to recognize the principles and practice of mathematical modelling. 3. The student is able to identify the essential mechanisms governing a physical phenomenon and is able to translate these into mathematics and can include them into a mathematical model. Specific 1. Is able to make a partial differential equation non-dimensional and can apply scaling techniques to a partial differential equation. 2. Understands conservation principles and constitutive relations. 3. Is able to linearise partial differential equations in order to perform a stability analysis 4. Is able to develop a mathematical model for a practical problem and has the knowledge and maturity to analyse this model with the available mathematical tools. 5. Has the knowledge and skills for the use and assessment of mathematical models and is able to adapt the models for a specific application. 6. Is able to pose adequate questions and has a critical yet constructive attitude towards analysing and solving complex mathematical modelling problems 7. Is able to communicate verbally and in writing about mathematical research with specialists, non-specialists and other parties involved. 8. Has sufficient communicative skills to collaborate with others and discuss mathematical modelling problems and results.	
Education Method	The first lecture summarizes the mathematical modelling tools that might be useful for the projects. Furthermore during this lecture the projects are introduced by the supervisors and the groups of 2-4 students are formed. The main part of the course is offered as a one week "intensive course", fully devoted to the project. In most cases a computer program for simulating a phenomenon from physics of the life sciences has to be developed. At the end of the fourth quarter the written report of the project has to be handed in and the final presentations are scheduled.	
Literature and Study Materials	Many of the students will have a good background in continuum mechanics. The book Continuum Modeling in the Physical Sciences, E. van Groesen and J. Molenaar, Mathematical Modeling and Computation, Vol. 13, SIAM, 2007 (ISBN 978-0-898716-25-2) can be used as background material. Copies of this e-book can be borrowed from the TU-Delft library.	
Assessment	The final grade of the course consists of the following components: - individual homework (20%) - project (80%) Final grade calculation: $(0.2 * \text{homework}) + (0.8 * \text{project})$ Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - homework: resubmit deliverable - project report: resubmit deliverable - project presentation: resit presentation At the first lecture home assignments are given. The final deadline for submitting the home work is the start of the intensive week. The grade of the project is based on the written report of the group work and on the oral presentation about the project. The following aspects of the project work will have equal weights: -The development of the model -Mathematical depth	

	-Report and presentation
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Pop-up tekst bij intekenen	Welcome to Advanced Modelling!

WI4205	Applied Finite Elements	6
Responsible Instructor	D. Toshniwal	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The students should have had at least introductory courses on the following topics: * continuum mechanics * partial differential equations * tensor Calculus * linear algebra * numerical methods for ODEs	
Course Contents	Weak formulations (derivation, motivation for use, well-posedness) Abstract introduction to the finite element method (FEM) Theoretical analysis of FEM (e.g., of well-posedness, convergence) Practical implementation of FEM (e.g., for linear/non-linear steady/unsteady problems) in one and higher dimensions Verification and validation of FEM	
Study Goals	After this course, students should be able to: * derive a weak formulation for the give PDE * show the well-posedness of the derived weak formulations * construct isoparametric finite elements in one and higher dimensions * discretize the weak problem using isoparametric finite elements * show the well-posedness of the discretized problem * implement the finite element method * perform basic verification of the implementation correctness	
Education Method	Lectures.	
Literature and Study Materials	The following texts will serve as supplementary references to the class notes: * Course reader (downloadable from Brightspace) * T.J.R. Hughes. The Finite Element Method: Linear Static and Dynamic Finite Element Analysis, 1987. * P. Wriggers. Nonlinear Finite Element Methods, 2008. Other references will be provided during the course.	
Assessment	Course will be assessed based on assignments (40% of final grade) and a written exam (60% of final grade). Note: Some of the assignments may require the students to study material not covered in class. Similarly, some assignments will ask the students to work on small extensions or generalisations of the concepts covered in class. To pass the course, the following is mandatory: * Assignments: - The assignments (written and programming exercises) must be completed with an average grade greater than 5.5. - The assignments will have deadlines and late submissions will be penalised. * Exam: - The written exam must be completed with a grade greater than 5.5. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Group assignments: resubmit deliverable - written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	For more information see the Brightspace course page.	

WI4475	Data Assimilation	6
Responsible Instructor	Prof.dr.ir. M. Verlaan	
Instructor	Prof.dr.ir. A.W. Heemink	
Contact Hours / Week x/x/x/x	0/0/2/0 lecture and 0/0/0/2 group work	
Education Period	3 4	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Partial differential equations and some knowledge on statistics	
Course Contents	Data assimilation methods are used to combine the results of a large scale numerical model with the measurement information available in order to obtain an optimal reconstruction of the dynamic behavior of the model state. An well-known example is weather forecasting where the numerical model of the atmosphere is continuous updated using the data that becomes available. The course gives an overview of the various methods used for data assimilation and discusses the underlying mathematical aspects.	
Study Goals	Students will understand the theory of data assimilation, will have an overview of the various techniques that are available, and will be able to use some of these techniques in practical applications. Specific Study goals 1. The student is able to rewrite a mathematical model in state space representation. 2. The student is able to formulate the data assimilation problem. 3. The student is familiar with variational data assimilation. 4. The student is able to solve parameter estimation problems using variational data assimilation. 5. The student is familiar with Kalman filtering for large scale systems 6. The student is able to implement and use ensemble modelling. 7. The student is able to implement and apply recursive data assimilation using ensemble Kalman filtering.	
Education Method	Lectures, homework, and a project. The course consists of seven 2-hour lectures. There are two homework sets. The project is handed out at the last lecture and continues until June.	
Literature and Study Materials	The following book will be used: Dynamic Data Assimilation: A least squares approach, John M. Lewis, S. Lakshmivarahan and S.K. Dhall, Encyclopedia of Mathematics and its Applications, Cambridge University Press, 2006.	
Assessment	The final grade consists of the following parts: * assignments (20% of final grade); * report about the project (50%); * oral exam (30%). Final grade calculation: $(0.2 * \text{assignments}) + (0.3 * \text{project report}) + (0.5 * \text{oral exam})$ Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Assignments: resubmit deliverable - Project report: resubmit deliverable - Oral exam: oral resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Recommended Courses

IN4049TU	Introduction to High Performance Computing	6
Responsible Instructor	Dr.ir. H.X. Lin	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Required: Linear algebra (matrix and vector operations), numerical analysis (solution of a system of linear equations) Strongly advised: some experience with a programming language (e.g., C) is preferred.	
Course Contents	This course is intended for students who are interested in computing-intensive research. In the course, a number of algorithms that are being used within a diversity of research areas are considered. The scaling behavior of these algorithms in case of increasing problem size and/or an increasing number of processors is analyzed. Attention is paid to those aspects of computer architectures that are important to understand the resulting performance, such as the memory hierarchy and the interconnection network. By analyzing a number of case studies (applications) with respect to their computing-intensive character, possible bottlenecks will be determined. Based on performance analysis, it will be indicated how the effect of those bottlenecks can be reduced. The goal is to learn how to get high performance with the available hardware architecture. The lab exercises will be done on the supercomputer at TU Delft. The emphasis will be on designing efficient parallel algorithms and the necessary optimization for high performance. During the lab exercises, the following types of problems will be elaborated on: a parallel Poisson solver, a parallel finite element simulation, a parallel N-body simulation, and matrix computation on GPUs. More information, such as handouts and slides, can be found the Brightspace.	
Course Contents Continuation	High Performance Computing, parallel programming, parallel algorithm	
Study Goals	1. Is able to categorize high-performance computer systems 2. Is able to explain and classify parallel and distributed architectures, and programming models; 3. Is able to explain the concepts of data decomposition and parallel algorithms; 4. Is able to apply various parallelization methods to benefit from the potential parallel computer systems; 5. Capable to implement parallel programs (using MPI) on parallel computers and GPU (using Cuda); 6. Capable to evaluate and analyze the performance of parallel programs.	
Education Method	Lectures, computer lab exercise using MPI. As an option, answers to the bi-weekly quizzes can be handed in, and a maximum of one bonus point to the exam grade can be obtained.	
Computer Use	Lab exercises (mandatory): implementing (small) parallel programs with C, MPI and Cuda.	
Literature and Study Materials	Will be made available throughout the course and can be downloaded from the Brightspace.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 50%) Lab work (weighting 50%) Quizzes, bonus of max 1 point can be obtained (only valid on the first examination occasion after the lectures) Final grade calculation: $0.5 * \text{Written exam} + 0.5 * \text{lab work}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: Resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Lab work: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Enrolment / Application	Via Brightspace	

WI4011-17	Computational Fluid Dynamics	6
Responsible Instructor	Dr. A. Heinlein	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	The students should have had at least introductory courses on the following topics: + continuum mechanics + partial differential equations + tensor calculus linear algebra + basic concepts of the finite element method + basic concepts of iterative solvers and preconditioning The students are also expected to be familiar with programming (e.g., in MATLAB or Python).	
Course Contents	Part I: Numerical methods for the convection-diffusion equation; Stability, consistency and convergence of the numerical methods; Extensions to incompressible Navier-Stokes equations; Part II: Basic iterative solvers and preconditioners for the Poisson problem; Schwarz domain decomposition preconditioners for the Poisson problem; Extension of Schwarz preconditioners to saddle point problems (Stokes and Navier-Stokes equations)	
Study Goals	Learning objectives of the course: Upon completion, the student should be able to: Part I: 1. Derive the equations of fluid dynamics using multivariable calculus and physical conservation laws 2. Explain the singularly-perturbed behavior of solutions to the equations of fluid dynamics 3. Discretize the convection-diffusion equation and derive discretization errors 4. Choose discretizations suitable for convection-dominated flows 5. Discretize the incompressible Navier-Stokes equations Part II: 6. Construct and implement Schwarz domain decomposition methods 7. Explain the concept of the abstract theory of Schwarz methods 8. Apply the abstract Schwarz theory to one- and two-level Schwarz preconditioners to derive condition number bounds for the Poisson problem 9. Apply nonlinear and linear solution techniques to (Navier-)Stokes equations 10. Extend Schwarz methods to (Navier-)Stokes equations	
Education Method	The course comprises 14 lectures (7 lectures for Part I taught by S. Jain, 7 lectures for Part II taught by A. Heinlein).	
Course Relations	Some concepts discussed in the course WI4201 'Scientific Computing' are assumed as prior knowledge for the course WI4011-17. Even though parts are reviewed very briefly to be adapted to computational fluid dynamics problems, students are strongly recommended to take the course WI4201 before WI4011-17.	
Literature and Study Materials	The following texts will serve as supplementary references to the class notes: + "The Finite Element Method" by T. J. R. Hughes + "Domain Decomposition Methods - Algorithms and Theory" by A. Toselli and O. B. Widlund + "Finite Element Methods for Incompressible Flow Problems" by V. John + "Principles of Computational Fluid Dynamics" by P. Wesseling Additional current topics, which are not covered in standard text books yet, will also be discussed. Further references will be given during the course.	
Assessment	The final grade of the course consists of the following components: - Individual assignments (50%) - Project presentation - Project report (together 50%) Final grade calculation: $(0.5 * \text{assignments}) + (0.5 * \text{presentation and report})$ Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Individual assignments: resubmit deliverable - Presentation: resit presentation - Report: resubmit improved report To pass the course, the following is mandatory: + The assignments (written and programming exercises) must be completed with an average grade greater than 5.5. + The projects must be completed with grades greater than 5.5. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	This course focuses on the mathematical background, analysis, and motivation of numerical methods for the challenging applications of the incompressible Navier-Stokes equations and convection-diffusion equations. It follows a bottom-up approach, developing the numerical techniques from scratch rather than building a complex numerical framework for simulating realistic fluid problems, as is often done in more engineering focused courses.	
Co-Instructor	Dr.ir. S. Jain	

WI4207	Continuous Optimization	6
Responsible Instructor	D. de Laat	
Contact Hours / Week x/x/x/x	1e semester Utrecht maandag	
Education Period	1 2	
Start Education	1	
Exam Period	2	
Course Language	English	
Expected prior knowledge	The student should have a solid knowledge of linear algebra and multivariable calculus. The student should also have knowledge of linear programming (including linear programming duality) and convex analysis to the level of being able to follow the text and do the exercises from: Chapters 1 and 2 (including the exercises) from the book 'Linear Programming, A Concise Introduction, Thomas S. Ferguson' https://www.math.ucla.edu/~tom/LP.pdf Exercises 2.1, 2.2, 2.12, 3.1, 3.3, 3.5, and 3.7 from the book 'Convex Optimization, Stephen Boyd and Lieven Vandenbergh' http://stanford.edu/~boyd/cvxbook	
Course Contents	Continuous optimization is the branch of optimization where we optimize a (differentiable) function over continuous (as opposed to discrete) variables. Here the variables can be subject to equality and inequality constraints. Optimization problems of this form are common in science and engineering, machine learning, and as relaxations of discrete optimization problems. One can use linear algebra, multivariable calculus, and convex geometry to study the properties of continuous optimization problems and to design and analyze efficient algorithms. In this course, we study theory, algorithms, and some applications of continuous optimization. In the theory part we discuss convexity, Lagrangian duality, optimality conditions, and conic programming. In the algorithmic part we discuss derivative-free methods, first-order optimization methods, neural networks/supervised learning, second-order optimization methods, interior-point methods, and support vector machines. For some of the algorithms, we analyze the convergence.	
Study Goals	See the Mastermath page.	
Education Method	There will be 14 lectures.	
Literature and Study Materials	- Convex Optimization by Stephen Boyd and Lieven Vandenbergh (http://stanford.edu/~boyd/cvxbook) - Additional notes distributed by the lecturer	
Assessment	The course will have a written exam that counts for 100% of the grade. The resit will also be a written exam.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	This is a LNMB Mastermath course taught by David de Laat.	

WI4209	Systems and Control	6
Responsible Instructor	Prof.dr.ir. M.B. van Gijzen	
Contact Hours / Week x/x/x/x	1st semester intensive course	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Remarks	This course is part of MasterMath. Please consult the MasterMath website https://elo.mastermath.nl for information on this course	

WI4212	Advanced Numerical Methods	6
Responsible Instructor	Dr.ir. S. Jain	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Introductory numerical analysis, Introductory partial differential equations, Introductory continuum mechanics.	
Course Contents	This course is an introduction to hyperbolic partial differential equations and a powerful class of numerical methods for approximating their solution, including both linear problems and nonlinear conservation laws. These equations describe a wide range of wave propagation and transport phenomena arising in nearly every scientific and engineering discipline. Several applications are described in a self-contained manner, along with much of the mathematical theory of hyperbolic problems. High-resolution versions of Godunov's method are developed, in which Riemann problems are solved to determine the local wave structure and limiters are then applied to eliminate numerical oscillations. These methods were originally designed to capture shock waves accurately, but are also useful tools for studying linear wave-propagation problems, particularly in heterogeneous material.	
Study Goals	apply numerical methods to hyperbolic systems, study convergence, stability, monotonicity, know methods for 2 dimensional hyperbolic systems	
Education Method	Lectures	
Literature and Study Materials	Finite volume methods for hyperbolic problems R.J. LeVeque Cambridge, UK: Cambridge University Press, 2002. # ISBN-10: 0521009243 # ISBN-13: 978-0521009249	
Assessment	The final grade of the course consists of the following components: - Homework assignments - Group assignment - oral exam Final grade calculation: Provided that G_3 is larger than or equal to 6, the final grade is computed by the formula: $(G_1 + G_2 + 2 \cdot G_3)/4$. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - homework assignments: resubmit deliverable - Group assignment: resubmit deliverable - Oral exam: oral resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. C.A. Urzúa Torres	

WI4221	Control of Discrete-Time Stochastic Systems	6
Responsible Instructor	Prof.dr.ir. J.H. van Schuppen	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	- Required: Mathematical analysis as provided at the bachelor level; and linear algebra of the bachelor level; - Strongly advised: (1) knowledge of elementary probability theory; (2) control and system theory of deterministic linear systems.	
Course Contents	Discrete-time stochastic systems, distributions and invariant measures. Stochastic realization. Control with complete observations, optimal control theory, dynamic programming for finite and infinite horizons. Kalman filtering and special cases of filtering of stochastic systems. Control with partial observations, separation property. Control of networked stochastic systems and decentralized control.	
Study Goals	By the end of the course, a course participant should be able: LO1. To analyse the conditional expectation of a Gaussian random variable conditioned on another such variable, of a finite-valued random variable conditioned on another such variable; and the concept of conditional independence. LO2. To analyse elementary properties of stochastic processes. LO3. To evaluate properties of a Gaussian stochastic system, of a finite stochastic system, and of a general stochastic system, in particular the forward and the backward system representations of such systems and their relations; and the concepts of stochastic observability and of stochastic co-observability. LO4. To analyse the results of a stochastic realization of a stationary Gaussian process, in particular the equivalence conditions of minimality of such a realization and the formulation of the set of equivalent minimal realizations; the results of a realization of a deterministic linear system, and the concept of dissipativity of such a system. LO5. To analyse a stochastic control system, in particular the stochastic controllability of such a system. LO6. To evaluate the dynamic programming procedure, the value function, and the optimal control law, for optimal control of a completely-observable stochastic system, both on a finite horizon and on an infinite horizon. LO7. To evaluate the Kalman filter of a Gaussian stochastic system, the Kalman predictor of a Gaussian stochastic system, and the filter of a finite stochastic system. LO8. To evaluate the dynamic programming procedure, the value function, and the optimal control law, for control of a partially-observable stochastic system both on a finite horizon and on an infinite horizon. LO9. To understand several of the research issues of control of a distributed or a decentralized stochastic control system.	
Education Method	Lectures/exercises	
Literature and Study Materials	Notes for this course can be obtained from the instructor.	
Assessment	The final grade of the course consists of the following components: - Weekly homework sets (50%); - An oral exam to be carried out after the last lecture of the course (50%). Final grade calculation: $(0.5 * \text{the average grade of the weekly homework sets}) + (0.5 * \text{the grade of the oral exam})$ The homework sets count for 50% if solutions of all sets were sent to the lecturer (proportionally if fewer solutions were provided). Repair possibilities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5,	

	sub 5., for: - Homework sets: submit 1 extra homework set. - Oral exam: A second oral exam in the next quarter. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Elective	Yes
Tags	Mathematics Signals and Systems Stochastics

WI4260TU	Scientific Programming for Engineers	3
Responsible Instructor	Dr.ir. H.X. Lin	
Practical Coordinator	Ir. C.W.J. Lemmens	
Contact Hours / Week x/x/x/x	0/0/2/0 + lab	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Required: experience with a programming language (e.g., Python or C), and linear algebra Strongly advised: experience with linux is a plus (but not a prerequisite).	
Course Contents	The course tries to bring students to a level where they are able to change algorithms from e.g., numerical analysis into efficient and robust programs. It comprises 1. Introduction to programming in general; 2. Floating point number rounding-off errors and numerical stability; 3. (Numerical) Software design; 4. Data Structures; 5. Testing, debugging, and profiling; 6. Efficiency issues in computing time and memory usage; 7. Optimization and dynamic memory allocation; 8. Scientific software sources and libraries. P.S. (i) This is not a general introductory course to learn how to write a computer program. Furthermore, the course concentrates mainly on sequential programming and only briefly introduces parallel programming (MPI and OpenMP). More advanced topics like threads or parallel (MPI/GPU) programming on supercomputers are not covered by this course (they are covered by other courses, e.g., Introduction to High-Performance Computing). (ii) This is a course aimed at master students, however, if you have followed the course Scientific Programming in the BSc CSE-minor, then you should not select this course as a Master elective course due to the large overlap.	
Study Goals	1. Is able to explain the problems related to the use of floating-point numbers in scientific computing; 2. Is able to implement, compile, and debug programs in a Linux programming environment, e.g., use of a makefile. 3. Is able to explain the impact of data structures on the performance of a program, e.g., data locality, cache and memory access. 4. 3. Is able to explain the impact of the program on the performance of a program, e.g., temporal locality, pipelining, and loop-carried dependency. 5. Is able to analyze the performance of a program, and can apply optimization, debugging, and profiling tools to improve of the program. 6. Is able to make the transition from a scientific model to a structured program;	
Education Method	Weekly there are 2-hour lectures and 3-hour lab sessions. Participation in a minimum of five lab sessions is required, with the optional homework, a bonus of max. 1 point can be obtained.	
Books	Textbook: Writing Scientific Software - A guide to good style, by Suely Oliveira and David Stewart, Cambridge University Press, ISBN-13 978-0-521-67595-6	
Assessment	1. Written exam: 3-hour exam comprising theory and practical (lab) questions. (90%) 2. Lab homework (10%) Final grade calculation: $(0.9 \cdot \text{Exam} + 0.1 \cdot \text{Lab homework})$ Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Written exam: resit in next quarter - Homework assignments: resubmit deliverable	
Tags	Disclaimer: information may change depending on unforeseen circumstances or measures (see TER Art 29, sub 4). Numeric Methods Programming	

WI4450	Special Topics in Computational Science and Engineering	6
Responsible Instructor	Prof.dr.ir. M.B. van Gijzen	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course discusses the state-of-the-art of a selected topic from computational science and engineering. This topic changes yearly. The first part of the course gives an introduction into the subject through lectures in Q3. In the second part of the course (in Q4) a project will be carried out to apply the theory and methods taught in the first part of the course. The topic of the course of year 2023/2024 is Nonlinear Dynamics and Computational Model Reduction. It will be given by Shobhit Jain and Martin van Gijzen Numerical models relevant for applied and science and engineering often deal with increasingly intricate physical processes, leading to complex governing equations that are often high-dimensional or only partially understood. Furthermore, most of these phenomena are dynamic and intrinsically nonlinear, which makes computational predictions challenging, if not entirely infeasible. In this course, a student will learn the essential concepts in dynamical systems relevant for obtaining rigorous reduced-order models for very high-dimensional systems. In addition to classical techniques which are provably effective for reducing linear systems, we will discuss the latest research and related computational tools that enable robust model reduction for nonlinear systems.	
Study Goals	At the end of the course the student: - Will have a good overview of the current research on the course topic; - Is able to read and understand the recent literature; - Is able to perform research on the topic.	
Education Method	Introductory lectures, lectures by external specialists, project	
Literature and Study Materials	Lecture notes, relevant research papers and a list of recommended background material will be made available via Brightspace.	
Prerequisites	The course WI4201 is prerequisite knowledge.	
Assessment	The final grade of the course consists of the following components: - Project presentation (50%) - Report (50%) Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Presentation: resit presentation - Report: resubmit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI4520	Nonlinear Analysis and Partial Differential Equations	6
Responsible Instructor	Dr. M.V. Gnann	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	required: courses in the bachelor in analysis and differential equations. The most important courses are: * Ordinary Differential Equations (AM2030) * Partial Differential Equations (AM2070) * Real Analysis (AM2090) recommended (useful but not necessary) courses in the bachelor and master in applied mathematics are: * Applied Functional Analysis (WI4203) * Dynamical Systems and Chaos (WI4660) * Fourier Analysis (AM3570) * Nonlinear Differential equations (WI4019) * Perturbation and Variational Methods for Partial Differential Equations (WI4655) * Scientific Computing (WI4201) * Spectral Theory of Linear Operators (WI4046)	
Course Contents	In the first part of the course, we will treat the Stokes and Navier-Stokes equations and construct weak and strong solutions to the latter. This will rely on spectral methods, embedding theorems for Sobolev spaces, interpolation methods, and compactness arguments. Thus, we will prove existence and uniqueness of global weak and strong solutions to the (time-dependent) Stokes equations, existence of global weak solutions to the Navier-Stokes equations, global uniqueness for the 2D Navier-Stokes equations, and existence and uniqueness of strong solutions to the Navier-Stokes equations for short times or small initial data. A discussion on the third Millennium Prize Problem rounds up the first half of the lecture. In the second part of this course we give an introduction to stability theory for nonlinear partial differential equations (PDE). Determining the stability of solutions is of central importance when analyzing PDE arising in applications, as it is typically the stable solutions that are observed in practice. This analysis is particularly challenging when the models are nonlinear, and often relies on the presence of symmetries and corresponding conservation laws, which are a common phenomenon in physical models. We will draw inspiration from linearization and energy methods for stability theory in finite-dimensional dynamical systems and show how these concepts can be adapted to the infinite-dimensional setting when studying nonlinear Hamiltonian PDE. We will apply the theory to several equations modeling wave propagation, such as the Klein-Gordon equation, the Nonlinear Schrödinger equation and the Korteweg-deVries equation.	
Study Goals	After the first part of the course the student can prove existence, (short-time) uniqueness, and regularity of solutions to equations from fluid dynamics, namely * the time-independent Stokes equations * the time-dependent Stokes equations * the Navier-Stokes equations. After the second part of the course the student can * formulate some nonlinear dispersive equations as Hamiltonian systems; * derive conservation laws from symmetries of Hamiltonian system; * characterize traveling waves as critical points of the Hamiltonian functional; * prove that (constrained) minimizers of this functional are (non)linearly stable; * analyse when saddle points of this functional are (non)linearly unstable. * apply the theory to prove (in)stability of solutions of model equations	
Education Method	Lectures	
Assessment	In each of the two quarters, a grade (Q_1/Q_2) is determined based on an oral exam (O_1/O_2), counting 80%, and solutions to exercise sheets (E_1/E_2), counting 20%. The arithmetic mean of these grades determines the final grade (F). The course is passed if the average grade is at least 6 points and if in each quarter a grade of at least 5 points is obtained. Rounding is done according to TU Delft regulations. It is possible to retake the oral exam, and if necessary the lectures and exercises in the following year, for each quarter independently until the passing criterion is fulfilled. In formulas: $Q_1 = 0.8 \cdot O_1 + 0.2 \cdot E_1$ $Q_2 = 0.8 \cdot O_2 + 0.2 \cdot E_2$ $F = (Q_1 + Q_2) / 2$ Passing criterion: $F \geq 6 \text{ and } Q_1 \geq 5 \text{ and } Q_2 \geq 5$ Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	This course is offered over two quarters. It is a master elective for the specializations Computational Science and Engineering (CSE) and in partial differential equations (PDE).	
Co-Instructor	Dr. A. Geyer	

WI4635	Linear Algebra and Optimization for Machine Learning	6
Responsible Instructor	Dr. A. Heinlein	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	The course material and literature are mostly self-contained, but prior knowledge equivalent to contents of the courses AM2010 Linear Algebra 2 and AM2020 Optimization from the Bachelor of Applied Mathematics programme is strongly recommended. Programming experience in Python or Matlab is expected.	
Course Contents	State-of-the-art machine learning algorithms combine techniques from many different areas of mathematics. In this course, we will discuss algorithmic foundations from the areas of linear algebra and optimization. This also includes aspects of their theory and efficient implementation. Among others, we will study + matrix decomposition and factorization techniques, + regression and classification problems and regularization, + iterative and continuous optimization methods, + hyper parameter optimization, + graph-based algorithms and clustering, + basic introduction to neural networks and algorithmic aspects neural networks (e.g. computational graphs and back-propagation)	
Study Goals	Upon completion of the course, the student should be able to: + Categorize typical data sets based on their structure + Derive suitable mathematical formulations for common supervised and unsupervised machine learning tasks + Choose suitable algorithmic building blocks for solving mathematical formulations of common machine learning tasks + Analyze machine learning algorithms with respect to their properties, such as efficiency and robustness + Implement machine learning algorithms in a computationally efficient way	
Education Method	The course comprises 14 lectures and 14 exercise classes.	
Literature and Study Materials	Lecture notes, slides, and the main literature: + Charu Aggarwal. Linear algebra and optimization for machine learning. Springer, 2020. + Gilbert Strang. Linear algebra and learning from data. Wellesley-Cambridge Press Cambridge, 2019. + Jeremy Watt, Reza Borhani, and Aggelos K Katsaggelos. Machine learning refined: Foundations, algorithms, and applications. Cambridge University Press, 2020.	
Assessment	The final grade of the course consists of the following components: - Two group projects (0.4) - Written exam (0.6) Final grade calculation: $(0.4 * \text{group projects} + 0.6 * \text{written exam})$ Note: For the written exam, a score of at least 5.0 must be obtained. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Group projects: resubmit deliverable - Written exam: There will be a resit for the written exam at a later point in the same academic year Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	D. de Laat	

WI4655	Perturbation and Variational Methods for Partial Differential Equations	6
Responsible Instructor	Dr. H.M. Schuttelaars	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Course Contents	In the course two different approaches to analyse PDEs will be presented. The first method will focus on approximating solutions to PDEs. In the lectures we elaborate on asymptotic approximations, matched asymptotic expansions, multiple scales, WKB method, bifurcation and stability analysis. In the second approach, a modern introduction to the field of elliptic PDEs is given. In the lectures we consider weak solutions to elliptic equations (such as Laplace equation), Sobolev spaces. Modern PDEs uses tools from functional analysis and measure theory. The precise regularity and behavior of solutions of elliptic equations is one of the main features in the lectures. For non-linear PDEs such results are fundamental. The course might be useful for student in analysis, PDE and numerics, or anyone who wants to know more about Sobolev spaces and second order equations. (Key words: elliptic, weak derivatives, Sobolev spaces extension theorem, trace theorem, Sobolev inequalities, Hilbert techniques, compactness theorems, second order elliptic equations, Weak solution, energy estimates, regularity, eigenvalue techniques).	
Study Goals	Global objectives for the Approximation approach: After following this course the student will be able to apply fundamental and advanced analytical techniques and methods to study properties of solutions of ordinary and partial differential equations. The mathematical methods and techniques that the student is able to use are mentioned under "course contents". After studying the lecture notes and after making the exercises successfully the student can start to work on research problems in the field of nonlinear differential equations. Global objectives for the Variational approach: After the course the student - knows what a weak derivative is - understands the definition of L^p and Sobolev spaces. - can apply basic Sobolev space techniques (approximation, extension, trace, Sobolev embedding, compactness) - can give the definition of a weak solution of a second order elliptic PDE - can apply Sobolev space techniques to prove existence and uniqueness of weak solutions of second order elliptic PDEs - can derive properties (regularity/estimates) of solutions of second order elliptic PDEs Objective per topic: Sobolev spaces the student - knows the definition of Hölder spaces - knows the definition of the weak derivative - can calculate weak derivatives - knows the definition of Sobolev spaces - can approximate Sobolev functions by smooth functions - can explain the extension theorem - can explain the trace theorem - knows Sobolev inequalities - can apply Sobolev inequalities - can derive Sobolev inequalities - knows the compactness theorem - can apply the compactness theorem - understands what the dual space of a Sobolev space is Second order elliptic equations the student - knows the definition of a weak solution. - can give energy estimates - can apply the Lax-Milgram theorem to obtain existence of weak solutions - knows what regularity is - can state regularity results for second order equations (interior and boundary) - can apply eigenvalue techniques.	
Education Method	Perturbation approach: Introduction to Perturbation Methods (second edition) by M.H. Holmes Variational approach: lecture notes. Optional literature: Partial Differential Equations, by Lawrence C. Evans, AMS, 1998, 662 pp., Hardcover, ISBN 0-8218-0772-2. Assessment	
Assessment	The final grade of the course consists of the following components: - Take home assignments (two per quarter) - final oral exam Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Take home assignments: resubmit deliverable - Oral exam: oral resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr.ir. M.C. Veraar	

WI4680	Applications in Partial Differential Equations	6
Responsible Instructor	Dr. H.M. Schuttelaars	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Ordinary differential equations, Numerical methods, Programming (Python or other)	
Course Contents	Many ecosystems and physical systems, such as the climate, can display strongly nonlinear behaviour, where the properties of the system suddenly change with changes in environmental conditions. A well-known and urgent example of this is the discussion about tipping of the climate, thought to set off a dramatic change in the Worlds climate if global temperatures start to exceed some threshold. Mathematically, tipping is an example of a so-called bifurcation of the underlying system of equations. We will focus on the bifurcations in systems of ordinary differential equations (ODEs) and partial differential equations (PDEs) representing real-world systems, which can often only be solved numerically. Hence, this course focusses on numerical bifurcation theory: the study of understanding parameter sensitivity and bifurcations in numerical solutions. The course covers mathematical theory and papers applying the theory to real-life applications including climate science and ecosystems. We build on theory of ODEs, numerical methods and mathematical modelling. Basic programming experience is required for the practical assignments.	
Study Goals	After following this course the student is able to 1. Describe various non-linear phenomena appearing in systems of ordinary and partial differential equations. These include tipping, appearance of periodic solutions, pattern formation, flickering and critical slowing down. 2. Apply numerical continuation, stability analysis and branch switching around bifurcations for common elementary bifurcations (transcritical, pitchfork, limit point, Hopf) in the context of a simple model with physical applications. 3. Explain practical problems with respect to computation time, computer memory and numerical accuracy in the above techniques and various solutions to these problems	
Education Method	Lectures and practical assignments The course consists of 14 lectures during which theory and papers will be discussed. The discussion of the papers is led by the students and hence active participation in the lectures is required. In the practical assignments, the students gain hands-on experience working with the learned techniques and revisit theory on ODEs and numerical methods. The assignments feature a significant amount of programming.	
Literature and Study Materials	Seydel, R. Practical bifurcation and stability analysis. Vol. 5. Springer Science & Business Media, 2009. Various papers available on Brightspace.	
Assessment	The final grade of the course consists of the following components: - Take home assignments - oral exam Final grade calculation: $(0.5 * \text{take home assignments}) + (0.5 * \text{oral exam})$ Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Take home assignments: resubmit deliverable - Oral exam: oral resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI4771TU	Object Oriented Scientific Programming with C++	3
Responsible Instructor	Dr. M. Möller	
Contact Hours / Week x/x/x/x	0/2/0/0 & 0/4/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic programming knowledge in at least one programming language (e.g. Matlab, C/C++, Fortran)	
Course Contents	Introduction to C++ programming including new concepts from the C++11/14/17/20 standard. Object-oriented scientific programming.	
Study Goals	The student will be able to apply modern software design patterns and state-of-the-art programming techniques to implement (numerical) algorithms in C++ (using the latest standards C++11/14/17/20). The student will build up practical experience in turning a textbook (numerical) algorithm into an efficient and maintainable C++ code. The student will be able to develop larger software projects in teams and test the implementation systematically.	
Education Method	Hands-on lectures and practical sessions in Q2. Problem-oriented teaching approach starting from a concrete (mathematical) problem (taken from introductory numerical analysis course) and building up knowledge in C++/scientific programming up to the level required to solve the problem. Further self-study is required to build up in-depth practical experience. Implementation of final project (in groups of 1-3 students) starting mid Q2 and with submission deadline end Q2 / beginning Q3.	
Literature and Study Materials	Complete course material (slides and source code of demo applications) will be provided. Though not mandatory, the book "A Tour of C++" by Bjarne Stroustrup, C++ In-Depth Series, Addison-Wesley, ISBN-13 978-0-321-95831-0 gives a good overview of the covered material.	
Prerequisites	Basic programming knowledge in at least one programming language (e.g. Matlab, C/C++, Fortran)	
Assessment	The final grade of the course consists of the following components: - Homework programming assignments (10 in total) for each session during Q2 (1/3 of final grade). - Group programming project (group work) starting mid Q2 with submission deadline end Q2 (2/3 of final grade). Final grade calculation: $(1/3 * \text{homework assignments}) + (2/3 * \text{group programming project})$ but it is only valid when all homework programming assignments have been completed with sufficient results. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Homework programming assignments: submit new set of homework - Group programming project: submit improved version	
Co-Instructor	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). K. Cools	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Specialisation Financial Engineering AM 2023

Introduction 1

Specialisation Financial Engineering

Compulsory orientation course:

WI4675 Introduction to Financial Mathematics

Compulsory specialisation courses:

WI4430 Martingales, Brownian Motion

WI4201 Scientific Computing

WI4079 Financial Mathematics

WI4154 Computational Finance

Recommended specialisation courses:

WI4019 Nonlinear Differential Equations

WI4230 Time series and Extreme Value Theory

WI4052 Risk Analysis

WI4156(TU) Game theory

WI4205 Applied Finite Elements

WI4224 Special Topics in Financial Engineering

WI4505 Quantitative Risk Management

WI4614 Stochastic Simulation

WI4425 Financial Markets Theory

WI4635 Linear Algebra and Optimisation for Machine Learning

WI4630 Statistical Learning

WI4771TU Object Oriented Scientific Programming with C++

WI4615 Stochastic Calculus

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Compulsory Courses

WI4079	Financial Mathematics	6
Responsible Instructor	Dr. A.F.F. Derumigny	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	This is a compulsory course of the Financial Engineering Specialization. For other students, this course can be chosen as an elective, but it must be stressed that this is a highly mathematical course. Therefore, students from other faculties are welcome, no need to write in advance, but they should be aware that this is a highly formalized and difficult course, belonging to the Master of Science in Applied Mathematics: the treatment of financial topics is at a high level of formalization. All students following this course need to be familiar with the contents of "Martingales, Brownian Motion, and Stochastic Processes" (WI4430) and with the contents of "Introduction to Financial Mathematics" (WI4675). In particular, knowledge of the notion of probability measure is fundamental.	
Course Contents	The aim of the course is to introduce the students to the probabilistic study of the classical theory of mathematical finance. The first part of the course is centered around stochastic calculus. In this first part, we introduce a detailed theory of stochastic integration that will be the basis for the second part of the course (which will be more financial and rely on the mathematical theory as studied on the first part). In the second part of the course, we will then cover arbitrage theory, perfect markets, risk-neutral pricing of options and other derivatives, Black-Scholes-Merton model, models for exchange rates and interest rates. For each topic and model we will stress its points of strength, but also its points of weakness.	
Study Goals	At the end of the course, the students will be able to: - explain and apply the theory of stochastic integration with respect to martingales and semi-martingales (Ito calculus) and reproduce the proofs seen during the lectures; - compute the price of a derivative; - choose relevant changes of measure for derivative pricing and explain the impact of such an operation in pricing and in financial mathematics more generally; - explain the importance of risk-neutrality; - model basic exchange and interest rates.	
Education Method	Lectures, exercises and possibly case studies. Note that this is an in-person course and that online learning is not supported.	
Literature and Study Materials	TBD. However, the book by Shreve (2004), Stochastic calculus for finance II: continuous models (Springer, New York) gives a good indication of contents and level of the course.	
Prerequisites	Students need to have completed both of the following courses: - "Martingales, Brownian Motion, and Stochastic Processes" (WI4430) - "Introduction to Financial Mathematics" (WI4675)	
Assessment	The final grade of the course consists of the following components: - Written or oral exam (100%). The type of examination (written or oral) will be determined based on the number of students. In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5 Written or oral exam: written or oral resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI4154	Computational Finance	6
Responsible Instructor	F. Fang	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic knowledge of Ito's lemma, partial differential equations (PDEs), linear algebra and the basic of numeric analysis such as numerical integration rule, computing tool: Python.	
Course Contents	The course aims at introducing the basic option pricing theory and various computational methods to price options in practice. More precisely, the derivation of pricing PDE and Feynman-Kac theory, derivation of the characteristic function for affine processes, Finite difference method to solve the PDE numerically, Monte-Carlo method and Fourier method to price the options. An overview of the financial products traded nowadays will also be covered.	
Study Goals	For different mathematical models, for products from the financial industry, different numerical mathematics techniques are required. The students learn about the numerical techniques used in finance. Finite differences for PDEs, Monte Carlo techniques for complicated integrals and Fourier-based integration are taught for the numerical side; For the financial product modelling side, we discuss asset pricing processes and a variety of options, from basic European options to options with early-exercise opportunities and so-called exotic options. Furthermore, details about financial products are given.	
Education Method	Lectures	
Literature and Study Materials	The course will be taught based on slides and some background material. The book "Mathematical Modeling and Computation in Finance", by C.W. Oosterlee and L.A. Grzelak, World Scientific Press, 2019, is used as background information.	
Assessment	The final grade of the course consists of the following components: - two sets of exercises, which together count for 20% of the final score. - One intermediate take-home exam will take place in the middle of the course, and will count for 30% of the final score. - There will be a final, oral exam at the end, which accounts for 50% of the final score. Final grade calculation: $(0.2 * \text{exercises}) + (0.3 * \text{take-home exam}) + (0.5 * \text{oral exam})$ Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - exercises: resubmit 1 set of exercises if the grade of the 2 sets is insufficient - take-home exam: resubmit deliverable - oral exam: oral resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	The course is linked to WI4079 (Financial Mathematics'), where the stochastic concepts of finance are introduced.	

WI4201	Scientific Computing	6
Responsible Instructor	Prof.dr.ir. C. Vuik	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	A basic knowledge on partial differential equations (PDEs), on numerical methods for solving ODEs/PDEs, and on linear algebra.	
Course Contents	During the course, the important steps towards the solution of real-life applications dealing with partial differential equations will be outlined. Based on a well-known basic partial differential equation, which is representative for different application areas, we treat and discuss direct and iterative solution methods from numerical linear algebra in great detail. The discretization of the equation will result in a large system of discrete equations, which can be represented by a sparse matrix. After a discussion of direct solution methods, the iterative solution of such systems of equations is an important step during numerical simulation. Emphasis is laid upon the so-called Krylov subspace methods, like the Conjugate Gradient Methods.	
Study Goals	A. General Study Objectives 1. After having completed this course, the student is able to use finite differences to approximate the solution of a partial differential equation and use an efficient and robust method to compute or approximate the solution of the resulting large system of linear equations. 2. The student will be able to assess the reliability in terms of the accuracy and stability of the numerical approximation of the solution. 3. The student will be able to use methods to approximate eigenvalues of large matrices. B. Specific Study Objectives 1. Finite Difference Method: Boundary Conditions, Grid, Numbering of the unknowns, Error Analysis - The student will be able to discretize 2-dimensional Poisson type problems. The student is able to implement the boundaries in various ways. - The student will be able to analyze the properties of the resulting matrix, as there are symmetry, positive definiteness, M-matrix etc. Also the sparseness of the matrix can be investigated and a notion of band and profile matrices should be known. 2. Direct Solution Methods Gaussian elimination, pivoting, iterative refinement, band and profile methods - The student will be able to understand and use direct solution methods based on Gaussian elimination and LU decomposition. - The student will be able to estimate the error due to rounding errors in the answer and to use pivoting to eliminate most of the errors. - The student will be able to estimate the amount of flops for direct solution methods and how to use more efficient variants as there are Cholesky method, band and profile methods. 3. Basic Iterative Methods - The student will be able to construct Basic Iterative Methods based on the splitting of the matrix. Furthermore an analysis of the convergence could be made. - A comparison of the various method should be possible together with the dependence on the gridsize. - The student will be able to choose suitable starting vectors and stopping criteria. 4. Krylov Subspace methods CG, SPD, preconditioning, general matrices, CGNR, Bi-CGSTAB and GMRES - The student will be able to use and understand the conjugate gradient method, together with a good understanding of the (super)linear convergence. - The student will be able to make a good choice for Krylov Subspace methods in the non-SPD case. - The student will be able to combine Krylov Subspace methods with preconditioners and know three classes of preconditioners: diagonal scaling, BIM and incomplete decompositions. 5. Eigenvalue methods Power method, Lanczos method, Arnoldi method - The student will be able to use the power method and is able to investigate its convergence. - The student will be able to choose suitable starting vectors and stopping criteria. - The student will be able to use the Lanczos and Arnoldi method.	
Education Method	Lectures/computer exercises	
Literature and Study Materials	Lecture notes, for further reading the book Matrix Computations, G.H. Golub and C.F. van Loan, the Johns Hopkins University, Baltimore, 2013, can be used.	
Assessment	The final grade of the course consists of the following components: - homework exercises deadline start of Q2 leads to grade G1, - take home exam deadline half of January grade G2 - written exam grade G3. The final grade is $(G1+G2+2*G3)/4$, provided that all grades are larger than or equal to 5. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - homework exercises: resubmit deliverable - take home exam: resubmit deliverable - written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Prof.dr.ir. M.B. van Gijzen	

WI4430	Martingales, Brownian Motion	6
Responsible Instructor	Prof.dr. F.H.J. Redig	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	1. Conditional expectation. 2. Martingales, stopping times, optional sampling. 3. Martingale convergence theorems, inequalities. 4. Elementary continuous-time stochastic processes: Poisson process, continuous-time Markov chains. 5. Brownian motion: definition, construction, basic properties. Martingales associated to Brownian motion, brownion motion with drift. 6. Elementary introduction to the stochastic integral, Ito isometry	
Study Goals	1. Understanding the concept of conditional expectation. Being able to compute conditional expectations in concrete examples, and being apply the tower property, and the take-out-what-is-known property. 2. Being able to recognize whether a process is a martingale, sub- or super martingale. 3. Being able to use martingales in computations such as gambler's ruin and distributions of hitting times. 4. Being able to apply the martingale convergence theorems and martingale inequalities in concrete contexts such as law of large numbers, branching process. 5. Being able to perform concrete computations with Brownian motion and its martingales (such as computation of hitting time distribution, recurrence transience). 6. Understanding the definition and construction of the stochastic integral and being able to perform elementary computations with it. 7. Being able to reproduce correctly definitions, statements and proofs and the main definitions and theorems as indicated in a list (this concerns the so-called theory question).	
Education Method	Lectures and problem sessions.	
Literature and Study Materials	1. Lecture notes based on chapter 10 of A. Gut ``A graduate course in probability" for the martingale part of the course. 2. Lecture notes for Brownian motion part of the course. The lecture notes are provided via Brightspace (course content).	
Reader	1. Lecture notes based on chapter 10 of A. Gut ``A graduate course in probability". 2. For the material on Brownian motion we will also sometimes use P. Morters and Y. Peres, Brownian Motion, (2010) freely available at http://people.bath.ac.uk/maspm/book.pdf . Also lecture notes will be available for that part. 3. Other recommended literature: Le Gall: Brownian motion, martingales and stochastic calculus, Springer 2016.	
Assessment	The final grade of the course consists of the following components: - Two graded homeworks - a written exam A homework grade h is computed as the average of the scores on the two graded homeworks. Course grade is the final exam grade f or $0.6f+0.4h$, whichever is larger, provided $f > 5$. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - homework assignments and written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). The homework grade does not count for the resit.	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Recommended Courses

WI4019-SP	Non-linear Differential Equations	6
Responsible Instructor	Dr.ir. W.T. van Horssen	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Introductory course on Differential Equations e.g. Boyce and DiPrima Chapters 2-7, 9.	
Course Contents	Existence and uniqueness of solutions of initial value problems, Gronwall's lemma, Autonomous systems, Critical points, Periodic solutions, Stability theory, Linear systems and Floquet theory, Perturbation theory and asymptotic methods, Poincare-Lindstedt method, Averaging method, Multiple time-scales method, Elementary bifurcations.	
Study Goals	After following this course the student will be able to apply fundamental and advanced, analytical techniques and methods to study properties of solutions of nonlinear ordinary differential equations. The mathematical methods and techniques are mentioned under "course contents". After studying the lecture notes and after making the exercises successfully the student can start to work on research problems in the field of nonlinear differential equations.	
Education Method	Lectures and a MAXIMA practical work. In total 14 lectures of 2 hours each will be given. The MAXIMA practical work is about 4 hours work. Following and preparing the lectures and the MAXIMA practical work takes about 60 to 70 hours. Making the 30 take-home exercises takes about 60 hours. Preparing the 45 minutes oral exam takes about 30 hours.	
Literature and Study Materials	F. Verhulst, Nonlinear Differential Equations and Dynamical Systems, Springer-Verlag 2nd edition, ISBN 3-540-60934-2.	
Assessment	The final grade of the course consists of the following components: - Take home exam (50%) - oral exam (50%) Final grade calculation: $(0.5 * \text{take home exam}) + (0.5 * \text{oral exam})$ The take-home exam consists of 30 exercises. The oral exams will be held at the end of the 2nd quarter, and the student is only allowed to take part in this oral exam when at least 70% of the take-home exercises are made correctly and when the MAXIMA practical work is followed successfully. In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5 take home exam: resubmit deliverable oral exam: oral resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI4052	Risk Analysis	6
Responsible Instructor	Dr. D. Kurowicka	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic Probability and Statistics	
Summary	Risk analysis course will treat problems often appearing in applications. Some generic methods that are applied in many different areas of applications will be discussed. We will not concentrate on any particular application, rather we will discuss important assumptions made in different models and their applicability for different real life problems. Moreover we show how parameters of these models can be obtained from data and discuss methods of validation of the obtained model.	
Course Contents	Risk modelling, life distributions, event trees, fault trees, Bayesian Belief Nets, reliability data bases, dependence modelling, software reliability, expert judgment.	
Course Contents Continuation	Introduction & Uncertainty Probability Multivariate, Copula Stochastic processes Bayesian, classical estimations & Testing Fault trees Bayesian belief nets Dependent failures Expert Judgment Reliability data bases Software reliability	
Study Goals	Students must be able to justify use of probabilistic methods in risk analysis and apply techniques like fault trees and Bayesian belief nets. They should know how to estimate parameters of models for dependent failure rates. Moreover they should be able to perform competing risks analysis for simple reliability data sets. Students should understand basic techniques of incorporating expert knowledge into risk models.	
Study Goals continuation	The goal of the course is to expose students to problems appearing in applications. During lectures the theory will be treated and exercises will be provided. Students will be required to complete the assignment where some practical problems have to be solved. The theoretical questions are not required. However making them allows to get a feedback and extra explanation in how they should be solved. These exercises together with course material will constitute basis for the exam in January. Practical exercises are required and will form part of your final grade.	
Education Method	Lectures and Assignments	
Literature and Study Materials	Parts of the book "Probabilistic Risk Analysis", Bedford & Cooke, Cam.U. press, 2001. Extra material on the Brightspace	
Reader	Brightspace	
Assessment	Written exam (100%). In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5 Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	Calculator and self made A4 with formulas.	
Tags	Mathematics Stochastics	

WI4156(TU)	Game theory	6
Responsible Instructor	Dr. R.J. Fokkink	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Required for	Open to all MSc students with a strong command of mathematics. Ideally suited for math students with an interest in finance.	
Expected prior knowledge	Standard BSc math courses: calculus, linear algebra, probability.	
Course Contents	This is a course in "mathematical" game theory, which is the counterpart of "algorithmic" game theory (agent based modelling, this part of game theory is taught in the computer science department). We cover all subjects from combinatorial games to cooperative games.	
Study Goals	Understanding the mathematical theory of rational behaviour/conflicts/negotiations/social choice. Game theory is one of the most prominent economic theories: three recent Nobel prizes have been awarded to game theorists. We will cover a very wide area, ranging from pure mathematics to economics, with an emphasis on the mathematical aspects. This is a course in the Finance Track of the Applied Maths programme, but it is open to all MSc students (math and non-math).	
Education Method	Lectures	
Literature and Study Materials	Game Theory - Thomas Ferguson - UCLA: find it on the net.	
Books	T.S. Ferguson, A course in game theory, World Scientific, 2020	
Assessment	The final grade of the course consists of the following components: written exam (100%) Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI4205	Applied Finite Elements	6
Responsible Instructor	D. Toshniwal	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The students should have had at least introductory courses on the following topics: * continuum mechanics * partial differential equations * tensor Calculus * linear algebra * numerical methods for ODEs	
Course Contents	Weak formulations (derivation, motivation for use, well-posedness) Abstract introduction to the finite element method (FEM) Theoretical analysis of FEM (e.g., of well-posedness, convergence) Practical implementation of FEM (e.g., for linear/non-linear steady/unsteady problems) in one and higher dimensions Verification and validation of FEM	
Study Goals	After this course, students should be able to: * derive a weak formulation for the give PDE * show the well-posedness of the derived weak formulations * construct isoparametric finite elements in one and higher dimensions * discretize the weak problem using isoparametric finite elements * show the well-posedness of the discretized problem * implement the finite element method * perform basic verification of the implementation correctness	
Education Method	Lectures.	
Literature and Study Materials	The following texts will serve as supplementary references to the class notes: * Course reader (downloadable from Brightspace) * T.J.R. Hughes. The Finite Element Method: Linear Static and Dynamic Finite Element Analysis, 1987. * P. Wriggers. Nonlinear Finite Element Methods, 2008. Other references will be provided during the course.	
Assessment	Course will be assessed based on assignments (40% of final grade) and a written exam (60% of final grade). Note: Some of the assignments may require the students to study material not covered in class. Similarly, some assignments will ask the students to work on small extensions or generalisations of the concepts covered in class. To pass the course, the following is mandatory: * Assignments: - The assignments (written and programming exercises) must be completed with an average grade greater than 5.5. - The assignments will have deadlines and late submissions will be penalised. * Exam: - The written exam must be completed with a grade greater than 5.5. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Group assignments: resubmit deliverable - written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	For more information see the Brightspace course page.	

WI4224	Special topics in Financial Engineering	6
Responsible Instructor	Dr. A. Papapantoleon	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course will deal with current topics in financial mathematics and financial engineering, such as the applications of machine learning methods in finance.	
Study Goals	- To understand the applications of machine learning methods in mathematical finance - To know the strengths and the weaknesses of all the discussed models. - To acquire useful and practical knowledge in these methods.	
Education Method	Lectures from the instructor and presentations from the students.	
Literature and Study Materials	# M. F. Dixon, I. Halperin, P. Bilokon: Machine Learning in Finance: From Theory to Practice. Springer, 2020. # H. Ni, G. Yu, J. Zheng and X. Dong, An Introduction to Machine Learning and Quantitative Finance. World Scientific, 2021.	
Prerequisites	"Introduction to Financial Mathematics" (WI4675), "Financial Mathematics" (WI4079), "Martingales, Brownian Motion" (WI4430), and "Computational Finance" (WI4154).	
Assessment	The final grade of the course consists of the following components: - Presentation (100%) Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Presentation: new presentation Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. F. Yu	

WI4230	Time Series & Extreme Value Theory	6
Responsible Instructor	Dr. P. Chen	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Knowledge of probability and statistics at the level of the BSc Applied Mathematics courses on these topics (AM1080, AM2080).	
Course Contents	The course consists of two parts: Extreme Value Theory and Time Series. The course considers questions such as: What is the chance that the price of oil will be higher than \$120/Barrel coming week? Can we expect that UK interest rates will rise next year? How can we assess if global warming is occurring? What is the chance that the daily loss of the stock price of ABN-AMRO is larger than, say 20%, which has never occurred before? How to determine the height of a dike such that the probability of a flood, i.e., the water level exceeding the dike, in a given year is 1/10,000? What is the impact of a dramatic decline in the US stock market on DAX 30 index? In this course students will build up some theoretical basis, mainly probabilistic results within the area of time series and extreme value theory. Moreover, students will learn statistical methods for dealing with non-stationary data, multivariate data and rare events and assessing extreme risks for data from various fields such as finance, actuarial science, hydrology and environmental science. Furthermore the student is capable to perform a statistical analysis with the statistical package R. The Extreme Value Theory part considers topics such as max-domain attractions, peak over threshold approach, estimations of extreme value index, tail probability and high quantile, and tail dependence and independence. The Time Series part will cover topics such as ARMA models, spectral analysis, non-stationarity, GARCH models, state space models. Implementations of statistical methods in R will be part of the course. Examples of real applications from environmental science and finance will also be discussed.	
Study Goals	Specific learning objectives for the Extreme Value Theory part are: 1. Validate the max-domain attraction conditions, that is to show if a given distribution is in the max domain of attraction or not. 2. Compute the extreme value index of a given distribution. 3. Develop estimations for tail probabilities, high quantiles, m-year return level, using different methods. 4. Prove some asymptotic results (consistency) of the estimators. 5. Evaluate the performance of estimation methods by simulation study; and apply the estimation methods to a real data set. 6. Establish the properties of exponent measure and spectral measure. Establish the link between the bivariate extreme value distribution and these two measures. 7. Characterize the extremal dependence for bivariate distributions. 8. Develop the estimation for a bivariate tail probability (failure probability). Specific learning objectives for the Time Series part are: 1. Discern whether a given time series data set can be associated to a stationary time series, and if not, perform suitable transformations to so that the resulting data corresponds to a stationary time series. 2. Understand and determine interpret theoretical properties of AR(I)MA time series models, such as the existence and uniqueness of solutions, causality and invertibility. 3. Understand and determine descriptive statistics of time series models, such as best linear predictors, autocorrelation function, partial autocorrelation function, estimators for mean and variance, and spectral density function. 4. Have general fluency in dealing with standard manipulations in time series, such as working with linear filters, computing conditional expectations and solving recurrence relations. 5. Fit a suitable time series model to time series data and assess the quality of a time series fit. 6. Have basic knowledge of advanced time series models used in financial mathematics, such as GARCH and stochastic volatility models.	
Education Method	Lectures/exercises/computer assignment	
Books	The lectures will be mainly based on: - Time Series Analysis and Its Applications: With R Examples By Robert H. Shumway, David S. Stoffer - Time Series: Theory and Methods By Brockwell, Peter J. and Davis, Richard A - Extreme Value Theory: An Introduction By Laurens de Haan and Ana Ferreira (Chapter 1, 3, 4, 6 and 7)	
Assessment	There will be a final assessment for the extreme value theory part at the end of Q3 and a final (written or oral) assessment for the time series part at the end of Q4. (If there are at least 10 students we will organize a written exam.) There will also be assignments handed out during the course. The final grade can be determined by a weighed sum, with the following weights: 0.15 average over assignments extreme value theory 0.35 exam extreme value theory 0.15 average over assignments time series	

0.35 exam time series

Resit/ Repair opportunities:

In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for:

- Individual assignments: resubmit deliverable
- Written exam: written resit OR
- Oral exam: oral resit

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

WI4425	Financial Markets Theory	6
Responsible Instructor	Dr. J.H.M. Anderluh	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	Choices under Uncertainty, General Equilibrium Theory, Pareto Optimality, Expected Utility Theory, Risk Aversion Choices under risk : Portfolio Problem, Insurance Demand en Prudence, Stochastic Dominance, Mean Variance Analysis Portfolio frontier, Mutual Funds separation, Risk sharing and Pareto optimality Asset Markets, Intertemporal Consumption, the F. theory of asset pricing, CAPM CAPM, APT, Portfolio Choice Multi Period Markets Information and Financial Markets Uncertainty, Rationality and Heterogeneity Financial Markets Microstructure Corporate Finance	
Study Goals	The main goal is to enable the student to quantify and in some cases optimise the relation between return and risk in a financial markets situation. The student will be able to build a mathematical model of the situation at hand with ingredients like equilibrium concepts, no-arbitrage concepts.	
Education Method	One block of 2 hours lectures every week during the 1st semester. A written exam afterwards.	
Books	Emilio Barucci, Financial Markets Theory	
Assessment	The final grade of the course consists of the following components: - Written exam (100%) Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI4505	Quantitative Risk Management	6
Responsible Instructor	F. Barsotti	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The implementation of sound and reliable quantitative risk models is of vital interest for all banks and financial institutions, both from an economic and a legal point of view. This course provides a rather comprehensive treatment of the theoretical concepts and modeling tools of quantitative risk management. At the end of the course, students will master both theory and practice, and they will be able to solve real world problems. We will cover the modeling of important risks like: market risk, operational risk and model risk. All models will be analyzed in their points of strength and, most importantly, their points of weakness, to address the main deficiencies of current practice.	
Study Goals	- To understand risk, epistemologically, statistically and in practice. - To be able to use the main risk measures available and, in case, to develop new ones. - To know the basics of EVT. - To be able to model market, credit and operational risks, using the most recent techniques. - To be able to estimate most of the models we will discuss together on real data. - To know the strengths and the weaknesses of all the discussed models, i.e. to understand the importance of model risk. - To acquire useful and practical knowledge in risk management.	
Education Method	Front lectures, as necessary supplemented by online materials including videos, case studies.	
Literature and Study Materials	McNeil, Frey and Embrechts: Quantitative Risk Management, Princeton University Press, 2015.	
Prerequisites	"Financial Mathematics" (WI4079) and "Computational Finance" (WI4154).	
Assessment	The final grade of the course consists of the following components: - Written / Oral exam (100%). Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Written exam: written resit OR - Oral exam: oral resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	Standard non graphical calculator.	
Special Information	If you are not a mathematics student, feel free to join the course, no need to ask, but be aware that this course is highly formalized and relies on good mathematical foundation knowledge. This is a course of the master in applied mathematics!	

WI4614	Stochastic Simulation	6
Responsible Instructor	Dr.ir. L.E. Meester	
Contact Hours / Week x/x/x/x	0/0/2/2 lecture & 0/0/2/2 projects.	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Martingales, Brownian Motion and Stochastic processes (wi4430 or equivalent course), in particular measure theoretic probability, and basic statistics, including linear regression. Familiarity with R, Matlab, or equivalent software for simulation, is recommended.	
Course Contents	Stochastic simulation: algorithms and analysis. Emphasis in this course is not in the first place on doing simulations, but on understanding (the workings of) simulation techniques and how simulation can be used to provide insight into stochastic problems; the most important aspect of the "doing simulations" part is that you can derive the simulation steps/algorithm that results when one of the covered methods is applied to a problem. Since the key in efficient simulation almost always lies with the stochastic specifics of the problem, we focus on the stochastic methods that play a role in this process. Some simulations will be done in the mini-projects, to supplement and illustrate the theory. Here, you are encouraged to "bring your own problem to work on". Subjects: introduction, general aspects of stochastic simulation; generating random objects, univariate and multivariate random variables, and stochastic processes; analysis of simulation output: how to obtain estimates for quantities of interest, as well as confidence intervals; bias and small sample issues; variance reduction, especially their stochastic background and optimization; in addition, a selection of some of the following: Quasi-Monte Carlo, discretization methods, estimating derivatives via simulation.	
Study Goals	After completing this course successfully, you will be able to 1. explain the general principles of pseudo random-number generators; 2. mention some of their important properties and how they may be assessed; 3. explain and apply the inverse transform and acceptance-rejection methods; 4. simulate, i.e., generate realizations of, random variables and random vectors; 5. do the same for some stochastic processes, among others Brownian motion and the Brownian bridge; 6. process simulation output into estimates plus standard errors/CIs; both for simple and more complex simulation schemes; 7. explain the principles of a number of variance reduction methods: control variates, antithetic variates, stratified sampling, importance sampling; 8. and apply them to problems of moderate size and complexity; 9. explain what quasi-Monte Carlo methods are and how the van der Corput and Halton sequences are generated; 10. explain some of the advantages and disadvantages of QMC methods, and what can be done about the disadvantages; 11. explain three methods of estimating parameter sensitivities or derivative estimation: finite-difference approximation, pathwise derivative estimates, the likelihood ratio method; 12. for each of them, work out the specific details of their application for small problems; In addition: 1. you will acquire some simulation experience, first-hand, including processing and presenting the results; 2. and second-hand by sharing others experience; 3. you will practice developing a critical attitude towards presented simulation results, 4. as well as asking the right questions; 5. you will acquire some insight into Where might I use/deploy what (method or trick) from the simulation toolbox to my advantage?	
Education Method	Lectures and a bi-weekly discussion/presentation/problem solving session; homework assignments;self-study.	
Literature and Study Materials	Paul Glasserman, Monte Carlo methods in Financial Engineering, 2003, ISBN 978-0-387-00451-8. Additional reference: Stochastic Simulation by Søren Asmussen and Peter Glynn, ISBN 978-0-387-69033-9 (somewhat higher level, but also more depth).	
Prerequisites	Martingales, Brownian Motion and Stochastic processes (wi4430 or equivalent course), in particular measure theoretic probability, and basic statistics, including linear regression. Familiarity with R, Matlab, or equivalent software for simulation, is recommended.	
Assessment	The final grade of the course consists of the following components: - Mini-projects (25%); - written exam (closed book; 75%); the grade for the final must be 6 or higher to pass the course. Final grade calculation: $(0.25 * \text{mini-projects}) + (0.75 * \text{written exam})$ Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Mini-projects: resubmit deliverable - Written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI4615	Stochastic Calculus	6
Responsible Instructor	Dr. K. Kirchner	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Basics of measure theory and analysis (as given in the course ``AM2090 Real Analysis") Martingales, Brownian motion (as taught in the master course ``W14430 Martingales, Brownian Motion").	
Course Contents	In this course the following subjects will be studied: 1. Ito integration 2. Ito calculus including Ito's formula 3. Stochastic differential equations 4. Feynman Kac formula 5. Introduction to numerical methods for stochastic differential equations	
Study Goals	After the course, you should be able to LO1. Explain the rigorous construction of the stochastic Ito integral. LO2. Demonstrate the derivation of Ito's formula. LO3. Calculate explicit stochastic Ito integrals by a) using the construction of the Ito integral, b) applying Ito's formula. LO4. Prove the existenceuniqueness theorem for stochastic ordinary differential equations (SODEs). LO5. Assess solvability of concrete SODEs by means of a) verifying all conditions of the existenceuniqueness theorem, b) explicitly calculating solutions. LO6. Establish links between Markov processes and partial differential equations.	
Education Method	Lectures (2 hours per week).	
Literature and Study Materials	The course will be based on the following sources: 1. P.J.C. Sprey: stochastic integration lecture notes available at https://staff.fnwi.uva.nl/p.j.c.spreij/onderwijs/master/si.pdf 2. Le Gall, Martingales, Brownian motion and stochastic calculus, Springer 2016. 3. Schilling, Partzch, Brownian Motion, An Introduction to Stochastic Processes, De Gruyter, 2010. 4. Karatzas, Shreve, Brownian Motion and Stochastic Calculus, Springer 1991. 5. Øksendal, Stochastic Differential Equations, Springer 2003. 6. Revuz, Yor, Continuous Martingales and Brownian Motion, Springer 1999.	
Assessment	These books can be freely downloaded via a library link The final grade of the course consists of the following components: - Depending on the number of students, there will be an oral or a written exam at the end of Q4 which counts for 100%. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Written exam: written resit OR - Oral exam: oral resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	The exam is closed book.	

WI4630	Statistical Learning	6
Responsible Instructor	Dr.ir. G.N.J.C. Bierkens	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Required: * Bachelor level knowledge and ability in calculus, probability, statistics, optimization, linear algebra. * Programming skills in Python (AM1090). If you do not have this background, consult the book Think Python, https://greenteapress.com/wp/think-python/, and familiarize yourself with the Numpy and Matplotlib packages. Recommended: * Linear Algebra and Optimization for Machine Learning (WI4635) * Statistical Inference (WI4455) * Knowledge of measure theory (as, for example, in AM2090)	
Course Contents	Statistical learning provides a statistical perspective on machine learning. The overarching goal of the course is to develop methods for estimating (or 'learning') an unknown function from data or making predictions for unseen function outputs. The course aims to empower the student to make a justified decision in adopting machine learning approaches for practical problems and even design their own machine learning methodology using sound mathematical principles. This is achieved by studying key ideas, models, algorithms and theories related to the subject of statistical learning. In the first half of the course a collection of essential models, algorithms and techniques is introduced. In the second half of the course we adopt a Bayesian perspective on machine learning and study associated methods for analysing models and developing computational methodology. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Study Goals	By the end of the course, a student should be able to: 1. Explain the challenges of statistical learning (e.g., curse of dimensionality, overfitting, bias-variance trade-off). 2. Implement machine learning algorithms and computational methods on artificial and real-world data sets (e.g., linear regression, logistic regression, sparse regression, neural networks, decision trees). 3. Compare the theoretical properties and practical aspects of machine learning approaches. 4. Validate machine learning approaches on arbitrary data sets. 5. Design tailor-made computational approaches for new learning problems by adapting, combining and/or extending methods and/or by employing a Bayesian approach.	
Education Method	Lectures, exercises, practical assignments, practice exam	
Literature and Study Materials	Lecture notes and slides will be made available. The main supporting reference is: Murphy, Kevin P. Probabilistic machine learning: an introduction. MIT press, 2022. which is publicly available here: https://probml.github.io/pml-book/book1.html Alternative background reading: * Hastie, T., Tibshirani, R., & Friedman, J. (2009). The elements of statistical learning (Second). Springer, New York. https://web.stanford.edu/~hastie/ElemStatLearn/ * Bishop, C. M. (2006). Pattern recognition and machine learning. Springer, New York. https://www.microsoft.com/en-us/research/publication/pattern-recognition-machine-learning/	
Assessment	The final grade of the course consists of the following components: * Practical assignments during the course (30%) * Final written exam (70%) Final grade calculation: $(0.3 * \text{practical assignments} + 0.7 * \text{final written exam})$ Note: For both components, a sufficient grade (5.8) is required. Repair possibilities: * Assignments: resubmit deliverable * Written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI4635	Linear Algebra and Optimization for Machine Learning	6
Responsible Instructor	Dr. A. Heinlein	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	The course material and literature are mostly self-contained, but prior knowledge equivalent to contents of the courses AM2010 Linear Algebra 2 and AM2020 Optimization from the Bachelor of Applied Mathematics programme is strongly recommended. Programming experience in Python or Matlab is expected.	
Course Contents	State-of-the-art machine learning algorithms combine techniques from many different areas of mathematics. In this course, we will discuss algorithmic foundations from the areas of linear algebra and optimization. This also includes aspects of their theory and efficient implementation. Among others, we will study + matrix decomposition and factorization techniques, + regression and classification problems and regularization, + iterative and continuous optimization methods, + hyper parameter optimization, + graph-based algorithms and clustering, + basic introduction to neural networks and algorithmic aspects neural networks (e.g. computational graphs and back-propagation)	
Study Goals	Upon completion of the course, the student should be able to: + Categorize typical data sets based on their structure + Derive suitable mathematical formulations for common supervised and unsupervised machine learning tasks + Choose suitable algorithmic building blocks for solving mathematical formulations of common machine learning tasks + Analyze machine learning algorithms with respect to their properties, such as efficiency and robustness + Implement machine learning algorithms in a computationally efficient way	
Education Method	The course comprises 14 lectures and 14 exercise classes.	
Literature and Study Materials	Lecture notes, slides, and the main literature: + Charu Aggarwal. Linear algebra and optimization for machine learning. Springer, 2020. + Gilbert Strang. Linear algebra and learning from data. Wellesley-Cambridge Press Cambridge, 2019. + Jeremy Watt, Reza Borhani, and Aggelos K Katsaggelos. Machine learning refined: Foundations, algorithms, and applications. Cambridge University Press, 2020.	
Assessment	The final grade of the course consists of the following components: - Two group projects (0.4) - Written exam (0.6) Final grade calculation: $(0.4 * \text{group projects} + 0.6 * \text{written exam})$ Note: For the written exam, a score of at least 5.0 must be obtained. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Group projects: resubmit deliverable - Written exam: There will be a resit for the written exam at a later point in the same academic year Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	D. de Laat	

WI4771TU	Object Oriented Scientific Programming with C++	3
Responsible Instructor	Dr. M. Möller	
Contact Hours / Week x/x/x/x	0/2/0/0 & 0/4/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic programming knowledge in at least one programming language (e.g. Matlab, C/C++, Fortran)	
Course Contents	Introduction to C++ programming including new concepts from the C++11/14/17/20 standard. Object-oriented scientific programming.	
Study Goals	The student will be able to apply modern software design patterns and state-of-the-art programming techniques to implement (numerical) algorithms in C++ (using the latest standards C++11/14/17/20). The student will build up practical experience in turning a textbook (numerical) algorithm into an efficient and maintainable C++ code. The student will be able to develop larger software projects in teams and test the implementation systematically.	
Education Method	Hands-on lectures and practical sessions in Q2. Problem-oriented teaching approach starting from a concrete (mathematical) problem (taken from introductory numerical analysis course) and building up knowledge in C++/scientific programming up to the level required to solve the problem. Further self-study is required to build up in-depth practical experience. Implementation of final project (in groups of 1-3 students) starting mid Q2 and with submission deadline end Q2 / beginning Q3.	
Literature and Study Materials	Complete course material (slides and source code of demo applications) will be provided. Though not mandatory, the book "A Tour of C++" by Bjarne Stroustrup, C++ In-Depth Series, Addison-Wesley, ISBN-13 978-0-321-95831-0 gives a good overview of the covered material.	
Prerequisites	Basic programming knowledge in at least one programming language (e.g. Matlab, C/C++, Fortran)	
Assessment	The final grade of the course consists of the following components: - Homework programming assignments (10 in total) for each session during Q2 (1/3 of final grade). - Group programming project (group work) starting mid Q2 with submission deadline end Q2 (2/3 of final grade). Final grade calculation: $(1/3 * \text{homework assignments}) + (2/3 * \text{group programming project})$ but it is only valid when all homework programming assignments have been completed with sufficient results. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Homework programming assignments: submit new set of homework - Group programming project: submit improved version	
Co-Instructor	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). K. Cools	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Specialisation Discrete Mathematics and Optimization AM 2023

Introduction 1

Specialisation Discrete Mathematics and Optimization

Compulsory orientation course:
WI4227-14 Discrete Optimisation

Compulsory specialisation course:
MM Semidefinite Optimization

Compulsory specialisation courses (choose 2 out of 4 courses):
WI4410 Advanced Discrete Optimization
WI4515 Relaxations and Heuristics
WI4670 Extremal Combinatorics
WI4635 Linear Algebra and Optimisation for Machine Learning

All of the compulsory specialisation courses can also be chosen as recommended specialisation course.

Recommended specialisation courses:

WI4207 Continuous Optimization
MM Machine Learning Theory
MM Quantum Computing
MM Quantum Information Theory
MM Scheduling
WI4630 Statistical Learning
ET4030 Error Correcting Codes
WI4156(TU)Game Theory
MM Additive Combinatorics
MM Graph Symmetries and Combinatorial Designs

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Compulsory Courses

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Compulsory Courses (select 2 out of 4)

WI4410	Advanced Discrete Optimization	6
Responsible Instructor	Dr.ir. L.J.J. van Iersel	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Linear optimization: basic solutions and their relation to extreme points, simplex method, duality and complementary slackness Integer optimization: basic cutting planes, branch-and-bound Computational complexity: Complexity classes P and NP and NP-complete, polynomial reductions	
Course Contents	Integer optimization: Polyhedral theory; Cutting planes; Gomory mixed-integer cuts, split cuts and their relation; Lift-and-project; IP in fixed dimension: Lattices and basis reduction; IP in fixed dimension: Lenstra's algorithm and extensions Parameterized complexity: fixed-parameter tractability, bounded search, kernelization, dynamic programming, treewidth Applications in phylogenetics: phylogenetic trees, agreement forests, SPR distance, phylogenetic networks	
Study Goals	To understand and be able to use the methods and concepts described in the course. The student should also be able to prove results similar to the ones described in the course.	
Education Method	There will be 2 lectures combined with exercise classes per week.	
Literature and Study Materials	Lecture slides, notes, papers and book chapters will be provided on Brightspace.	
Assessment	The final grade of the course consists of the following components: - Written exam (100%) Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Written exam: written resit	
Co-Instructor	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). Dr. M.E.L. Jones	

WI4515	Relaxations and Heuristics	6
Responsible Instructor	Dr.ir. J.T. van Essen	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	* Programming skills (Python/C/C++) * Linear algebra * Linear programming, simplex method, LP duality (covered in TW2020) * Integer linear programming, branch-and-bound (covered in TW2020) * Complexity theory (covered in TW2020)	
Course Contents	Many real-life optimization problems are NP-hard. For these problems, it is not known whether there exist efficient algorithms to determine an optimal solution within a reasonable amount of time. In this course, we will focus on providing lower and upper bounds for the objective function values of such problems by studying relaxations and heuristics. On the one hand, several methods are discussed which are able to obtain dual bounds (upper bounds on the optimal value of maximization problems) and, on the other hand, we will discuss ways to find good feasible solutions that provide primal (lower) bounds. We will discuss the following topics: * ILP formulations * Branch-and-bound * Cutting planes, valid inequalities, polyhedra & facets * Column generation * Local search heuristics, metaheuristics, heuristics based on integer programming * Approximation algorithms	
Study Goals	At the end of the course, you should be able to: 1. Formulate real-life optimization problems using integer linear programming. 2. Use a solver to find a (near)-optimal solution to an ILP model. 3. Apply relaxation techniques to derive and improve dual bounds. 4. Implement heuristics to find (good) primal bounds. 5. Solve real-life optimization problems by integrating dual and primal bounds techniques in a branch-and-cut framework.	
Education Method	There will be 2 hours of lectures each week. Throughout the course, the students will work on a real-life problem to which they can apply the discussed theory.	
Literature and Study Materials	* Laurence A. Wolsey, Integer Programming, Wiley, 2021 * Additional lecture notes * Prerecorded videos	
Assessment	The final grade consists of the following parts: * Final written exam (70%) * Project (30%) For both the exam and project, your grade should be a 5 or higher to pass the course. Final grade calculation: $(0.7 * \text{written exam}) + (0.3 * \text{project})$ Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - written exam: written resit - Project: resubmit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. F.M. de Oliveira Filho	

WI4635	Linear Algebra and Optimization for Machine Learning	6
Responsible Instructor	Dr. A. Heinlein	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	The course material and literature are mostly self-contained, but prior knowledge equivalent to contents of the courses AM2010 Linear Algebra 2 and AM2020 Optimization from the Bachelor of Applied Mathematics programme is strongly recommended. Programming experience in Python or Matlab is expected.	
Course Contents	State-of-the-art machine learning algorithms combine techniques from many different areas of mathematics. In this course, we will discuss algorithmic foundations from the areas of linear algebra and optimization. This also includes aspects of their theory and efficient implementation. Among others, we will study + matrix decomposition and factorization techniques, + regression and classification problems and regularization, + iterative and continuous optimization methods, + hyper parameter optimization, + graph-based algorithms and clustering, + basic introduction to neural networks and algorithmic aspects neural networks (e.g. computational graphs and back-propagation)	
Study Goals	Upon completion of the course, the student should be able to: + Categorize typical data sets based on their structure + Derive suitable mathematical formulations for common supervised and unsupervised machine learning tasks + Choose suitable algorithmic building blocks for solving mathematical formulations of common machine learning tasks + Analyze machine learning algorithms with respect to their properties, such as efficiency and robustness + Implement machine learning algorithms in a computationally efficient way	
Education Method	The course comprises 14 lectures and 14 exercise classes.	
Literature and Study Materials	Lecture notes, slides, and the main literature: + Charu Aggarwal. Linear algebra and optimization for machine learning. Springer, 2020. + Gilbert Strang. Linear algebra and learning from data. Wellesley-Cambridge Press Cambridge, 2019. + Jeremy Watt, Reza Borhani, and Aggelos K Katsaggelos. Machine learning refined: Foundations, algorithms, and applications. Cambridge University Press, 2020.	
Assessment	The final grade of the course consists of the following components: - Two group projects (0.4) - Written exam (0.6) Final grade calculation: $(0.4 * \text{group projects} + 0.6 * \text{written exam})$ Note: For the written exam, a score of at least 5.0 must be obtained. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Group projects: resubmit deliverable - Written exam: There will be a resit for the written exam at a later point in the same academic year Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	D. de Laat	

WI4670	Extremal Combinatorics	6
Responsible Instructor	Dr. A. Bishnoi	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Linear algebra, graph theory, basic probability theory	
Course Contents	In Extremal Combinatorics we study problems of the form "How large (or small) can a parameter associated to a finite structure be under certain restrictions"? In this course we will study several problems of this kind and learn some exciting theorems from Ramsey theory, extremal set theory, etc. There will be an equal emphasis on the methods used to solve these problems. In particular, we will learn the probabilistic method, polynomial method and topological method.	
Study Goals	By the end of this course you will be able to: 1. Explain the proofs of some of the central results in extremal combinatorics. 2. Solve problems in extremal combinatorics using probabilistic, algebraic and topological methods. 3. Apply these methods to problems in other areas of mathematics and computer science.	
Education Method	Lectures	
Literature and Study Materials	Digital lecture notes will be provided on Brightspace	
Books	Extremal Combinatorics by Stasys Jukna Linear Algebra Methods in Combinatorics by László Babai and Péter Frankl A course in combinatorics by J. H. van Lint and R. M. Wilson	
Assessment	The final grade of the course consists of the following components: - Written exam (100%) In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	Pen and paper only	
Co-Instructor	Prof.dr. D.C. Gijswijt	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Recommended Courses

ET4030	Error Correcting Codes	4
Responsible Instructor	Dr.ir. J.H. Weber	
Contact Hours / Week x/x/x/x	0/0/0/3	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Basic linear algebra and probability theory, as being treated in B.Sc. programmes like Electrical Engineering, Computer Science, and Mathematics	
Course Contents	Introduction into error-correcting codes; mathematical basics; block codes fundamentals; cyclic codes; co-operating codes; soft-decision decoding; convolutional codes; iterative decoding (turbo codes, LDPC codes); applications.	
Study Goals	The general overall objectives of this course are that the students can - reproduce the error control concepts as presented in the lecture notes; - explain the meaning and usefulness of these concepts; - apply the concepts to transmission and storage systems, including basic calculations, at the level of the exercises in the lecture notes and previous exams. More specifically, the students are able to - discuss the basic trade-offs between efficiency, reliability, and complexity in designing coding schemes for transmission and storage systems; - apply finite fields theory in order to construct (cyclic) codes; - encode data into code sequences that are resistant to a certain number of random and/or burst errors; - decode received sequences using (iterative) hard or soft decision techniques.	
Education Method	Lectures; expected workload is 20 hours attending lectures, 60 hours preparing for the lectures, studying the lecture notes, and making suggested exercises, and 32 hours for preparing and making the exam.	
Literature and Study Materials	Lecture notes "Error-Correcting Codes" by J.H. Weber, available on Brightspace. Also previous exams as well as demos are available via Brightspace.	
Assessment	The final grade will be fully determined by a scheduled written exam, which will be held at the end of Q4. The resit will take place as an oral exam in Q5, on appointment with the lecturer. The exam format is closed-book in any case. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	Actual course information available on Brightspace.	

WI4156(TU)	Game theory	6
Responsible Instructor	Dr. R.J. Fokkink	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Required for	Open to all MSc students with a strong command of mathematics. Ideally suited for math students with an interest in finance.	
Expected prior knowledge	Standard BSc math courses: calculus, linear algebra, probability.	
Course Contents	This is a course in "mathematical" game theory, which is the counterpart of "algorithmic" game theory (agent based modelling, this part of game theory is taught in the computer science department). We cover all subjects from combinatorial games to cooperative games.	
Study Goals	Understanding the mathematical theory of rational behaviour/conflicts/negotiations/social choice. Game theory is one of the most prominent economic theories: three recent Nobel prizes have been awarded to game theorists. We will cover a very wide area, ranging from pure mathematics to economics, with an emphasis on the mathematical aspects. This is a course in the Finance Track of the Applied Maths programme, but it is open to all MSc students (math and non-math).	
Education Method	Lectures	
Literature and Study Materials	Game Theory - Thomas Ferguson - UCLA: find it on the net.	
Books	T.S. Ferguson, A course in game theory, World Scientific, 2020	
Assessment	The final grade of the course consists of the following components: written exam (100%) Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI4207	Continuous Optimization	6
Responsible Instructor	D. de Laat	
Contact Hours / Week x/x/x/x	1e semester Utrecht maandag	
Education Period	1 2	
Start Education	1	
Exam Period	2	
Course Language	English	
Expected prior knowledge	The student should have a solid knowledge of linear algebra and multivariable calculus. The student should also have knowledge of linear programming (including linear programming duality) and convex analysis to the level of being able to follow the text and do the exercises from: Chapters 1 and 2 (including the exercises) from the book 'Linear Programming, A Concise Introduction, Thomas S. Ferguson' https://www.math.ucla.edu/~tom/LP.pdf Exercises 2.1, 2.2, 2.12, 3.1, 3.3, 3.5, and 3.7 from the book 'Convex Optimization, Stephen Boyd and Lieven Vandenbergh' http://stanford.edu/~boyd/cvxbook	
Course Contents	Continuous optimization is the branch of optimization where we optimize a (differentiable) function over continuous (as opposed to discrete) variables. Here the variables can be subject to equality and inequality constraints. Optimization problems of this form are common in science and engineering, machine learning, and as relaxations of discrete optimization problems. One can use linear algebra, multivariable calculus, and convex geometry to study the properties of continuous optimization problems and to design and analyze efficient algorithms. In this course, we study theory, algorithms, and some applications of continuous optimization. In the theory part we discuss convexity, Lagrangian duality, optimality conditions, and conic programming. In the algorithmic part we discuss derivative-free methods, first-order optimization methods, neural networks/supervised learning, second-order optimization methods, interior-point methods, and support vector machines. For some of the algorithms, we analyze the convergence.	
Study Goals	See the Mastermath page.	
Education Method	There will be 14 lectures.	
Literature and Study Materials	- Convex Optimization by Stephen Boyd and Lieven Vandenbergh (http://stanford.edu/~boyd/cvxbook) - Additional notes distributed by the lecturer	
Assessment	The course will have a written exam that counts for 100% of the grade. The resit will also be a written exam. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	This is a LNMB Mastermath course taught by David de Laat.	

WI4630	Statistical Learning	6
Responsible Instructor	Dr.ir. G.N.J.C. Bierkens	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Required: * Bachelor level knowledge and ability in calculus, probability, statistics, optimization, linear algebra. * Programming skills in Python (AM1090). If you do not have this background, consult the book Think Python, https://greenteapress.com/wp/think-python/, and familiarize yourself with the Numpy and Matplotlib packages. Recommended: * Linear Algebra and Optimization for Machine Learning (WI4635) * Statistical Inference (WI4455) * Knowledge of measure theory (as, for example, in AM2090)	
Course Contents	Statistical learning provides a statistical perspective on machine learning. The overarching goal of the course is to develop methods for estimating (or 'learning') an unknown function from data or making predictions for unseen function outputs. The course aims to empower the student to make a justified decision in adopting machine learning approaches for practical problems and even design their own machine learning methodology using sound mathematical principles. This is achieved by studying key ideas, models, algorithms and theories related to the subject of statistical learning. In the first half of the course a collection of essential models, algorithms and techniques is introduced. In the second half of the course we adopt a Bayesian perspective on machine learning and study associated methods for analysing models and developing computational methodology. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Study Goals	By the end of the course, a student should be able to: 1. Explain the challenges of statistical learning (e.g., curse of dimensionality, overfitting, bias-variance trade-off). 2. Implement machine learning algorithms and computational methods on artificial and real-world data sets (e.g., linear regression, logistic regression, sparse regression, neural networks, decision trees). 3. Compare the theoretical properties and practical aspects of machine learning approaches. 4. Validate machine learning approaches on arbitrary data sets. 5. Design tailor-made computational approaches for new learning problems by adapting, combining and/or extending methods and/or by employing a Bayesian approach.	
Education Method	Lectures, exercises, practical assignments, practice exam	
Literature and Study Materials	Lecture notes and slides will be made available. The main supporting reference is: Murphy, Kevin P. Probabilistic machine learning: an introduction. MIT press, 2022. which is publicly available here: https://probml.github.io/pml-book/book1.html Alternative background reading: * Hastie, T., Tibshirani, R., & Friedman, J. (2009). The elements of statistical learning (Second). Springer, New York. https://web.stanford.edu/~hastie/ElemStatLearn/ * Bishop, C. M. (2006). Pattern recognition and machine learning. Springer, New York. https://www.microsoft.com/en-us/research/publication/pattern-recognition-machine-learning/	
Assessment	The final grade of the course consists of the following components: * Practical assignments during the course (30%) * Final written exam (70%) Final grade calculation: $(0.3 * \text{practical assignments} + 0.7 * \text{final written exam})$ Note: For both components, a sufficient grade (5.8) is required. Repair possibilities: * Assignments: resubmit deliverable * Written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Specialisation Stochastics AM 2023

Introduction 1

Specialisation Stochastics

Compulsory orientation course:

WI4430 Martingales, Brownian Motion

Compulsory specialisation course:

WI4455 Statistical Inference

Compulsory specialisation courses (choose 2 out of 6 courses):

WI4230 Time series and Extreme Value Theory

WI4052 Risk Analysis

WI4665 Advanced Topics in Statistics

WI4615 Stochastic Calculus

WI4465 Advanced Topics in Probability

WI4630 Statistical Learning

All of the compulsory specialisation courses can also be chosen as recommended specialisation course.

Recommended specialisation courses:

MM Asymptotic Statistics

WI4050 Uncertainty and Sensitivity Analysis

WI4138 Decision Theory/Expert Judgment

WI4156(TU) Game theory

WI4640 High Dimensional Probability

WI4614 Stochastic Simulation

MM Stochastic Processes

MM Statistics for Stochastic Processes

WI4505 Quantitative Risk Management

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Compulsory Courses

WI4455	Statistical Inference	6
Responsible Instructor	Prof.dr.ir. G. Jongbloed	
Co-responsible for assignments	T.F.W. van der Jagt	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	- Introductory statistics and probability course at bachelor level (such as: various probability distributions, central limit theorem, estimators, maximum likelihood estimation, frequentist hypothesis testing). - Familiarity with concepts from measure theoretic probability.	
Course Contents	This course aims to provide a concise but comprehensive account of the essential elements of statistical inference and theory. The course focusses on a clear presentation of the main concepts and results underlying different frameworks of inference.	
Study Goals	Study goals Statistical Inference WI4455 The topics correspond to chapters in the lecture notes. 1. Preliminaries: - The student is able to reproduce the definition of a statistical model, possibly defined in terms of densities. - The student is able to explain the unifying concept of exponential families (the precise form of the density will be given at the exam) - The student is able to assess fundamental differences between classical- and Bayesian approaches towards problems in the given examples. - The student is able to identify with the different notions of stochastic convergence and is able to apply the Delta-method. 2. Topics from classical statistics: - The student is able to prove that a statistic is sufficient. The factorisation theorem can be proved in case of discrete random variables. - The student is able to reproduce the definition of Fisher information and Kullback-Leiber information and can prove properties of these (as in the lecture notes). - The student is able to check whether an estimator is UMVU and can prove the Lehmann-Scheffe and Cramer-Rao bound theorems. - The student can state the limiting distribution of the MLE in regular cases and can give examples of cases where the MLE is inconsistent. - The student can compute tests, p-values, power-curves of tests and errors of type I and II of (randomised) tests in simple settings. The student can prove the Neyman-Pearson lemma. - The student can state criticism to the use of p-values. - The student can derive confidence sets using (asymptotic) pivots. 3. The likelihood principle: - The student can state Birnbaums result. - The student is able to explain the implications of the sufficiency-, conditionality- and likelihood principle. 4. Bayesian analysis: - The student is able to explain the Bayesian approach towards statistical inference and can apply Bayes' rule to do Bayesian computations in case a closed form solution is available. - The student is able to describe the concept of exchangeability and its relation to justifying Bayesian analysis. - The student can derive Jeffreys prior in a given problem setting. - The student can construct hierarchical models. - The students can explain how empirical Bayes methods are defined, and can compute empirical Bayes estimators in a given problem setting. - The student can state the Bernstein-Von Mises theorem and its implications. 5. Markov Chain Monte Carlo methods: - The student can derive the steps of a MCMC scheme for drawing from the posterior, also in case a closed form solution for drawing from this posterior is unavailable. This scheme may include Gibbs steps. 6. Statistical decision theory: - The student is able to describe the basics of statistical decision theory including admissibility, minimax optimality, Bayes optimality, and connections between these topics. The student can prove results that state interrelations between these topics. - The student is able to state Steins example of inadmissibility of the MLE. - The student can derive minimax and Bayes estimators in tractable and finite-decision problems. - The student is able to derive Bayesian tests and compute a Bayes factor for a given statistical model.	
Education Method	Each week there will be a 2-hours lecture and a scheduled meeting where exercises from the lecture notes are discussed.	
Literature and Study Materials	Lecture notes	
Reader	Lecture notes will be available on brightspace.	
Assessment	At the end of the course there will be a written on campus exam (3 hours) which counts for 100%. The assignments in the lecture notes are representative for the level and type of questions at the exam. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Compulsory Courses (select 2 out of 6)

WI4052	Risk Analysis	6
Responsible Instructor	Dr. D. Kurowicka	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic Probability and Statistics	
Summary	Risk analysis course will treat problems often appearing in applications. Some generic methods that are applied in many different areas of applications will be discussed. We will not concentrate on any particular application, rather we will discuss important assumptions made in different models and their applicability for different real life problems. Moreover we show how parameters of these models can be obtained from data and discuss methods of validation of the obtained model.	
Course Contents	Risk modelling, life distributions, event trees, fault trees, Bayesian Belief Nets, reliability data bases, dependence modelling, software reliability, expert judgment.	
Course Contents Continuation	Introduction & Uncertainty Probability Multivariate, Copula Stochastic processes Bayesian, classical estimations & Testing Fault trees Bayesian belief nets Dependent failures Expert Judgment Reliability data bases Software reliability	
Study Goals	Students must be able to justify use of probabilistic methods in risk analysis and apply techniques like fault trees and Bayesian belief nets. They should know how to estimate parameters of models for dependent failure rates. Moreover they should be able to perform competing risks analysis for simple reliability data sets. Students should understand basic techniques of incorporating expert knowledge into risk models.	
Study Goals continuation	The goal of the course is to expose students to problems appearing in applications. During lectures the theory will be treated and exercises will be provided. Students will be required to complete the assignment where some practical problems have to be solved. The theoretical questions are not required. However making them allows to get a feedback and extra explanation in how they should be solved. These exercises together with course material will constitute basis for the exam in January. Practical exercises are required and will form part of your final grade.	
Education Method	Lectures and Assignments	
Literature and Study Materials	Parts of the book "Probabilistic Risk Analysis", Bedford & Cooke, Cam.U. press, 2001. Extra material on the Brightspace	
Reader	Brightspace	
Assessment	Written exam (100%). In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5 Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	Calculator and self made A4 with formulas.	
Tags	Mathematics Stochastics	

WI4230	Time Series & Extreme Value Theory	6
Responsible Instructor	Dr. P. Chen	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Knowledge of probability and statistics at the level of the BSc Applied Mathematics courses on these topics (AM1080, AM2080).	
Course Contents	The course consists of two parts: Extreme Value Theory and Time Series. The course considers questions such as: What is the chance that the price of oil will be higher than \$120/Barrel coming week? Can we expect that UK interest rates will rise next year? How can we assess if global warming is occurring? What is the chance that the daily loss of the stock price of ABN-AMRO is larger than, say 20%, which has never occurred before? How to determine the height of a dike such that the probability of a flood, i.e., the water level exceeding the dike, in a given year is 1/10,000? What is the impact of a dramatic decline in the US stock market on DAX 30 index? In this course students will build up some theoretical basis, mainly probabilistic results within the area of time series and extreme value theory. Moreover, students will learn statistical methods for dealing with non-stationary data, multivariate data and rare events and assessing extreme risks for data from various fields such as finance, actuarial science, hydrology and environmental science. Furthermore the student is capable to perform a statistical analysis with the statistical package R. The Extreme Value Theory part considers topics such as max-domain attractions, peak over threshold approach, estimations of extreme value index, tail probability and high quantile, and tail dependence and independence. The Time Series part will cover topics such as ARMA models, spectral analysis, non-stationarity, GARCH models, state space models. Implementations of statistical methods in R will be part of the course. Examples of real applications from environmental science and finance will also be discussed.	
Study Goals	Specific learning objectives for the Extreme Value Theory part are: 1. Validate the max-domain attraction conditions, that is to show if a given distribution is in the max domain of attraction or not. 2. Compute the extreme value index of a given distribution. 3. Develop estimations for tail probabilities, high quantiles, m-year return level, using different methods. 4. Prove some asymptotic results (consistency) of the estimators. 5. Evaluate the performance of estimation methods by simulation study; and apply the estimation methods to a real data set. 6. Establish the properties of exponent measure and spectral measure. Establish the link between the bivariate extreme value distribution and these two measures. 7. Characterize the extremal dependence for bivariate distributions. 8. Develop the estimation for a bivariate tail probability (failure probability). Specific learning objectives for the Time Series part are: 1. Discern whether a given time series data set can be associated to a stationary time series, and if not, perform suitable transformations to so that the resulting data corresponds to a stationary time series. 2. Understand and determine interpret theoretical properties of AR(I)MA time series models, such as the existence and uniqueness of solutions, causality and invertibility. 3. Understand and determine descriptive statistics of time series models, such as best linear predictors, autocorrelation function, partial autocorrelation function, estimators for mean and variance, and spectral density function. 4. Have general fluency in dealing with standard manipulations in time series, such as working with linear filters, computing conditional expectations and solving recurrence relations. 5. Fit a suitable time series model to time series data and assess the quality of a time series fit. 6. Have basic knowledge of advanced time series models used in financial mathematics, such as GARCH and stochastic volatility models.	
Education Method	Lectures/exercises/computer assignment	
Books	The lectures will be mainly based on: - Time Series Analysis and Its Applications: With R Examples By Robert H. Shumway, David S. Stoffer - Time Series: Theory and Methods By Brockwell, Peter J. and Davis, Richard A - Extreme Value Theory: An Introduction By Laurens de Haan and Ana Ferreira (Chapter 1, 3, 4, 6 and 7)	
Assessment	There will be a final assessment for the extreme value theory part at the end of Q3 and a final (written or oral) assessment for the time series part at the end of Q4. (If there are at least 10 students we will organize a written exam.) There will also be assignments handed out during the course. The final grade can be determined by a weighed sum, with the following weights: 0.15 average over assignments extreme value theory 0.35 exam extreme value theory 0.15 average over assignments time series	

0.35 exam time series

Resit/ Repair opportunities:

In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for:

- Individual assignments: resubmit deliverable
- Written exam: written resit OR
- Oral exam: oral resit

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

WI4465	Advanced Topics in Probability	6
Responsible Instructor	Prof.dr. F.H.J. Redig	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	A first course in probability theory (discrete random variables, law of large numbers, central limit theorem) suffices. Knowledge of stochastic processes like Markov processes and martingales is helpful. General mathematical maturity (e.g. Bachelor in (Applied) Mathematics) is highly advised.	
Course Contents	Even years: Random graphs and complex networks. Random graphs have become one of the most studied models in probability theory. Understanding the behaviour of large networks (eg. social media, financial transactions, communication networks) has become of paramount importance for practitioners and scientists alike. In this course, we aim at providing a general theory to answer the most immediate questions about large random graphs. We will begin by recalling some tools of discrete probability (moment methods Markov's and Chebyshev's inequality), and learn some new tools: coupling and stochastic dominations. We will then study ErdosRényi graphs, one of the first examples of random networks in probability. We will then move on to more modern models of complex networks: configuration model, preferential attachment graphs, random geometric graphs. Odd years: Interacting Particle Systems This course provides an introduction in the theory of interacting particle systems, a family of interacting Markov processes used to model many real-world phenomena such as the spread of an infection, the evolution of opinions in a population, traffic, the transport phenomena of interacting molecules, temperature-dependent behavior of magnetic systems. The course treats the following subjects 1. Markov processes in continuous time, generators, semigroups, martingales. 2. Basic techniques of interacting particle systems: a) Coupling, monotonicity and positive correlations. b) Duality. 3. These techniques will be applied in the study of the exclusion process, a basic interacting particle system used in non-equilibrium statistical physics, biology and modelling of traffic. Course Contents Continuation Random Graphs part: The following topics will be discussed. Note that this is only a tentative schedule and the pace at which the material will be covered will depend on the class as well. Real life complex networks and their properties, main modelling approaches. Probabilistic toolkit: Notions of convergence, coupling, stochastic ordering, Probabilistic bounds (1 lecture) Branching processes (0.5-1 lecture) Phase transition in the Erdos-Renyi random graph (1-2 lectures) Configuration model: (intro 0.5. lecture) loops and multiple edges (0.5 lecture) erased configuration model (0.5 lecture) generating simple graphs connection to number of simple graphs with prescribed degree sequence (0.5 lecture) ultra-small world property with power-law degrees. (1 lecture) Inhomogeneous random graphs (1 lectures) Preferential attachment model The model (0.5 lecture) degree distribution (1.5 lecture) Lindeberg's Central limit theorem (0.5 lecture) component sizes with Lindeberg's theorem (1 lecture) Additional topics if time allows might include small world phenomenon, etc. New models GIRGs, scale-free percolation, etc.	
Study Goals	Random Graphs part: The general goal is to make the student familiar with the probabilistic toolkit of studying random graphs and branching processes, and to get to learn the most commonly used network models. 0) The student is able to apply the following probabilistic tools in the setting of random graphs: introducing indicator variables, Markov's and Chebyshev's inequality, first and second moment method, notions of convergence of random variables, coupling and stochastic domination. 1) The student knows the definition of Galton--Watson trees and is able to derive properties of the graphs by identifying its regime (critical, subcritical, supercritical). 2) The students can recognise an Erdos--Rényi random graph and determine the size of its largest components. S/he is able to use stochastic domination and branching processes to compute expectations and variances of observables of the graph. 3) The student recognizes a configuration model, and knows efficient algorithms to generate them. They can determine and estimate basic observables: number of edges, self loops, probability of simplicity, and the relation to the number of simple graphs with given degree sequence. 4) The student knows the definition of preferential attachment models. They can apply martingale methods to determine the growth of individual degrees of vertices and give a recursion-based proof to determine the degree distribution. Interacting particle systems part: Learning objectives 1. Learn the basic techniques of interacting particle systems and Markov process theory: generators, semigroups, ergodicity, monotonicity, duality, coupling, graphical representations. 2. Being able to apply these techniques in the symmetric exclusion process and related models. 3. Being able to read a research paper with up to date techniques in this area and give a presentation about it.	

Study Goals continuation

Random Graphs, Continuation:

40% of the final grade will come from a 'project'. The projects can be worked out in pairs or triplets. The possible projects are posted on bright space and will be distributed among the students on the first class.

Each of the projects requires a deeper understanding and simulation of a random graph model (or a process on it). Each project has a minimal component that is required for passing grade, but beyond that, the students have the freedom to choose which direction of the project to explore. In particular, they can run simulations, or go deeper into the theory and study some papers, do calculations themselves on (partly) open questions related to random graph models.

The students have to hand in a report at the end of the course, at least 5 but no longer than 12 pages. The students also present their work at the end of the course in a 10 minute presentation. (Each additional page above 12 constitutes to -0.25 grade). Projects are graded based on content and professional (writing) skills. An empty evaluation form can be found on bright space. Presentations will be (partially) peer-graded.

Education Method

Random Graphs: One lecture per week, to discuss the theory. We will spend every third or fourth lecture on the exercises. Exercises are provided to get familiar with the material. Beyond this, students are expected to work in pairs or triplets on their project, write a 10 page report and give a 10 min presentation (in pairs or triplets)

Interacting Particle Systems:

One lecture per week, to discuss theory.

Literature and Study Materials

Random Graphs part: R. van der Hofstad, "Random graphs and complex networks", volume I, available at <http://www.win.tue.nl/~rhofstad/NotesRGCN.pdf>

Interacting Particle Systems Part: Lecture notes written by Frank Redig

Assessment

Random graphs: Written exam at the end of the semester for 60% of the grade. 40% of the grade comes from a hand-in project, executed in pairs, with a written report of max 10 pages, and a 10 min presentation.

Interacting Particle Systems:

Assessment: presentation or oral exam (students may choose which one)

Resit/ Repair opportunities:

In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for:

- presentation/oral exam: oral resit

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

Co-Instructor

J. Komjáthy

WI4615	Stochastic Calculus	6
Responsible Instructor	Dr. K. Kirchner	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Basics of measure theory and analysis (as given in the course ``AM2090 Real Analysis'') Martingales, Brownian motion (as taught in the master course ``WI4430 Martingales, Brownian Motion'').	
Course Contents	In this course the following subjects will be studied: 1. Ito integration 2. Ito calculus including Ito's formula 3. Stochastic differential equations 4. Feynman Kac formula 5. Introduction to numerical methods for stochastic differential equations	
Study Goals	After the course, you should be able to LO1. Explain the rigorous construction of the stochastic Ito integral. LO2. Demonstrate the derivation of Ito's formula. LO3. Calculate explicit stochastic Ito integrals by a) using the construction of the Ito integral, b) applying Ito's formula. LO4. Prove the existenceuniqueness theorem for stochastic ordinary differential equations (SODEs). LO5. Assess solvability of concrete SODEs by means of a) verifying all conditions of the existenceuniqueness theorem, b) explicitly calculating solutions. LO6. Establish links between Markov processes and partial differential equations.	
Education Method	Lectures (2 hours per week).	
Literature and Study Materials	The course will be based on the following sources: 1. P.J.C. Spreij: stochastic integration lecture notes available at https://staff.fnwi.uva.nl/p.j.c.spreij/onderwijs/master/si.pdf 2. Le Gall, Martingales, Brownian motion and stochastic calculus, Springer 2016. 3. Schilling, Partzch, Brownian Motion, An Introduction to Stochastic Processes, De Gruyter, 2010. 4. Karatzas, Shreve, Brownian Motion and Stochastic Calculus, Springer 1991. 5. Øksendal, Stochastic Differential Equations, Springer 2003. 6. Revuz, Yor, Continuous Martingales and Brownian Motion, Springer 1999.	
Assessment	These books can be freely downloaded via a library link The final grade of the course consists of the following components: - Depending on the number of students, there will be an oral or a written exam at the end of Q4 which counts for 100%. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Written exam: written resit OR - Oral exam: oral resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	The exam is closed book.	

WI4630	Statistical Learning	6
Responsible Instructor	Dr.ir. G.N.J.C. Bierkens	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Required: * Bachelor level knowledge and ability in calculus, probability, statistics, optimization, linear algebra. * Programming skills in Python (AM1090). If you do not have this background, consult the book Think Python, https://greenteapress.com/wp/think-python/, and familiarize yourself with the Numpy and Matplotlib packages. Recommended: * Linear Algebra and Optimization for Machine Learning (WI4635) * Statistical Inference (WI4455) * Knowledge of measure theory (as, for example, in AM2090)	
Course Contents	Statistical learning provides a statistical perspective on machine learning. The overarching goal of the course is to develop methods for estimating (or 'learning') an unknown function from data or making predictions for unseen function outputs. The course aims to empower the student to make a justified decision in adopting machine learning approaches for practical problems and even design their own machine learning methodology using sound mathematical principles. This is achieved by studying key ideas, models, algorithms and theories related to the subject of statistical learning. In the first half of the course a collection of essential models, algorithms and techniques is introduced. In the second half of the course we adopt a Bayesian perspective on machine learning and study associated methods for analysing models and developing computational methodology. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Study Goals	By the end of the course, a student should be able to: 1. Explain the challenges of statistical learning (e.g., curse of dimensionality, overfitting, bias-variance trade-off). 2. Implement machine learning algorithms and computational methods on artificial and real-world data sets (e.g., linear regression, logistic regression, sparse regression, neural networks, decision trees). 3. Compare the theoretical properties and practical aspects of machine learning approaches. 4. Validate machine learning approaches on arbitrary data sets. 5. Design tailor-made computational approaches for new learning problems by adapting, combining and/or extending methods and/or by employing a Bayesian approach.	
Education Method	Lectures, exercises, practical assignments, practice exam	
Literature and Study Materials	Lecture notes and slides will be made available. The main supporting reference is: Murphy, Kevin P. Probabilistic machine learning: an introduction. MIT press, 2022. which is publicly available here: https://probml.github.io/pml-book/book1.html Alternative background reading: * Hastie, T., Tibshirani, R., & Friedman, J. (2009). The elements of statistical learning (Second). Springer, New York. https://web.stanford.edu/~hastie/ElemStatLearn/ * Bishop, C. M. (2006). Pattern recognition and machine learning. Springer, New York. https://www.microsoft.com/en-us/research/publication/pattern-recognition-machine-learning/	
Assessment	The final grade of the course consists of the following components: * Practical assignments during the course (30%) * Final written exam (70%) Final grade calculation: $(0.3 * \text{practical assignments} + 0.7 * \text{final written exam})$ Note: For both components, a sufficient grade (5.8) is required. Repair possibilities: * Assignments: resubmit deliverable * Written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI4665	Advanced Topics in Statistics	6
Responsible Instructor	Prof.dr. A.W. van der Vaart	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	The subtitle of the course is Causality and Graphical Models A graphical model expresses dependence relations between variables. Each node of a graph represents a variable and each edge or arrow a relationship. Next to this structure there is a quantitative model in the form of a joint probability distribution of all variables. The graph expresses conditional (in)dependencies between the variables given other variables in the sense of probability. Graphical models with directed graphs are also called Bayesian networks, as Bayes rule gives the conditional probabilities. They are the basis of probabilistic expert systems, as applied in many contexts. In practice a graphical model must be learned from data, guided by prior information, and hence there is a strong statistical content. The graph helps to form a likelihood, and one can apply various statistical methods, including maximum likelihood and Bayesian methods, to estimate the graph or parameters. Because there usually are many variables, statistical methods for high-dimensional data and models can be necessary. This includes for instance efficiency theory for semiparametric models and penalized likelihood methods. A main application considered in this course is the discovery and estimation of causal effects from data. Cause and effect relationships between variables can be graphically represented as arrows in a graphical model for the variables. Causal discovery is concerned with inferring such a graph from observed data, and causal estimation with quantifying the effects. The challenge is to perform this with data that is not produced by a designed experiment, but observational, a situation that abounds in epidemiological and social science research. As correlation is not causation, corrections must be made for other possible explanations that might cause a correlation. Causal reasoning can alternatively be cast in terms of "potential outcomes", the effect that would have realised had a random individual or unit received a treatment of interest. Statistical methods for estimating causal effects can be justified in terms of potential outcomes. A given causal graph allows to define the relationships of such potential outcomes to other variables, and be the basis of verifying necessary assumptions on observed data. The subject of the course is at the intersection of statistics/probability and artificial intelligence. The style will be mathematical, although only a modest amount of specialised preknowledge is required, mostly a good grasp of probability and statistical methods, and of course linear algebra and analysis. We shall provide short introductions to conditioning, asymptotic methods, Bayesian methods, Gaussian models, etc. when needed.	
Study Goals	Understand statistical graphical models and their applications, in particular in causal inference.	
Education Method	Lectures. Homework problems.	
Assessment	The final grade of the course consists of the following components: - Written exam 70 % - Average grade of homework assignments 30 % Final grade calculation: $(0.7 * \text{written} + 0.3 * \text{homework})$ Repair possibilities: - Written exam: repeat exam. Final grade calculation: $(0.7 * \text{resit written} + 0.3 * \text{homework})$ Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Tags	Artificial intelligence Mathematics Stochastics	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Recommended Courses

WI4050	Uncertainty and Sensitivity Analysis	6
Responsible Instructor	Dr. D. Kurowicka	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic Probability and Statistics (containing knowledge about joint distributions)	
Course Contents	During this course we concentrated on the most challenging part of uncertainty analysis: dependence modelling. In particular, on copula models and their applications in finance and insurance. First part of the course is concerned with the theory of copulas and their use in regression, time series analysis and factor models. The second part is designed to apply these models to financial and insurance problems.	
Study Goals	Students must be able to explain on their own examples concepts such as: uncertainty, independence, and dependence. They should understand proofs of basic properties of different dependence measures and dependence models. In particular students should be able to apply concepts as univariate and multivariate uncertainty distributions, copulas. They should be able to justify which model should be used in the particular situation and explain what are pluses and minuses of different constructions from the applied point of view. Students will apply the theory studied in the first part of the course on an individual project.	
Education Method	Lectures and Projects	
Literature and Study Materials	Study material on the Brightspace	
Assessment	Oral exam after the first part/individual projects (report, presentation) the second part. In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5 Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI4138	Decision Theory/Expert Judgment	6
Responsible Instructor	G.F. Nane	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Probability and Statistics	
Course Contents	Representation of uncertainty as rational preference, subjective probability, utility, Savage's representation theorem, exchangeability, De Finetti's theorem, expert judgment, scoring rules, performance evaluation, applications.	
Study Goals	1. Can explain when and why expert judgement is needed 2. Acquire an in-depth understanding of rational decision theory and the mathematical foundations for the use of expert opinion in science. 3. Be capable of designing an expert elicitation study following the Classical Model setting 4. Be capable of conducting a real-life structured expert judgment elicitation 5. Be capable to analyze the expert data. 6. Be capable to summarize the findings in a concise and accessible report and presentation for peers.	
Education Method	Lectures, videos and online exercises (MOOC) Expected workload: 1. Attending lectures and follow the online material: 28 hours 2. Oral examination: 1 hour 3. Preparing for oral examination: 36 hours 4. Project: 70 hours 5. Oral presentation of projects: 5 h 6. Preparing for lectures, homework assignments: 20 hours 7. Reading papers: 8 hours	
Literature and Study Materials	Experts in Uncertainty, Oxford. U. Press 1991.	
Assessment	The final grade of the course consists of the following components: - Oral examination after Q3 (40% of the final grade) - Project and report (40% of the final grade) - oral presentation after Q4 (10% of the final grade) and short report comparing expert judgment methods in Q4 (10%). Final grade calculation: $(0.4 * \text{oral exam}) + (0.4 * \text{project report}) + (0.1 * \text{oral exam}) + (0.1 * \text{report})$ Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - oral exam: oral resit - reports: resubmit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI4156(TU)	Game theory	6
Responsible Instructor	Dr. R.J. Fokkink	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Required for	Open to all MSc students with a strong command of mathematics. Ideally suited for math students with an interest in finance.	
Expected prior knowledge	Standard BSc math courses: calculus, linear algebra, probability.	
Course Contents	This is a course in "mathematical" game theory, which is the counterpart of "algorithmic" game theory (agent based modelling, this part of game theory is taught in the computer science department). We cover all subjects from combinatorial games to cooperative games.	
Study Goals	Understanding the mathematical theory of rational behaviour/conflicts/negotiations/social choice. Game theory is one of the most prominent economic theories: three recent Nobel prizes have been awarded to game theorists. We will cover a very wide area, ranging from pure mathematics to economics, with an emphasis on the mathematical aspects. This is a course in the Finance Track of the Applied Maths programme, but it is open to all MSc students (math and non-math).	
Education Method	Lectures	
Literature and Study Materials	Game Theory - Thomas Ferguson - UCLA: find it on the net.	
Books	T.S. Ferguson, A course in game theory, World Scientific, 2020	
Assessment	The final grade of the course consists of the following components: written exam (100%) Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI4505	Quantitative Risk Management	6
Responsible Instructor	F. Barsotti	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The implementation of sound and reliable quantitative risk models is of vital interest for all banks and financial institutions, both from an economic and a legal point of view. This course provides a rather comprehensive treatment of the theoretical concepts and modeling tools of quantitative risk management. At the end of the course, students will master both theory and practice, and they will be able to solve real world problems. We will cover the modeling of important risks like: market risk, operational risk and model risk. All models will be analyzed in their points of strength and, most importantly, their points of weakness, to address the main deficiencies of current practice.	
Study Goals	- To understand risk, epistemologically, statistically and in practice. - To be able to use the main risk measures available and, in case, to develop new ones. - To know the basics of EVT. - To be able to model market, credit and operational risks, using the most recent techniques. - To be able to estimate most of the models we will discuss together on real data. - To know the strengths and the weaknesses of all the discussed models, i.e. to understand the importance of model risk. - To acquire useful and practical knowledge in risk management.	
Education Method	Front lectures, as necessary supplemented by online materials including videos, case studies.	
Literature and Study Materials	McNeil, Frey and Embrechts: Quantitative Risk Management, Princeton University Press, 2015.	
Prerequisites	"Financial Mathematics" (WI4079) and "Computational Finance" (WI4154).	
Assessment	The final grade of the course consists of the following components: - Written / Oral exam (100%). Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Written exam: written resit OR - Oral exam: oral resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	Standard non graphical calculator.	
Special Information	If you are not a mathematics student, feel free to join the course, no need to ask, but be aware that this course is highly formalized and relies on good mathematical foundation knowledge. This is a course of the master in applied mathematics!	

WI4614	Stochastic Simulation	6
Responsible Instructor	Dr.ir. L.E. Meester	
Contact Hours / Week x/x/x/x	0/0/2/2 lecture & 0/0/2/2 projects.	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Martingales, Brownian Motion and Stochastic processes (wi4430 or equivalent course), in particular measure theoretic probability, and basic statistics, including linear regression. Familiarity with R, Matlab, or equivalent software for simulation, is recommended.	
Course Contents	Stochastic simulation: algorithms and analysis. Emphasis in this course is not in the first place on doing simulations, but on understanding (the workings of) simulation techniques and how simulation can be used to provide insight into stochastic problems; the most important aspect of the "doing simulations" part is that you can derive the simulation steps/algorithm that results when one of the covered methods is applied to a problem. Since the key in efficient simulation almost always lies with the stochastic specifics of the problem, we focus on the stochastic methods that play a role in this process. Some simulations will be done in the mini-projects, to supplement and illustrate the theory. Here, you are encouraged to "bring your own problem to work on". Subjects: introduction, general aspects of stochastic simulation; generating random objects, univariate and multivariate random variables, and stochastic processes; analysis of simulation output: how to obtain estimates for quantities of interest, as well as confidence intervals; bias and small sample issues; variance reduction, especially their stochastic background and optimization; in addition, a selection of some of the following: Quasi-Monte Carlo, discretization methods, estimating derivatives via simulation.	
Study Goals	After completing this course successfully, you will be able to 1. explain the general principles of pseudo random-number generators; 2. mention some of their important properties and how they may be assessed; 3. explain and apply the inverse transform and acceptance-rejection methods; 4. simulate, i.e., generate realizations of, random variables and random vectors; 5. do the same for some stochastic processes, among others Brownian motion and the Brownian bridge; 6. process simulation output into estimates plus standard errors/CIs; both for simple and more complex simulation schemes; 7. explain the principles of a number of variance reduction methods: control variates, antithetic variates, stratified sampling, importance sampling; 8. and apply them to problems of moderate size and complexity; 9. explain what quasi-Monte Carlo methods are and how the van der Corput and Halton sequences are generated; 10. explain some of the advantages and disadvantages of QMC methods, and what can be done about the disadvantages; 11. explain three methods of estimating parameter sensitivities or derivative estimation: finite-difference approximation, pathwise derivative estimates, the likelihood ratio method; 12. for each of them, work out the specific details of their application for small problems; In addition: 1. you will acquire some simulation experience, first-hand, including processing and presenting the results; 2. and second-hand by sharing others experience; 3. you will practice developing a critical attitude towards presented simulation results, 4. as well as asking the right questions; 5. you will acquire some insight into Where might I use/deploy what (method or trick) from the simulation toolbox to my advantage?	
Education Method	Lectures and a bi-weekly discussion/presentation/problem solving session; homework assignments;self-study.	
Literature and Study Materials	Paul Glasserman, Monte Carlo methods in Financial Engineering, 2003, ISBN 978-0-387-00451-8. Additional reference: Stochastic Simulation by Søren Asmussen and Peter Glynn, ISBN 978-0-387-69033-9 (somewhat higher level, but also more depth).	
Prerequisites	Martingales, Brownian Motion and Stochastic processes (wi4430 or equivalent course), in particular measure theoretic probability, and basic statistics, including linear regression. Familiarity with R, Matlab, or equivalent software for simulation, is recommended.	
Assessment	The final grade of the course consists of the following components: - Mini-projects (25%); - written exam (closed book; 75%); the grade for the final must be 6 or higher to pass the course. Final grade calculation: $(0.25 * \text{mini-projects}) + (0.75 * \text{written exam})$ Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Mini-projects: resubmit deliverable - Written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI4640	High-dimensional Probability	6
Responsible Instructor	Prof.dr. F.H.J. Redig	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Basic analysis and linear algebra Introduction to Probability and Statistics Basic measure theory (as taught in the course of real analysis)	
Course Contents	High dimensional probability is the study of random vectors in $\mathbb{R}^n$ where the dimension n is large. This modern research area is important in various applications such as - Large random structures: random matrices, random networks. - Statistics and machine learning - Statistical physics - Randomized algorithms - Mixing times and other phenomena in high-dimensional Markov chains An important aspect of this theory is non-asymptotic probabilistic bounds (concentration inequalities), i.e., inequalities which are or dimension free or where the role of the dimension is explicit. The course will mainly focus on various concentration inequalities and applications. Attention will also be given to Markov semigroups and interpolation methods based on them. Concrete subjects treated in the course are - Azuma Hoeffding inequality for martingale differences - Mc Diarmid inequality, bounded difference inequality - Sub-Gaussian random variables - Sub-exponential random variables - Efron-Stein inequality and tensorization of variance - Poincare inequality and the Ornstein Uhlenbeck semigroup - Log-Sobolev inequality and the Orstein Uhlenbeck semigroup - Tensorization of entropy	
Study Goals	1. Getting acquainted with various concentration inequalities and their use in random graphs, randomized algorithms, data science. (tested via the homework) 2. Being able to study, make a report and a presentation on a research paper or a chapter of a book on theoretical or applied aspects of the theory provided during the course (tested via the presentation).	
Education Method	Quarter 3: lectures based on the books of Vershynin, Von Handel. Lecture notes are provided. Quarter 4: self-study of the material of the presentation, assistance is provided in the form of offices hours (around the usual lecture time slot).	
Literature and Study Materials	The basic material of the course will be based on the following two sources: 1. Vershynin: High-Dimensional Probability An Introduction with Applications in Data Science (Draft available at the homepage of the author) 2. Von Handel: Probability in High Dimension. Lecture notes available at the homepage of the author. Besides, various papers and books will be made available for the presentations of the students (these will appear on the brightspace page of the course).	
Assessment	The final grade of the course consists of the following components: - Homework - presentation (or oral exam). The weighting is 0.4 homework+ 0.6 presentation (or oral exam) For the presentation the evaluation criteria are 0.4 content 0.4 clarity 0.2 presentation skills Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Homework: resubmit deliverable - Presentation/oral exam: oral resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Specialisation Mathematics of Data Science AM 2022

Introduction 1

Specialisation Mathematics of Data Science

Compulsory orientation course:

WI4635 Linear Algebra and Optimisation for Machine Learning

Compulsory specialisation course:

WI4630 Statistical Learning

Compulsory specialisation courses (choose 2 out of 4 courses):

WI4203 Applied Functional Analysis

WI4640 High Dimensional Probability

WI4455 Statistical Inference

WI4614 Stochastic Simulation

All of the compulsory specialisation courses can also be chosen as recommended specialisation course.

Recommended specialisation courses:

MM Machine Learning Theory

WI4207 Continuous Optimization

WI4410 Advanced Discrete Optimization

WI4230 Time series and Extreme Value Theory

CS4230 Machine Learning 2

WI4665 Advanced Topics in Statistics

MM Semidefinite Optimization

EE4685 Machine Learning: a Bayesian Perspective

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Compulsory Courses

WI4630	Statistical Learning	6
Responsible Instructor	Dr.ir. G.N.J.C. Bierkens	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Required: * Bachelor level knowledge and ability in calculus, probability, statistics, optimization, linear algebra. * Programming skills in Python (AM1090). If you do not have this background, consult the book Think Python, https://greenteapress.com/wp/think-python/, and familiarize yourself with the Numpy and Matplotlib packages. Recommended: * Linear Algebra and Optimization for Machine Learning (WI4635) * Statistical Inference (WI4455) * Knowledge of measure theory (as, for example, in AM2090)	
Course Contents	Statistical learning provides a statistical perspective on machine learning. The overarching goal of the course is to develop methods for estimating (or 'learning') an unknown function from data or making predictions for unseen function outputs. The course aims to empower the student to make a justified decision in adopting machine learning approaches for practical problems and even design their own machine learning methodology using sound mathematical principles. This is achieved by studying key ideas, models, algorithms and theories related to the subject of statistical learning. In the first half of the course a collection of essential models, algorithms and techniques is introduced. In the second half of the course we adopt a Bayesian perspective on machine learning and study associated methods for analysing models and developing computational methodology. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Study Goals	By the end of the course, a student should be able to: 1. Explain the challenges of statistical learning (e.g., curse of dimensionality, overfitting, bias-variance trade-off). 2. Implement machine learning algorithms and computational methods on artificial and real-world data sets (e.g., linear regression, logistic regression, sparse regression, neural networks, decision trees). 3. Compare the theoretical properties and practical aspects of machine learning approaches. 4. Validate machine learning approaches on arbitrary data sets. 5. Design tailor-made computational approaches for new learning problems by adapting, combining and/or extending methods and/or by employing a Bayesian approach.	
Education Method	Lectures, exercises, practical assignments, practice exam	
Literature and Study Materials	Lecture notes and slides will be made available. The main supporting reference is: Murphy, Kevin P. Probabilistic machine learning: an introduction. MIT press, 2022. which is publicly available here: https://probml.github.io/pml-book/book1.html Alternative background reading: * Hastie, T., Tibshirani, R., & Friedman, J. (2009). The elements of statistical learning (Second). Springer, New York. https://web.stanford.edu/~hastie/ElemStatLearn/ * Bishop, C. M. (2006). Pattern recognition and machine learning. Springer, New York. https://www.microsoft.com/en-us/research/publication/pattern-recognition-machine-learning/	
Assessment	The final grade of the course consists of the following components: * Practical assignments during the course (30%) * Final written exam (70%) Final grade calculation: $(0.3 * \text{practical assignments} + 0.7 * \text{final written exam})$ Note: For both components, a sufficient grade (5.8) is required. Repair possibilities: * Assignments: resubmit deliverable * Written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Compulsory Courses (select 2 out of 4)

WI4203	Applied Functional Analysis	6
Responsible Instructor	Prof.dr. J.M.A.M. van Neerven	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	The student is expected to have firm knowledge in real and complex analysis and linear algebra. Students can use the checklist below to see if he/she is well-prepared: From set theory: standard set-theoretic notions and notations, mappings and relations, injective and surjective mappings, equivalence relations and partial orderings, supremum and infimum, countable and uncountable sets. From analysis: convergence and limits, Cauchy sequences, theorem of Bolzano-Weierstrass, completeness of the fields of real and complex numbers, continuity, differentiability, convergence and absolute convergence of series, power series. From Topology: metric spaces, open and closed sets, denseness, completeness, compactness. From Integration Theory: theory of the Lebesgue integral, monotone and dominated convergence theorem, Fubini's theorem. From (linear) algebra: vector spaces over the real and complex numbers, linear mappings, subspaces, direct sum of subspaces, kernel and range of a linear mapping, linear independence, linear span, bases, dimension, polynomials, continuous functions on an interval as a vector space, inner product spaces, Cauchy-Schwarz inequality. From Complex Variables: holomorphic functions, Cauchy's integral formula and integral theorem, power series representation of holomorphic functions, analytic continuation.	
Course Contents	The course introduces those notions and results of functional analysis that are relevant for applications to, e.g., partial differential equations, numerical analysis, signal processing, probability theory, stochastic analysis, and quantum theory. More precisely, the course covers the following subjects: - Banach spaces: norms, duality, Hahn-Banach theorems. - Hilbert spaces: best approximation, orthogonality, orthocomplementation, orthonormal bases, Riesz representation theorem. - $C(K)$-spaces and L^p-spaces: completeness, duality, compactness, approximation techniques. - Bounded linear operators: uniform boundedness theorem, open mapping theorem, closed graph theorem, compact operators, adjoint operators, spectrum and resolvent, spectral projections, Riesz-Schauder theory, Fredholm operators. - Hilbert space operators: orthogonal projections, unitary operators, self-adjoint operators, normal operators, spectral theorem for compact normal operators. - weak derivatives, Sobolev spaces, applications to boundary value problems - densely defined operators, closed operators - semigroups of operators, generator, resolvent, Hille-Yosida theorem.	
Study Goals	After completing this course, the student is able to: 1) reproduce the definitions and statements of the main theorems of functional analysis as covered in the course 2) give correct and rigorous proofs of those results whose proofs do not go beyond standard reasoning from first principles 3) apply standard functional analytic proof techniques in new situations 4) name and describe some applications of functional analysis in other areas of mathematics, such as partial differential equations and quantum theory 5) communicate and present mathematical proofs clearly and understandably in a brightspace presentation	
Education Method	Lectures and Exercises/Assignments. See Brightspace for details.	
Literature and Study Materials	The course is based on my textbook "Functional Analysis", Cambridge University Press, 2022. A corrected printing will appear in the Summer of 2023. The e-book version of it is downloadable free of charge via the TUD library and the CUP website for IP addresses at the TUD. In preprint form, the book is available via my personal website.	
Assessment	Written exam, in combination with graded exercises. Details will be announced via Brightspace. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	None	

WI4455	Statistical Inference	6
Responsible Instructor	Prof.dr.ir. G. Jongbloed	
Co-responsible for assignments	T.F.W. van der Jagt	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	- Introductory statistics and probability course at bachelor level (such as: various probability distributions, central limit theorem, estimators, maximum likelihood estimation, frequentist hypothesis testing). - Familiarity with concepts from measure theoretic probability.	
Course Contents	This course aims to provide a concise but comprehensive account of the essential elements of statistical inference and theory. The course focusses on a clear presentation of the main concepts and results underlying different frameworks of inference.	
Study Goals	Study goals Statistical Inference WI4455 The topics correspond to chapters in the lecture notes. 1. Preliminaries: - The student is able to reproduce the definition of a statistical model, possibly defined in terms of densities. - The student is able to explain the unifying concept of exponential families (the precise form of the density will be given at the exam) - The student is able to assess fundamental differences between classical- and Bayesian approaches towards problems in the given examples. - The student is able to identify with the different notions of stochastic convergence and is able to apply the Delta-method. 2. Topics from classical statistics: - The student is able to prove that a statistic is sufficient. The factorisation theorem can be proved in case of discrete random variables. - The student is able to reproduce the definition of Fisher information and Kullback-Leiber information and can prove properties of these (as in the lecture notes). - The student is able to check whether an estimator is UMVU and can prove the Lehmann-Scheffe and Cramer-Rao bound theorems. - The student can state the limiting distribution of the MLE in regular cases and can give examples of cases where the MLE is inconsistent. - The student can compute tests, p-values, power-curves of tests and errors of type I and II of (randomised) tests in simple settings. The student can prove the Neyman-Pearson lemma. - The student can state criticism to the use of p-values. - The student can derive confidence sets using (asymptotic) pivots. 3. The likelihood principle: - The student can state Birnbaums result. - The student is able to explain the implications of the sufficiency-, conditionality- and likelihood principle. 4. Bayesian analysis: - The student is able to explain the Bayesian approach towards statistical inference and can apply Bayes' rule to do Bayesian computations in case a closed form solution is available. - The student is able to describe the concept of exchangeability and its relation to justifying Bayesian analysis. - The student can derive Jeffreys prior in a given problem setting. - The student can construct hierarchical models. - The students can explain how empirical Bayes methods are defined, and can compute empirical Bayes estimators in a given problem setting. - The student can state the Bernstein-Von Mises theorem and its implications. 5. Markov Chain Monte Carlo methods: - The student can derive the steps of a MCMC scheme for drawing from the posterior, also in case a closed form solution for drawing from this posterior is unavailable. This scheme may include Gibbs steps. 6. Statistical decision theory: - The student is able to describe the basics of statistical decision theory including admissibility, minimax optimality, Bayes optimality, and connections between these topics. The student can prove results that state interrelations between these topics. - The student is able to state Steins example of inadmissibility of the MLE. - The student can derive minimax and Bayes estimators in tractable and finite-decision problems. - The student is able to derive Bayesian tests and compute a Bayes factor for a given statistical model.	
Education Method	Each week there will be a 2-hours lecture and a scheduled meeting where exercises from the lecture notes are discussed.	
Literature and Study Materials	Lecture notes	
Reader	Lecture notes will be available on brightspace.	
Assessment	At the end of the course there will be a written on campus exam (3 hours) which counts for 100%. The assignments in the lecture notes are representative for the level and type of questions at the exam. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI4614	Stochastic Simulation	6
Responsible Instructor	Dr.ir. L.E. Meester	
Contact Hours / Week x/x/x/x	0/0/2/2 lecture & 0/0/2/2 projects.	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Martingales, Brownian Motion and Stochastic processes (wi4430 or equivalent course), in particular measure theoretic probability, and basic statistics, including linear regression. Familiarity with R, Matlab, or equivalent software for simulation, is recommended.	
Course Contents	Stochastic simulation: algorithms and analysis. Emphasis in this course is not in the first place on doing simulations, but on understanding (the workings of) simulation techniques and how simulation can be used to provide insight into stochastic problems; the most important aspect of the "doing simulations" part is that you can derive the simulation steps/algorithm that results when one of the covered methods is applied to a problem. Since the key in efficient simulation almost always lies with the stochastic specifics of the problem, we focus on the stochastic methods that play a role in this process. Some simulations will be done in the mini-projects, to supplement and illustrate the theory. Here, you are encouraged to "bring your own problem to work on". Subjects: introduction, general aspects of stochastic simulation; generating random objects, univariate and multivariate random variables, and stochastic processes; analysis of simulation output: how to obtain estimates for quantities of interest, as well as confidence intervals; bias and small sample issues; variance reduction, especially their stochastic background and optimization; in addition, a selection of some of the following: Quasi-Monte Carlo, discretization methods, estimating derivatives via simulation.	
Study Goals	After completing this course successfully, you will be able to 1. explain the general principles of pseudo random-number generators; 2. mention some of their important properties and how they may be assessed; 3. explain and apply the inverse transform and acceptance-rejection methods; 4. simulate, i.e., generate realizations of, random variables and random vectors; 5. do the same for some stochastic processes, among others Brownian motion and the Brownian bridge; 6. process simulation output into estimates plus standard errors/CIs; both for simple and more complex simulation schemes; 7. explain the principles of a number of variance reduction methods: control variates, antithetic variates, stratified sampling, importance sampling; 8. and apply them to problems of moderate size and complexity; 9. explain what quasi-Monte Carlo methods are and how the van der Corput and Halton sequences are generated; 10. explain some of the advantages and disadvantages of QMC methods, and what can be done about the disadvantages; 11. explain three methods of estimating parameter sensitivities or derivative estimation: finite-difference approximation, pathwise derivative estimates, the likelihood ratio method; 12. for each of them, work out the specific details of their application for small problems; In addition: 1. you will acquire some simulation experience, first-hand, including processing and presenting the results; 2. and second-hand by sharing others experience; 3. you will practice developing a critical attitude towards presented simulation results, 4. as well as asking the right questions; 5. you will acquire some insight into Where might I use/deploy what (method or trick) from the simulation toolbox to my advantage?	
Education Method	Lectures and a bi-weekly discussion/presentation/problem solving session; homework assignments;self-study.	
Literature and Study Materials	Paul Glasserman, Monte Carlo methods in Financial Engineering, 2003, ISBN 978-0-387-00451-8. Additional reference: Stochastic Simulation by Søren Asmussen and Peter Glynn, ISBN 978-0-387-69033-9 (somewhat higher level, but also more depth).	
Prerequisites	Martingales, Brownian Motion and Stochastic processes (wi4430 or equivalent course), in particular measure theoretic probability, and basic statistics, including linear regression. Familiarity with R, Matlab, or equivalent software for simulation, is recommended.	
Assessment	The final grade of the course consists of the following components: - Mini-projects (25%); - written exam (closed book; 75%); the grade for the final must be 6 or higher to pass the course. Final grade calculation: $(0.25 * \text{mini-projects}) + (0.75 * \text{written exam})$ Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Mini-projects: resubmit deliverable - Written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI4640	High-dimensional Probability	6
Responsible Instructor	Prof.dr. F.H.J. Redig	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Basic analysis and linear algebra Introduction to Probability and Statistics Basic measure theory (as taught in the course of real analysis)	
Course Contents	High dimensional probability is the study of random vectors in $\mathbb{R}^n$ where the dimension n is large. This modern research area is important in various applications such as - Large random structures: random matrices, random networks. - Statistics and machine learning - Statistical physics - Randomized algorithms - Mixing times and other phenomena in high-dimensional Markov chains An important aspect of this theory is non-asymptotic probabilistic bounds (concentration inequalities), i.e., inequalities which are or dimension free or where the role of the dimension is explicit. The course will mainly focus on various concentration inequalities and applications. Attention will also be given to Markov semigroups and interpolation methods based on them. Concrete subjects treated in the course are - Azuma Hoeffding inequality for martingale differences - Mc Diarmid inequality, bounded difference inequality - Sub-Gaussian random variables - Sub-exponential random variables - Efron-Stein inequality and tensorization of variance - Poincare inequality and the Ornstein Uhlenbeck semigroup - Log-Sobolev inequality and the Orstein Uhlenbeck semigroup - Tensorization of entropy	
Study Goals	1. Getting acquainted with various concentration inequalities and their use in random graphs, randomized algorithms, data science. (tested via the homework) 2. Being able to study, make a report and a presentation on a research paper or a chapter of a book on theoretical or applied aspects of the theory provided during the course (tested via the presentation).	
Education Method	Quarter 3: lectures based on the books of Vershynin, Von Handel. Lecture notes are provided. Quarter 4: self-study of the material of the presentation, assistance is provided in the form of offices hours (around the usual lecture time slot).	
Literature and Study Materials	The basic material of the course will be based on the following two sources: 1. Vershynin: High-Dimensional Probability An Introduction with Applications in Data Science (Draft available at the homepage of the author) 2. Von Handel: Probability in High Dimension. Lecture notes available at the homepage of the author. Besides, various papers and books will be made available for the presentations of the students (these will appear on the brightspace page of the course).	
Assessment	The final grade of the course consists of the following components: - Homework - presentation (or oral exam). The weighting is 0.4 homework+ 0.6 presentation (or oral exam) For the presentation the evaluation criteria are 0.4 content 0.4 clarity 0.2 presentation skills Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Homework: resubmit deliverable - Presentation/oral exam: oral resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Recommended Courses

CS4230	Machine Learning 2	5
Responsible Instructor	Dr.ir. J.H. Krijthe	
Instructor	Dr. F.A. Oliehoek	
Instructor	T.J. Viering	
Instructor	Dr. D.M.J. Tax	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Required: This course is the more advanced and research oriented follow-up to Machine Learning 1 (CS4220). CS4220 or an equivalent machine learning course is therefore expected as prior knowledge.	
Course Contents	The course will treat a number of machine learning settings and theories spanning the breadth of machine learning research in detail and on an advanced level.	
Study Goals	Topics that are covered include (subject to change): learning theory, learning curves, latent variable models, causal inference and discovery, adversarial learning, online learning and neuromorphic computing. After completing the course, you should be able to: 1. Identify which setting beyond (un)supervised learning a problem belongs to 2. Explain the essential problem that (adversarial/causal/online/PAC/neuromorphic/...) learning addresses and how it differs from the classical ML setting 3. Evaluate methods that are proposed in these learning scenarios, both theoretically and empirically 4. Mathematically prove basic properties of methods in these scenarios 5. Apply common methods in these learning scenarios on real-world datasets 6. Develop machine learning techniques adapted for a particular learning task	
Education Method	Each week: 1 lecture and 1 Q&A session where the weekly exercise set will be discussed.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%) 4 combined assignments (pass/fail). There will be multiple small assignments (in weeks 2,4,6,8). Final grade calculation: $P/F * \text{assignments} + 1 * \text{Written exam}$ A passing final grade for the course can only be earned when for all components at least a 5.0/pass is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: 4 combined assignments: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4685	Machine Learning, a Bayesian Perspective	5
Responsible Instructor	Dr. J.H.G. Dauwels	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Good understanding of linear algebra and calculus (vector and matrix operations, gradient and Hessian), probability and statistics. It is assumed that the student has seen a basic course on machine learning that has presented an overview of the general concepts.	
Course Contents	This course provides a solid mathematical foundation for machine learning algorithms, presented from a statistical (Bayesian) and optimization point of view. It is considered to be a second (more advanced) course, after an introductory course that would have presented the general concepts. Machine Learning is an umbrella for a range of methods coming from different scientific communities, such as (EE) statistical learning, statistical signal processing, adaptive signal processing, image processing and analysis, system identification and control, and (CS) pattern recognition, data mining and information retrieval, computer vision, and computational learning. The name Machine Learning indicates what these disciplines have in common: to learn from data, and then make predictions. What one tries to learn from data is their underlying structure and regularities, via the development of a model, which can then be used to provide predictions. To this end, a number of diverse approaches have been developed, ranging from optimization of cost functions, whose goal is to optimize the deviation between what one observes from data and what the model predicts, to probabilistic models that attempt to model the statistical properties of the observed data.	
Course Contents Continuation	The course covers 3 introductory chapters and 9 regular chapters of a recently published rather exhaustive book. The introductory chapters summarize the mathematical language used in the course: conditional probability, distributions, random processes (as also seen in EE4C03), and repeat basic tools in Estimation (regression) and Detection (classification), as seen in ET4386. The course itself then presents three groups of techniques: - Bayesian learning: inference; conjugate priors; the EM algorithm; Gaussian mixture models; non-parametric models; variational bound approximation technique; sparse Bayesian learning; relevance vector machine (RVM). - Graphical models: Bayesian networks, factor graphs, message passing algorithms. Hidden Markov models. Learning graphical models. - Neural networks and deep learning: (probabilistic) CNN, variational autoencoder, GAN, learning deep networks.	
Study Goals	The objectives of the course are as follows: - Present the fundamentals of Bayesian learning, probabilistic graphical models, neural networks and deep learning in a unifying context. - For each of these, present compelling examples that demonstrate how the theory is applied in practice. Very often, this centers on the translation of a problem into a suitable model description, such that it becomes apparent which tools can be applied.	
Study Goals continuation	After the course, the student is able to - Translate a data analysis problem as a Bayesian machine learning problem through a 4-step process (problem, probabilistic model, inference algorithm, and solution). - Reason about the most appropriate tools to solve the data analysis problem. - Implement the most appropriate tools to solve the data analysis problem, e.g., for ECG or EEG analysis, radar, image analysis/denoising - Derive proofs that underly these tools and techniques. - Explore recent literature that builds upon the presented tools and techniques. - Compare the pros and cons of a variety of machine learning solutions in terms of societal aspects such as (but not limited to) quality control, ethics, robustness, and privacy.	
Education Method	Lectures; assignments. The course consists of 11 lectures of 2h per week theory, and 4 lectures/working sessions for example discussions. Each 2 weeks about 2 chapters are covered, centered around a compelling example/exercise/ data set that students take home to work on; the exercise is later discussed in depth during class. The examples are taken e.g. from biomedical applications (ECG, EEG data), time-frequency (spectral) data, radar, computer vision. Data sets will be provided.	
Course Relations	Although sufficiently independent, the course refers to EE4C03 Stat DSP, ET4386 Estimation and detection, EE4530 Applied convex optimization.	
Literature and Study Materials	Machine Learning, A Bayesian and Optimization Perspective by Sergios Theodoridis, Elsevier Academic Press, 2nd Edition (Feb 2020), hardcover ISBN: 9780128188033	
Assessment	The course is closed by a written exam (40% of final grade). There is one resit per year for the written exam. Several homework assignments count together towards 60% of the grade; the assignments involve solving machine learning problems, writing a report and essay, and giving a presentation. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

WI4207	Continuous Optimization	6
Responsible Instructor	D. de Laat	
Contact Hours / Week x/x/x/x	1e semester Utrecht maandag	
Education Period	1 2	
Start Education	1	
Exam Period	2	
Course Language	English	
Expected prior knowledge	The student should have a solid knowledge of linear algebra and multivariable calculus. The student should also have knowledge of linear programming (including linear programming duality) and convex analysis to the level of being able to follow the text and do the exercises from: Chapters 1 and 2 (including the exercises) from the book 'Linear Programming, A Concise Introduction, Thomas S. Ferguson' https://www.math.ucla.edu/~tom/LP.pdf Exercises 2.1, 2.2, 2.12, 3.1, 3.3, 3.5, and 3.7 from the book 'Convex Optimization, Stephen Boyd and Lieven Vandenbergh' http://stanford.edu/~boyd/cvxbook	
Course Contents	Continuous optimization is the branch of optimization where we optimize a (differentiable) function over continuous (as opposed to discrete) variables. Here the variables can be subject to equality and inequality constraints. Optimization problems of this form are common in science and engineering, machine learning, and as relaxations of discrete optimization problems. One can use linear algebra, multivariable calculus, and convex geometry to study the properties of continuous optimization problems and to design and analyze efficient algorithms. In this course, we study theory, algorithms, and some applications of continuous optimization. In the theory part we discuss convexity, Lagrangian duality, optimality conditions, and conic programming. In the algorithmic part we discuss derivative-free methods, first-order optimization methods, neural networks/supervised learning, second-order optimization methods, interior-point methods, and support vector machines. For some of the algorithms, we analyze the convergence.	
Study Goals	See the Mastermath page.	
Education Method	There will be 14 lectures.	
Literature and Study Materials	- Convex Optimization by Stephen Boyd and Lieven Vandenbergh (http://stanford.edu/~boyd/cvxbook) - Additional notes distributed by the lecturer	
Assessment	The course will have a written exam that counts for 100% of the grade. The resit will also be a written exam. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	This is a LNMB Mastermath course taught by David de Laat.	

WI4230	Time Series & Extreme Value Theory	6
Responsible Instructor	Dr. P. Chen	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Knowledge of probability and statistics at the level of the BSc Applied Mathematics courses on these topics (AM1080, AM2080).	
Course Contents	The course consists of two parts: Extreme Value Theory and Time Series. The course considers questions such as: What is the chance that the price of oil will be higher than \$120/Barrel coming week? Can we expect that UK interest rates will rise next year? How can we assess if global warming is occurring? What is the chance that the daily loss of the stock price of ABN-AMRO is larger than, say 20%, which has never occurred before? How to determine the height of a dike such that the probability of a flood, i.e., the water level exceeding the dike, in a given year is 1/10,000? What is the impact of a dramatic decline in the US stock market on DAX 30 index? In this course students will build up some theoretical basis, mainly probabilistic results within the area of time series and extreme value theory. Moreover, students will learn statistical methods for dealing with non-stationary data, multivariate data and rare events and assessing extreme risks for data from various fields such as finance, actuarial science, hydrology and environmental science. Furthermore the student is capable to perform a statistical analysis with the statistical package R. The Extreme Value Theory part considers topics such as max-domain attractions, peak over threshold approach, estimations of extreme value index, tail probability and high quantile, and tail dependence and independence. The Time Series part will cover topics such as ARMA models, spectral analysis, non-stationarity, GARCH models, state space models. Implementations of statistical methods in R will be part of the course. Examples of real applications from environmental science and finance will also be discussed.	
Study Goals	Specific learning objectives for the Extreme Value Theory part are: 1. Validate the max-domain attraction conditions, that is to show if a given distribution is in the max domain of attraction or not. 2. Compute the extreme value index of a given distribution. 3. Develop estimations for tail probabilities, high quantiles, m-year return level, using different methods. 4. Prove some asymptotic results (consistency) of the estimators. 5. Evaluate the performance of estimation methods by simulation study; and apply the estimation methods to a real data set. 6. Establish the properties of exponent measure and spectral measure. Establish the link between the bivariate extreme value distribution and these two measures. 7. Characterize the extremal dependence for bivariate distributions. 8. Develop the estimation for a bivariate tail probability (failure probability). Specific learning objectives for the Time Series part are: 1. Discern whether a given time series data set can be associated to a stationary time series, and if not, perform suitable transformations to so that the resulting data corresponds to a stationary time series. 2. Understand and determine interpret theoretical properties of AR(I)MA time series models, such as the existence and uniqueness of solutions, causality and invertibility. 3. Understand and determine descriptive statistics of time series models, such as best linear predictors, autocorrelation function, partial autocorrelation function, estimators for mean and variance, and spectral density function. 4. Have general fluency in dealing with standard manipulations in time series, such as working with linear filters, computing conditional expectations and solving recurrence relations. 5. Fit a suitable time series model to time series data and assess the quality of a time series fit. 6. Have basic knowledge of advanced time series models used in financial mathematics, such as GARCH and stochastic volatility models.	
Education Method	Lectures/exercises/computer assignment	
Books	The lectures will be mainly based on: - Time Series Analysis and Its Applications: With R Examples By Robert H. Shumway, David S. Stoffer - Time Series: Theory and Methods By Brockwell, Peter J. and Davis, Richard A - Extreme Value Theory: An Introduction By Laurens de Haan and Ana Ferreira (Chapter 1, 3, 4, 6 and 7)	
Assessment	There will be a final assessment for the extreme value theory part at the end of Q3 and a final (written or oral) assessment for the time series part at the end of Q4. (If there are at least 10 students we will organize a written exam.) There will also be assignments handed out during the course. The final grade can be determined by a weighed sum, with the following weights: 0.15 average over assignments extreme value theory 0.35 exam extreme value theory 0.15 average over assignments time series	

0.35 exam time series

Resit/ Repair opportunities:

In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for:

- Individual assignments: resubmit deliverable
- Written exam: written resit OR
- Oral exam: oral resit

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

WI4410	Advanced Discrete Optimization	6
Responsible Instructor	Dr.ir. L.J.J. van Iersel	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Linear optimization: basic solutions and their relation to extreme points, simplex method, duality and complementary slackness Integer optimization: basic cutting planes, branch-and-bound Computational complexity: Complexity classes P and NP and NP-complete, polynomial reductions	
Course Contents	Integer optimization: Polyhedral theory; Cutting planes; Gomory mixed-integer cuts, split cuts and their relation; Lift-and-project; IP in fixed dimension: Lattices and basis reduction; IP in fixed dimension: Lenstra's algorithm and extensions Parameterized complexity: fixed-parameter tractability, bounded search, kernelization, dynamic programming, treewidth Applications in phylogenetics: phylogenetic trees, agreement forests, SPR distance, phylogenetic networks	
Study Goals	To understand and be able to use the methods and concepts described in the course. The student should also be able to prove results similar to the ones described in the course.	
Education Method	There will be 2 lectures combined with exercise classes per week.	
Literature and Study Materials	Lecture slides, notes, papers and book chapters will be provided on Brightspace.	
Assessment	The final grade of the course consists of the following components: - Written exam (100%) Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. M.E.L. Jones	

WI4665	Advanced Topics in Statistics	6
Responsible Instructor	Prof.dr. A.W. van der Vaart	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	The subtitle of the course is Causality and Graphical Models A graphical model expresses dependence relations between variables. Each node of a graph represents a variable and each edge or arrow a relationship. Next to this structure there is a quantitative model in the form of a joint probability distribution of all variables. The graph expresses conditional (in)dependencies between the variables given other variables in the sense of probability. Graphical models with directed graphs are also called Bayesian networks, as Bayes rule gives the conditional probabilities. They are the basis of probabilistic expert systems, as applied in many contexts. In practice a graphical model must be learned from data, guided by prior information, and hence there is a strong statistical content. The graph helps to form a likelihood, and one can apply various statistical methods, including maximum likelihood and Bayesian methods, to estimate the graph or parameters. Because there usually are many variables, statistical methods for high-dimensional data and models can be necessary. This includes for instance efficiency theory for semiparametric models and penalized likelihood methods. A main application considered in this course is the discovery and estimation of causal effects from data. Cause and effect relationships between variables can be graphically represented as arrows in a graphical model for the variables. Causal discovery is concerned with inferring such a graph from observed data, and causal estimation with quantifying the effects. The challenge is to perform this with data that is not produced by a designed experiment, but observational, a situation that abounds in epidemiological and social science research. As correlation is not causation, corrections must be made for other possible explanations that might cause a correlation. Causal reasoning can alternatively be cast in terms of "potential outcomes", the effect that would have realised had a random individual or unit received a treatment of interest. Statistical methods for estimating causal effects can be justified in terms of potential outcomes. A given causal graph allows to define the relationships of such potential outcomes to other variables, and be the basis of verifying necessary assumptions on observed data. The subject of the course is at the intersection of statistics/probability and artificial intelligence. The style will be mathematical, although only a modest amount of specialised preknowledge is required, mostly a good grasp of probability and statistical methods, and of course linear algebra and analysis. We shall provide short introductions to conditioning, asymptotic methods, Bayesian methods, Gaussian models, etc. when needed.	
Study Goals	Understand statistical graphical models and their applications, in particular in causal inference.	
Education Method	Lectures. Homework problems.	
Assessment	The final grade of the course consists of the following components: - Written exam 70 % - Average grade of homework assignments 30 % Final grade calculation: $(0.7 * \text{written} + 0.3 * \text{homework})$ Repair possibilities: - Written exam: repeat exam. Final grade calculation: $(0.7 * \text{resit written} + 0.3 * \text{homework})$ Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Tags	Artificial intelligence Mathematics Stochastics	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Specialisation Partial Differential Equations AM 2023

Introduction 1

Specialisation Partial Differential Equations

Compulsory orientation course:

WI4655 Perturbation and Variational Methods for Partial Differential Equations

Compulsory specialisation courses (choose 2 out of 6 courses):

WI4203 Applied Functional Analysis

WI4019 Nonlinear Differential Equations

WI4520 Nonlinear Analysis and PDEs

WI4660 Dynamical Systems and Chaos

WI4430 Martingales, Brownian Motion

WI4201 Scientific Computing

All of the compulsory specialisation courses can also be chosen as recommended specialisation course.

Recommended specialisation courses:

WI4006 Special Functions and Representation Theory

WI4480 Internet Seminar on Evolution Equations

WI4485 Harmonic Analysis

WI4615 Stochastic Calculus

WI4046 Spectral Theory of Linear Operators

MM Calculus of Variations

WI4204 Advanced Modelling

WI4205 Applied Finite Elements

MM Introduction to Numerical Bifurcation Analysis of ODEs and Maps

MM Mathematical Biology

WI4211 Advanced Topics in Analysis

WI4680 Applications in Partial Differential Equations

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Compulsory Courses (select 2 out of 6)

WI4019-SP	Non-linear Differential Equations	6
Responsible Instructor	Dr.ir. W.T. van Horssen	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Introductory course on Differential Equations e.g. Boyce and DiPrima Chapters 2-7, 9.	
Course Contents	Existence and uniqueness of solutions of initial value problems, Gronwall's lemma, Autonomous systems, Critical points, Periodic solutions, Stability theory, Linear systems and Floquet theory, Perturbation theory and asymptotic methods, Poincare-Lindstedt method, Averaging method, Multiple time-scales method, Elementary bifurcations.	
Study Goals	After following this course the student will be able to apply fundamental and advanced, analytical techniques and methods to study properties of solutions of nonlinear ordinary differential equations. The mathematical methods and techniques are mentioned under "course contents". After studying the lecture notes and after making the exercises successfully the student can start to work on research problems in the field of nonlinear differential equations.	
Education Method	Lectures and a MAXIMA practical work. In total 14 lectures of 2 hours each will be given. The MAXIMA practical work is about 4 hours work. Following and preparing the lectures and the MAXIMA practical work takes about 60 to 70 hours. Making the 30 take-home exercises takes about 60 hours. Preparing the 45 minutes oral exam takes about 30 hours.	
Literature and Study Materials	F. Verhulst, Nonlinear Differential Equations and Dynamical Systems, Springer-Verlag 2nd edition, ISBN 3-540-60934-2.	
Assessment	The final grade of the course consists of the following components: - Take home exam (50%) - oral exam (50%) Final grade calculation: $(0.5 * \text{take home exam}) + (0.5 * \text{oral exam})$ The take-home exam consists of 30 exercises. The oral exams will be held at the end of the 2nd quarter, and the student is only allowed to take part in this oral exam when at least 70% of the take-home exercises are made correctly and when the MAXIMA practical work is followed successfully. In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5 take home exam: resubmit deliverable oral exam: oral resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI4201	Scientific Computing	6
Responsible Instructor	Prof.dr.ir. C. Vuik	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	A basic knowledge on partial differential equations (PDEs), on numerical methods for solving ODEs/PDEs, and on linear algebra.	
Course Contents	During the course, the important steps towards the solution of real-life applications dealing with partial differential equations will be outlined. Based on a well-known basic partial differential equation, which is representative for different application areas, we treat and discuss direct and iterative solution methods from numerical linear algebra in great detail. The discretization of the equation will result in a large system of discrete equations, which can be represented by a sparse matrix. After a discussion of direct solution methods, the iterative solution of such systems of equations is an important step during numerical simulation. Emphasis is laid upon the so-called Krylov subspace methods, like the Conjugate Gradient Methods.	
Study Goals	A. General Study Objectives 1. After having completed this course, the student is able to use finite differences to approximate the solution of a partial differential equation and use an efficient and robust method to compute or approximate the solution of the resulting large system of linear equations. 2. The student will be able to assess the reliability in terms of the accuracy and stability of the numerical approximation of the solution. 3. The student will be able to use methods to approximate eigenvalues of large matrices. B. Specific Study Objectives 1. Finite Difference Method: Boundary Conditions, Grid, Numbering of the unknowns, Error Analysis - The student will be able to discretize 2-dimensional Poisson type problems. The student is able to implement the boundaries in various ways. - The student will be able to analyze the properties of the resulting matrix, as there are symmetry, positive definiteness, M-matrix etc. Also the sparseness of the matrix can be investigated and a notion of band and profile matrices should be known. 2. Direct Solution Methods Gaussian elimination, pivoting, iterative refinement, band and profile methods - The student will be able to understand and use direct solution methods based on Gaussian elimination and LU decomposition. - The student will be able to estimate the error due to rounding errors in the answer and to use pivoting to eliminate most of the errors. - The student will be able to estimate the amount of flops for direct solution methods and how to use more efficient variants as there are Cholesky method, band and profile methods. 3. Basic Iterative Methods - The student will be able to construct Basic Iterative Methods based on the splitting of the matrix. Furthermore an analysis of the convergence could be made. - A comparison of the various method should be possible together with the dependence on the gridsize. - The student will be able to choose suitable starting vectors and stopping criteria. 4. Krylov Subspace methods CG, SPD, preconditioning, general matrices, CGNR, Bi-CGSTAB and GMRES - The student will be able to use and understand the conjugate gradient method, together with a good understanding of the (super)linear convergence. - The student will be able to make a good choice for Krylov Subspace methods in the non-SPD case. - The student will be able to combine Krylov Subspace methods with preconditioners and know three classes of preconditioners: diagonal scaling, BIM and incomplete decompositions. 5. Eigenvalue methods Power method, Lanczos method, Arnoldi method - The student will be able to use the power method and is able to investigate its convergence. - The student will be able to choose suitable starting vectors and stopping criteria. - The student will be able to use the Lanczos and Arnoldi method.	
Education Method	Lectures/computer exercises	
Literature and Study Materials	Lecture notes, for further reading the book Matrix Computations, G.H. Golub and C.F. van Loan, the Johns Hopkins University, Baltimore, 2013, can be used.	
Assessment	The final grade of the course consists of the following components: - homework exercises deadline start of Q2 leads to grade G1, - take home exam deadline half of January grade G2 - written exam grade G3. The final grade is $(G1+G2+2*G3)/4$, provided that all grades are larger than or equal to 5. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - homework exercises: resubmit deliverable - take home exam: resubmit deliverable - written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Prof.dr.ir. M.B. van Gijzen	

WI4203	Applied Functional Analysis	6
Responsible Instructor	Prof.dr. J.M.A.M. van Neerven	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	The student is expected to have firm knowledge in real and complex analysis and linear algebra. Students can use the checklist below to see if he/she is well-prepared: From set theory: standard set-theoretic notions and notations, mappings and relations, injective and surjective mappings, equivalence relations and partial orderings, supremum and infimum, countable and uncountable sets. From analysis: convergence and limits, Cauchy sequences, theorem of Bolzano-Weierstrass, completeness of the fields of real and complex numbers, continuity, differentiability, convergence and absolute convergence of series, power series. From Topology: metric spaces, open and closed sets, denseness, completeness, compactness. From Integration Theory: theory of the Lebesgue integral, monotone and dominated convergence theorem, Fubini's theorem. From (linear) algebra: vector spaces over the real and complex numbers, linear mappings, subspaces, direct sum of subspaces, kernel and range of a linear mapping, linear independence, linear span, bases, dimension, polynomials, continuous functions on an interval as a vector space, inner product spaces, Cauchy-Schwarz inequality. From Complex Variables: holomorphic functions, Cauchy's integral formula and integral theorem, power series representation of holomorphic functions, analytic continuation.	
Course Contents	The course introduces those notions and results of functional analysis that are relevant for applications to, e.g., partial differential equations, numerical analysis, signal processing, probability theory, stochastic analysis, and quantum theory. More precisely, the course covers the following subjects: - Banach spaces: norms, duality, Hahn-Banach theorems. - Hilbert spaces: best approximation, orthogonality, orthocomplementation, orthonormal bases, Riesz representation theorem. - $C(K)$-spaces and L^p-spaces: completeness, duality, compactness, approximation techniques. - Bounded linear operators: uniform boundedness theorem, open mapping theorem, closed graph theorem, compact operators, adjoint operators, spectrum and resolvent, spectral projections, Riesz-Schauder theory, Fredholm operators. - Hilbert space operators: orthogonal projections, unitary operators, self-adjoint operators, normal operators, spectral theorem for compact normal operators. - weak derivatives, Sobolev spaces, applications to boundary value problems - densely defined operators, closed operators - semigroups of operators, generator, resolvent, Hille-Yosida theorem.	
Study Goals	After completing this course, the student is able to: 1) reproduce the definitions and statements of the main theorems of functional analysis as covered in the course 2) give correct and rigorous proofs of those results whose proofs do not go beyond standard reasoning from first principles 3) apply standard functional analytic proof techniques in new situations 4) name and describe some applications of functional analysis in other areas of mathematics, such as partial differential equations and quantum theory 5) communicate and present mathematical proofs clearly and understandably in a brightspace presentation	
Education Method	Lectures and Exercises/Assignments. See Brightspace for details.	
Literature and Study Materials	The course is based on my textbook "Functional Analysis", Cambridge University Press, 2022. A corrected printing will appear in the Summer of 2023. The e-book version of it is downloadable free of charge via the TUD library and the CUP website for IP addresses at the TUD. In preprint form, the book is available via my personal website.	
Assessment	Written exam, in combination with graded exercises. Details will be announced via Brightspace. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	None	

WI4430	Martingales, Brownian Motion	6
Responsible Instructor	Prof.dr. F.H.J. Redig	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	1. Conditional expectation. 2. Martingales, stopping times, optional sampling. 3. Martingale convergence theorems, inequalities. 4. Elementary continuous-time stochastic processes: Poisson process, continuous-time Markov chains. 5. Brownian motion: definition, construction, basic properties. Martingales associated to Brownian motion, brownion motion with drift. 6. Elementary introduction to the stochastic integral, Ito isometry	
Study Goals	1. Understanding the concept of conditional expectation. Being able to compute conditional expectations in concrete examples, and being apply the tower property, and the take-out-what-is-known property. 2. Being able to recognize whether a process is a martingale, sub- or super martingale. 3. Being able to use martingales in computations such as gambler's ruin and distributions of hitting times. 4. Being able to apply the martingale convergence theorems and martingale inequalities in concrete contexts such as law of large numbers, branching process. 5. Being able to perform concrete computations with Brownian motion and its martingales (such as computation of hitting time distribution, recurrence transience). 6. Understanding the definition and construction of the stochastic integral and being able to perform elementary computations with it. 7. Being able to reproduce correctly definitions, statements and proofs and the main definitions and theorems as indicated in a list (this concerns the so-called theory question).	
Education Method	Lectures and problem sessions.	
Literature and Study Materials	1. Lecture notes based on chapter 10 of A. Gut ``A graduate course in probability" for the martingale part of the course. 2. Lecture notes for Brownian motion part of the course. The lecture notes are provided via Brightspace (course content).	
Reader	1. Lecture notes based on chapter 10 of A. Gut ``A graduate course in probability". 2. For the material on Brownian motion we will also sometimes use P. Morters and Y. Peres, Brownian Motion, (2010) freely available at http://people.bath.ac.uk/maspm/book.pdf . Also lecture notes will be available for that part. 3. Other recommended literature: Le Gall: Brownian motion, martingales and stochastic calculus, Springer 2016.	
Assessment	The final grade of the course consists of the following components: - Two graded homeworks - a written exam A homework grade h is computed as the average of the scores on the two graded homeworks. Course grade is the final exam grade f or $0.6f+0.4h$, whichever is larger, provided $f > 5$. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - homework assignments and written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). The homework grade does not count for the resit.	

WI4520	Nonlinear Analysis and Partial Differential Equations	6
Responsible Instructor	Dr. M.V. Gnann	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	required: courses in the bachelor in analysis and differential equations. The most important courses are: * Ordinary Differential Equations (AM2030) * Partial Differential Equations (AM2070) * Real Analysis (AM2090) recommended (useful but not necessary) courses in the bachelor and master in applied mathematics are: * Applied Functional Analysis (WI4203) * Dynamical Systems and Chaos (WI4660) * Fourier Analysis (AM3570) * Nonlinear Differential equations (WI4019) * Perturbation and Variational Methods for Partial Differential Equations (WI4655) * Scientific Computing (WI4201) * Spectral Theory of Linear Operators (WI4046)	
Course Contents	In the first part of the course, we will treat the Stokes and Navier-Stokes equations and construct weak and strong solutions to the latter. This will rely on spectral methods, embedding theorems for Sobolev spaces, interpolation methods, and compactness arguments. Thus, we will prove existence and uniqueness of global weak and strong solutions to the (time-dependent) Stokes equations, existence of global weak solutions to the Navier-Stokes equations, global uniqueness for the 2D Navier-Stokes equations, and existence and uniqueness of strong solutions to the Navier-Stokes equations for short times or small initial data. A discussion on the third Millennium Prize Problem rounds up the first half of the lecture. In the second part of this course we give an introduction to stability theory for nonlinear partial differential equations (PDE). Determining the stability of solutions is of central importance when analyzing PDE arising in applications, as it is typically the stable solutions that are observed in practice. This analysis is particularly challenging when the models are nonlinear, and often relies on the presence of symmetries and corresponding conservation laws, which are a common phenomenon in physical models. We will draw inspiration from linearization and energy methods for stability theory in finite-dimensional dynamical systems and show how these concepts can be adapted to the infinite-dimensional setting when studying nonlinear Hamiltonian PDE. We will apply the theory to several equations modeling wave propagation, such as the Klein-Gordon equation, the Nonlinear Schrödinger equation and the Korteweg-deVries equation.	
Study Goals	After the first part of the course the student can prove existence, (short-time) uniqueness, and regularity of solutions to equations from fluid dynamics, namely * the time-independent Stokes equations * the time-dependent Stokes equations * the Navier-Stokes equations. After the second part of the course the student can * formulate some nonlinear dispersive equations as Hamiltonian systems; * derive conservation laws from symmetries of Hamiltonian system; * characterize traveling waves as critical points of the Hamiltonian functional; * prove that (constrained) minimizers of this functional are (non)linearly stable; * analyse when saddle points of this functional are (non)linearly unstable. * apply the theory to prove (in)stability of solutions of model equations	
Education Method	Lectures	
Assessment	In each of the two quarters, a grade (Q_1/Q_2) is determined based on an oral exam (O_1/O_2), counting 80%, and solutions to exercise sheets (E_1/E_2), counting 20%. The arithmetic mean of these grades determines the final grade (F). The course is passed if the average grade is at least 6 points and if in each quarter a grade of at least 5 points is obtained. Rounding is done according to TU Delft regulations. It is possible to retake the oral exam, and if necessary the lectures and exercises in the following year, for each quarter independently until the passing criterion is fulfilled. In formulas: $Q_1 = 0.8 \cdot O_1 + 0.2 \cdot E_1$ $Q_2 = 0.8 \cdot O_2 + 0.2 \cdot E_2$ $F = (Q_1 + Q_2) / 2$ Passing criterion: $F \geq 6 \text{ and } Q_1 \geq 5 \text{ and } Q_2 \geq 5$ Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	This course is offered over two quarters. It is a master elective for the specializations Computational Science and Engineering (CSE) and in partial differential equations (PDE).	
Co-Instructor	Dr. A. Geyer	

WI4660	Dynamical Systems and Chaos	6
Responsible Instructor	Dr. K. Marynets	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Required: analysis, ordinary differential equations. Strongly advised: partial differential equations.	
Course Contents	Topics to be covered in the course include: - center manifolds for vector fields and maps; - normal forms for vector fields and maps; - local bifurcations for vector fields and maps; - fractional order dynamical systems and their stability; - continuous and discrete time chaotic systems; - Liapunov exponents; - Melnikovs method.	
Study Goals	The course aims to introduce the advanced analysis of continuous and discrete time applied dynamical systems in terms of their stability and chaotic behavior. At the end of this course the student will be able to: - analyse stability of a dynamical system, using the center manifold theory; - simplify dynamical systems using the normal forms' method; - analyse local bifurcations and bifurcation diagrams; - analyse stability of a fractional order dynamical system; - prove that the dynamical system is chaotic.	
Education Method	Lectures	
Books	The main books, used in the course are: - S. Wiggins, Introduction to Applied Nonlinear Dynamical Systems and Chaos, Springer-Verlag 2nd Edition, Hardcover ISBN 978-0-387-00177-7; eBook ISBN 978-0-387-21749-9. - I. Petras, Fractional order nonlinear systems: modeling, analysis and simulation; DOI: 10.1007/978-3-642-18101-6.	
Assessment	The final grade of the course consists of the following components: - 2 written assignments (AS1+AS2: 50% of the final grade): the passing grade of every assignment is 6.0, which corresponds to 70% of correctly made exercises; - oral exam (OE: 50% of the final grade): the passing grade of the oral exam is 6.0. Final grade calculation: $0.25 \cdot AS1 + 0.25 \cdot AS2 + 0.5 \cdot OE$. Repair possibilities: - written assignments: re-submit (parts of) the assignment during the course, the repair grade will be capped at 6.0; - oral exam: resit in next quarter. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Recommended Courses

WI4006	Special Functions and Representation Theory	6
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Basic knowledge of Group Theory and Complex Analysis.	
Course Contents	In this course we study special functions in connection with representation theory of specific Lie groups and Lie algebras. Topics include: - Gamma and Beta integrals - hypergeometric functions - representation theory of compact matrix Lie groups and their Lie algebras - representation theory of $SU(2)$ and $SO(3)$ - Jacobi polynomials and matrix elements of $SU(2)$ - tensor products and Clebsch-Gordan coefficients	
Study Goals	The student knows the material treated during the lectures and is able to apply this to several problems. The student should be able to explain the treated material.	
Education Method	Lectures	
Literature and Study Materials	Lecture notes will be available on Brightspace.	
Assessment	The final grade of the course consists of the following components: - Oral exam (70%) - mandatory homework assignments (30%). Final grade calculation: $(0.7 * \text{oral exam}) + (0.3 * \text{homework assignments})$ In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5, for: Oral exam: oral resit Homework assignments: resubmit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI4046	Spectral Theory of Linear Operators	6
Responsible Instructor	Dr. M.P.T. Caspers	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Elements of functional analysis: Banach and Hilbert spaces, bounded linear operators, compact operators, elementary properties of the spectrum of bounded operators, spectral theorem for compact self-adjoint operators on a Hilbert space. If you have followed Applied Functional Analysis, then you have the right prerequisites. A student will benefit from a course in topology, but we do not expect this as a prerequisite. We shall use the notions: compactness, Hausdorff and topological base. We will briefly discuss this and provide material for the basics that we shall need as self-study.	
Course Contents	We will certainly treat: Single operator theory and functional calculi -Holomorphic functional calculus (also known as Dunford calculus). -Gelfand-Naimark theorem for commutative unital C*-algebras. -Spectral theorem for bounded operators. -Unbounded operators and notions as: symmetric, self-adjoint, closed, domain, dense domain, deficiency indices, self-adjoint extensions. -Spectral theorem for unbounded self-adjoint operators. -Stone's theorem on 1-parameter groups of unitary transformations. -Fredholm operators and Fredholm index. Operator algebras -C*-algebras -Positive operators in a C*-algebra -States on a C*-algebra, and the GNS-construction -Elements of von Neumann algebra theory: double commutant theorem and classification of abelian von Neumann algebras	
Study Goals	-The student is able to explain the main notions that occur in all the topics of the course content. This concerns: holomorphic functional calculus, continuous functional calculus, the Gelfand-Naimark theorem, GNS-representations, spectral measures, the spectral theorem for bounded and unbounded operators, unbounded operators and when they are densely defined/symmetric/self-adjoint, Stone's theorem, Fredholm operators and index. -From the most important theorems the student is able to analyse the proofs. This concerns: the Gelfand-Naimark theorem and continuous functional calculus, the spectral theorem both for bounded and unbounded operators, the correspondence between deficiency indices and symmetric extensions of operators, Stone's theorem, the GNS-construction, the double commutant theorem and the characterization of abelian von Neumann algebras. -The student is able to apply this knowledge to solve problems in spectral theory, C*-algebras, von Neumann algebras and examples from physics such as the harmonic oscillator.	
Education Method	Classroom lectures + homework exercises.	
Literature and Study Materials	We will extensively make use of the book: John B. Conway - A course in functional analysis (Springer). You should find yourself a copy of this book to follow the course.	
Assessment	The final grade of the course consists of the following components: - take-home mid-term exam - Oral exam Final grade calculation: $\text{Max}(E, 0.15 \cdot M + 0.85 \cdot E)$. If the estimated number of students that takes the exam is above 15 then the exam may be written. This will be decided and announced during the first quarter of the course. In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5 Oral exam: oral resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Elective	Yes	
Tags	Analysis	

WI4204	Advanced Modeling	6
Responsible Instructor	Dr. J.L.A. Dubbeldam	
Contact Hours / Week x/x/x/x	2e semester intensieve cursus	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The participants will benefit from a solid knowledge of advanced calculus, basic knowledge of ordinary and partial differential equations and programming experience with Matlab (or an equivalent language).	
Course Contents	This modelling course is introducing the principles and practice of mathematical modelling. An important part of the course is to recognize the essential mechanisms governing a phenomenon. These mechanisms have to be translated into mathematical equations and formulas, and included into a mathematical model. This activity requires both a good understanding of the system under consideration and good mathematical skills. Mathematical modelling is essential for understanding processes in real life, complementing expensive experiments that sometimes cannot even be carried out at all. This course will concentrate on applications in physics and life sciences and focus on models in terms of partial differential equations. The course focuses on working in a small group on a specific project. The topics for project work may differ from year to year. For the academic year 2020-2021 some emphasis will be on continuum mechanics	
Study Goals	General 1. Has knowledge of mathematical modelling tools that are used for model applications in physics and for mathematical models formulated in terms of partial differential equations. 2. The student is able to recognize the principles and practice of mathematical modelling. 3. The student is able to identify the essential mechanisms governing a physical phenomenon and is able to translate these into mathematics and can include them into a mathematical model. Specific 1. Is able to make a partial differential equation non-dimensional and can apply scaling techniques to a partial differential equation. 2. Understands conservation principles and constitutive relations. 3. Is able to linearise partial differential equations in order to perform a stability analysis 4. Is able to develop a mathematical model for a practical problem and has the knowledge and maturity to analyse this model with the available mathematical tools. 5. Has the knowledge and skills for the use and assessment of mathematical models and is able to adapt the models for a specific application. 6. Is able to pose adequate questions and has a critical yet constructive attitude towards analysing and solving complex mathematical modelling problems 7. Is able to communicate verbally and in writing about mathematical research with specialists, non-specialists and other parties involved. 8. Has sufficient communicative skills to collaborate with others and discuss mathematical modelling problems and results.	
Education Method	The first lecture summarizes the mathematical modelling tools that might be useful for the projects. Furthermore during this lecture the projects are introduced by the supervisors and the groups of 2-4 students are formed. The main part of the course is offered as a one week "intensive course", fully devoted to the project. In most cases a computer program for simulating a phenomenon from physics of the life sciences has to be developed. At the end of the fourth quarter the written report of the project has to be handed in and the final presentations are scheduled.	
Literature and Study Materials	Many of the students will have a good background in continuum mechanics. The book Continuum Modeling in the Physical Sciences, E. van Groesen and J. Molenaar, Mathematical Modeling and Computation, Vol. 13, SIAM, 2007 (ISBN 978-0-898716-25-2) can be used as background material. Copies of this e-book can be borrowed from the TU-Delft library.	
Assessment	The final grade of the course consists of the following components: - individual homework (20%) - project (80%) Final grade calculation: $(0.2 * \text{homework}) + (0.8 * \text{project})$ Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - homework: resubmit deliverable - project report: resubmit deliverable - project presentation: resit presentation At the first lecture home assignments are given. The final deadline for submitting the home work is the start of the intensive week. The grade of the project is based on the written report of the group work and on the oral presentation about the project. The following aspects of the project work will have equal weights: -The development of the model -Mathematical depth	

	-Report and presentation
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Pop-up tekst bij intekenen	Welcome to Advanced Modelling!

WI4205	Applied Finite Elements	6
Responsible Instructor	D. Toshniwal	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The students should have had at least introductory courses on the following topics: * continuum mechanics * partial differential equations * tensor Calculus * linear algebra * numerical methods for ODEs	
Course Contents	Weak formulations (derivation, motivation for use, well-posedness) Abstract introduction to the finite element method (FEM) Theoretical analysis of FEM (e.g., of well-posedness, convergence) Practical implementation of FEM (e.g., for linear/non-linear steady/unsteady problems) in one and higher dimensions Verification and validation of FEM	
Study Goals	After this course, students should be able to: * derive a weak formulation for the give PDE * show the well-posedness of the derived weak formulations * construct isoparametric finite elements in one and higher dimensions * discretize the weak problem using isoparametric finite elements * show the well-posedness of the discretized problem * implement the finite element method * perform basic verification of the implementation correctness	
Education Method	Lectures.	
Literature and Study Materials	The following texts will serve as supplementary references to the class notes: * Course reader (downloadable from Brightspace) * T.J.R. Hughes. The Finite Element Method: Linear Static and Dynamic Finite Element Analysis, 1987. * P. Wriggers. Nonlinear Finite Element Methods, 2008. Other references will be provided during the course.	
Assessment	Course will be assessed based on assignments (40% of final grade) and a written exam (60% of final grade). Note: Some of the assignments may require the students to study material not covered in class. Similarly, some assignments will ask the students to work on small extensions or generalisations of the concepts covered in class. To pass the course, the following is mandatory: * Assignments: - The assignments (written and programming exercises) must be completed with an average grade greater than 5.5. - The assignments will have deadlines and late submissions will be penalised. * Exam: - The written exam must be completed with a grade greater than 5.5. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Group assignments: resubmit deliverable - written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	For more information see the Brightspace course page.	

WI4211	Advanced Topics in Analysis	6
Responsible Instructor	Prof.dr. J.M.A.M. van Neerven	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Measure theory, functional analysis, Fourier analysis (Fourier series, Fourier transform, Plancherel's theorem). (last modified: 20/2/2023)	
Course Contents	The scope of the course is to introduce the students to the applications of group representation theory in signal analysis. The course will cover the following subjects: - Locally compact groups: Haar measures, modular functions. - Representation theory of locally compact groups: irreducible representations, Schurs lemma, voice transforms, square integrable representations, Duflo-Moore operator, reproducing formulae. - Voice transforms in signal analysis: wavelet transform, short time Fourier transform, shearlet transform. (last modified: 20/2/2023)	
Study Goals	After completing this course, the student is expected to: - Reproduce the definitions and the statements of the theorems covered in the course. - Reproduce the proofs of the results covered in the course in a clear and rigorous way. - Apply theoretical results and proof techniques in solving new exercises. - Name and describe some applications of group representation theory in signal analysis. (last modified: 20/2/2023)	
Education Method	Lectures and assigned exercises. (last modified: 20/2/2023)	
Literature and Study Materials	The course is based on my lecture notes Representation Theory in Signal Analysis (which will be made available via Brightspace). In the background, the following books are relevant: - G. B. Folland. A course in abstract harmonic analysis. Textbooks in Mathematics. CRC Press, second edition, 2016. - S. Dahlke, F. De Mari, P. Grohs, and D. Labate. From group representations to signal analysis. In: Harmonic and Applied Analysis, Applied Numerical Harmonic Analysis, vol 68. Birkhäuser, 2015. - H. Führ. Abstract harmonic analysis of continuous wavelet transforms. Springer, 2005. - K. Gröchenig. Foundations of time-frequency analysis. Springer Science & Business Media, 2001. - V. S. Varadarajan. Geometry of quantum theory. Springer-Verlag, 2nd edition, 1985. - G. Kutyniok and D. Labate. Shearlets. Applied Numerical Harmonic Analysis, Birkhäuser/Springer, 2012. (last modified: 20/2/2023)	
Assessment	The final grade of the course consists of the following components: - Oral exam (100%) Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Oral exam: oral resit	
Special Information	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). The instructor for this course will be Francesca Bartolucci. (last modified: 20/2/2023)	
Co-Instructor	S.E. Zegers	

WI4480	Internet Seminar on Evolution Equations	9
Responsible Instructor	Dr.ir. M.C. Veraar	
Contact Hours / Week x/x/x/x	n/a	
Education Period	None (Self Study)	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The internet seminar is about an advanced topic within functional analysis which is usually related to evolution equations	
Study Goals	depend on year. In principle you will learn the approach to PDEs based on semigroup theory.	
Education Method	1st phase. weekly lecture notes for self-study. You have to hand in difficult assignments on weekly basis. There are weekly meetings where the difficulties are discussed. The 2nd phase is to study advanced literature on a sub-topic in a small international group. The third phase is a workshop where the theory of the 2nd phase is presented	
Assessment	based on assignment and performance during 2nd and 3rd phase Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. Ir. E. Lorist	

WI4485	Harmonic Analysis	6
Responsible Instructor	Dr.ir. M.C. Veraar	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Real Analysis, Functional Analysis, basic measure theory. Fourier Analysis can be helpful in some applications, but is not a prerequisite.	
Course Contents	This course gives an introduction into real harmonic analysis. Harmonic analysis has its origin in Fourier's work on the heat equation. Nowadays, it provides important tools in tackling a large variety of problems coming from elliptic and parabolic partial differential equation, in particular in boundary value problems. The theory has close links with operator theory and spectral theory. In this course, we will develop important concepts of real harmonic analysis, including: - covering lemmas and maximal functions - basic interpolation theory - Hilbert transform, Riesz transform and Cauchy transform - Calderón-Zygmund operators (singular integral operators) - Mikhlin multiplier theorem - Littlewood-Paley theory	
Study Goals	Global objectives: After this course students are able to - state definition and properties of fundamental objects of real harmonic analysis; - apply key techniques of harmonic analysis in a different context; - connect and apply the theory to the theory of partial differential equations. Objectives per topic: After this course students are able to - apply covering lemmas in various settings; - apply maximal operators in boundedness arguments; - apply real and complex interpolation; - verify if a given operator is a singular integral operator; - state the most important examples of singular integral operators; - show the L^p boundedness of singular integral operators; - give a proof of Mikhlin's multiplier theorem; - apply Mikhlin's multiplier theorem to elliptic PDEs; - explain the basics of Littlewood-Paley theory.	
Education Method	Lectures and exercises/assignments.	
Literature and Study Materials	lecture notes online available on brightspace	
Books	For further reading: P. Auscher, Real Harmonic Analysis, Lecture Notes, ANU. Online available at http://eview.anu.edu.au/harmonic/index.php. L. Grafakos, Classical and modern Fourier analysis. Pearson Education, Inc., Upper Saddle River, NJ, 2004. E. M. Stein, Singular integrals and differentiability properties of functions, Princeton University Press 1970. E. M. Stein, Harmonic Analysis: Real-variables methods, orthogonality and oscillatory integrals, Princeton University Press, 1993.	
Assessment	The final grade of the course consists of the following components: - Oral exam - homework assignments. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Oral exam: oral resit - Homework assignments: resubmit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. Ir. E. Lorist	

WI4615	Stochastic Calculus	6
Responsible Instructor	Dr. K. Kirchner	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Basics of measure theory and analysis (as given in the course ``AM2090 Real Analysis") Martingales, Brownian motion (as taught in the master course ``W14430 Martingales, Brownian Motion").	
Course Contents	In this course the following subjects will be studied: 1. Ito integration 2. Ito calculus including Ito's formula 3. Stochastic differential equations 4. Feynman Kac formula 5. Introduction to numerical methods for stochastic differential equations	
Study Goals	After the course, you should be able to LO1. Explain the rigorous construction of the stochastic Ito integral. LO2. Demonstrate the derivation of Ito's formula. LO3. Calculate explicit stochastic Ito integrals by a) using the construction of the Ito integral, b) applying Ito's formula. LO4. Prove the existenceuniqueness theorem for stochastic ordinary differential equations (SODEs). LO5. Assess solvability of concrete SODEs by means of a) verifying all conditions of the existenceuniqueness theorem, b) explicitly calculating solutions. LO6. Establish links between Markov processes and partial differential equations.	
Education Method	Lectures (2 hours per week).	
Literature and Study Materials	The course will be based on the following sources: 1. P.J.C. Sprey: stochastic integration lecture notes available at https://staff.fnwi.uva.nl/p.j.c.spreij/onderwijs/master/si.pdf 2. Le Gall, Martingales, Brownian motion and stochastic calculus, Springer 2016. 3. Schilling, Partzch, Brownian Motion, An Introduction to Stochastic Processes, De Gruyter, 2010. 4. Karatzas, Shreve, Brownian Motion and Stochastic Calculus, Springer 1991. 5. Øksendal, Stochastic Differential Equations, Springer 2003. 6. Revuz, Yor, Continuous Martingales and Brownian Motion, Springer 1999.	
Assessment	These books can be freely downloaded via a library link The final grade of the course consists of the following components: - Depending on the number of students, there will be an oral or a written exam at the end of Q4 which counts for 100%. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Written exam: written resit OR - Oral exam: oral resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	The exam is closed book.	

WI4680	Applications in Partial Differential Equations	6
Responsible Instructor	Dr. H.M. Schuttelaars	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Ordinary differential equations, Numerical methods, Programming (Python or other)	
Course Contents	Many ecosystems and physical systems, such as the climate, can display strongly nonlinear behaviour, where the properties of the system suddenly change with changes in environmental conditions. A well-known and urgent example of this is the discussion about tipping of the climate, thought to set off a dramatic change in the Worlds climate if global temperatures start to exceed some threshold. Mathematically, tipping is an example of a so-called bifurcation of the underlying system of equations. We will focus on the bifurcations in systems of ordinary differential equations (ODEs) and partial differential equations (PDEs) representing real-world systems, which can often only be solved numerically. Hence, this course focusses on numerical bifurcation theory: the study of understanding parameter sensitivity and bifurcations in numerical solutions. The course covers mathematical theory and papers applying the theory to real-life applications including climate science and ecosystems. We build on theory of ODEs, numerical methods and mathematical modelling. Basic programming experience is required for the practical assignments.	
Study Goals	After following this course the student is able to 1. Describe various non-linear phenomena appearing in systems of ordinary and partial differential equations. These include tipping, appearance of periodic solutions, pattern formation, flickering and critical slowing down. 2. Apply numerical continuation, stability analysis and branch switching around bifurcations for common elementary bifurcations (transcritical, pitchfork, limit point, Hopf) in the context of a simple model with physical applications. 3. Explain practical problems with respect to computation time, computer memory and numerical accuracy in the above techniques and various solutions to these problems	
Education Method	Lectures and practical assignments The course consists of 14 lectures during which theory and papers will be discussed. The discussion of the papers is led by the students and hence active participation in the lectures is required. In the practical assignments, the students gain hands-on experience working with the learned techniques and revisit theory on ODEs and numerical methods. The assignments feature a significant amount of programming.	
Literature and Study Materials	Seydel, R. Practical bifurcation and stability analysis. Vol. 5. Springer Science & Business Media, 2009. Various papers available on Brightspace.	
Assessment	The final grade of the course consists of the following components: - Take home assignments - oral exam Final grade calculation: $(0.5 * \text{take home assignments}) + (0.5 * \text{oral exam})$ Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Take home assignments: resubmit deliverable - Oral exam: oral resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Specialisation Mathematics of Quantum Technology and Computation AM	
Introduction 1	Specialisation Mathematics of Quantum Technology and Computation Compulsory orientation course: WI4645 Introduction to Quantum Information and Computing Compulsory specialisation courses (choose 2 out of 4 courses): WI4203 Applied Functional Analysis MM Semidefinite Optimization WI4006 Special Functions and Representation Theory WI4650 Applied Quantum Algorithms All of the compulsory specialisation courses can also be chosen as as recommended specialisation course. Recommended specialisation courses: MM Quantum Computing MM Quantum Information Theory WI4211 Advanced Topics in Analysis WI4046 Spectral Theory of Linear Operators MM Operator Algebras MM Lie Groups WI4635 Linear Algebra and Optimization for Machine Learning WI4207 Continuous Optimization AP3452 Quantum Error Correction CS4090 Quantum Communication and Cryptography QIST4300 Qubit Dynamics and Quantum Control QIST4320 Quantum Computation and Quantum Programming

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Compulsory Courses (select 2 out of 4)

WI4006	Special Functions and Representation Theory	6
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Basic knowledge of Group Theory and Complex Analysis.	
Course Contents	In this course we study special functions in connection with representation theory of specific Lie groups and Lie algebras. Topics include: - Gamma and Beta integrals - hypergeometric functions - representation theory of compact matrix Lie groups and their Lie algebras - representation theory of $SU(2)$ and $SO(3)$ - Jacobi polynomials and matrix elements of $SU(2)$ - tensor products and Clebsch-Gordan coefficients	
Study Goals	The student knows the material treated during the lectures and is able to apply this to several problems. The student should be able to explain the treated material.	
Education Method	Lectures	
Literature and Study Materials	Lecture notes will be available on Brightspace.	
Assessment	The final grade of the course consists of the following components: - Oral exam (70%) - mandatory homework assignments (30%). Final grade calculation: $(0.7 * \text{oral exam}) + (0.3 * \text{homework assignments})$ In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5, for: Oral exam: oral resit Homework assignments: resubmit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI4203	Applied Functional Analysis	6
Responsible Instructor	Prof.dr. J.M.A.M. van Neerven	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	The student is expected to have firm knowledge in real and complex analysis and linear algebra. Students can use the checklist below to see if he/she is well-prepared: From set theory: standard set-theoretic notions and notations, mappings and relations, injective and surjective mappings, equivalence relations and partial orderings, supremum and infimum, countable and uncountable sets. From analysis: convergence and limits, Cauchy sequences, theorem of Bolzano-Weierstrass, completeness of the fields of real and complex numbers, continuity, differentiability, convergence and absolute convergence of series, power series. From Topology: metric spaces, open and closed sets, denseness, completeness, compactness. From Integration Theory: theory of the Lebesgue integral, monotone and dominated convergence theorem, Fubini's theorem. From (linear) algebra: vector spaces over the real and complex numbers, linear mappings, subspaces, direct sum of subspaces, kernel and range of a linear mapping, linear independence, linear span, bases, dimension, polynomials, continuous functions on an interval as a vector space, inner product spaces, Cauchy-Schwarz inequality. From Complex Variables: holomorphic functions, Cauchy's integral formula and integral theorem, power series representation of holomorphic functions, analytic continuation.	
Course Contents	The course introduces those notions and results of functional analysis that are relevant for applications to, e.g., partial differential equations, numerical analysis, signal processing, probability theory, stochastic analysis, and quantum theory. More precisely, the course covers the following subjects: - Banach spaces: norms, duality, Hahn-Banach theorems. - Hilbert spaces: best approximation, orthogonality, orthocomplementation, orthonormal bases, Riesz representation theorem. - $C(K)$-spaces and L^p-spaces: completeness, duality, compactness, approximation techniques. - Bounded linear operators: uniform boundedness theorem, open mapping theorem, closed graph theorem, compact operators, adjoint operators, spectrum and resolvent, spectral projections, Riesz-Schauder theory, Fredholm operators. - Hilbert space operators: orthogonal projections, unitary operators, self-adjoint operators, normal operators, spectral theorem for compact normal operators. - weak derivatives, Sobolev spaces, applications to boundary value problems - densely defined operators, closed operators - semigroups of operators, generator, resolvent, Hille-Yosida theorem.	
Study Goals	After completing this course, the student is able to: 1) reproduce the definitions and statements of the main theorems of functional analysis as covered in the course 2) give correct and rigorous proofs of those results whose proofs do not go beyond standard reasoning from first principles 3) apply standard functional analytic proof techniques in new situations 4) name and describe some applications of functional analysis in other areas of mathematics, such as partial differential equations and quantum theory 5) communicate and present mathematical proofs clearly and understandably in a brightspace presentation	
Education Method	Lectures and Exercises/Assignments. See Brightspace for details.	
Literature and Study Materials	The course is based on my textbook "Functional Analysis", Cambridge University Press, 2022. A corrected printing will appear in the Summer of 2023. The e-book version of it is downloadable free of charge via the TUD library and the CUP website for IP addresses at the TUD. In preprint form, the book is available via my personal website.	
Assessment	Written exam, in combination with graded exercises. Details will be announced via Brightspace. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	None	

WI4650	Applied Quantum Algorithms	6
Responsible Instructor	Dr. M. Möller	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Good knowledge in linear algebra, numerical methods for differential equations and programming skills (Python, C/C++) are a must. Moreover, prior knowledge from the course "WI4645 Quantum Information and Computing" is highly recommended.	
Course Contents	This course introduces you to the fascinating field of quantum algorithms and their applications. We will discuss state-of-the-art quantum algorithms for solving linear systems of equations, linear and nonlinear differential equations, and continuous as well as combinatorial optimisation problems. The focus of this course is on quantum algorithms that are of practical interest which means that we will not only study the general working principles of these algorithms and their theoretical properties but also discuss the necessary pre- and post-processing steps (e.g. the encoding of system matrices into quantum-amenable format and the interpretation of probabilistic results) to integrate quantum algorithms into start-to-end quantum-assisted applications. Today's quantum computers are Noisy and Intermediate Scale Quantum (NISQ) devices that are very limited in their practical utility. This course will therefore focus on so-called variational hybrid classical-quantum algorithms (e.g. variational quantum linear solver (VQLS), quantum approximate optimization algorithm (QAOA), and quantum alternating operator ansatz (QAOA)) which combine quantum and classical computing and can be executed on real quantum hardware at least for small problem sizes. If time permits we will also be looking into the fascinating field of quantum machine learning (QML). The course will consist of two parts: - The first part will give an introduction into the subject through lectures and hands-on assignments in Q3. - In the second part (Q4) students will work in small teams on individual projects and apply the theory and methods taught in the first part of the course. As applied quantum computing is an emerging field, the content of the course might eventually change if novel promising approaches appear in recent literature or existing ones are demonstrated to be impractical.	
Study Goals	At the end of the course the student will 1. have a good overview of the current state-of-the-art in quantum algorithms and their applications; 2. be able to understand the recent literature on quantum algorithms; 3. be able to analyse the theoretical properties of quantum algorithms; 4. be able to implement quantum algorithms for practical applications; 5. be able to run quantum-assisted applications on quantum simulators and real NISQ devices; 6. be able to perform independent research on this topics, e.g. in a master project.	
Education Method	Lectures and hands-on assignments (theoretical analysis + code development) during Q3. Student team project in Q4 on topics related to the course's content and based on recent literature.	
Computer Use	A cloud-based programming environment for developing and testing quantum algorithms on multiple quantum simulator devices and real quantum hardware (if available) will be provided. It should be noted that free access to quantum hardware devices changes rapidly. We try hard to provide access to at least one (possibly more) real quantum hardware during the course.	
Literature and Study Materials	Lecture notes, relevant research papers and a list of recommended background material will be made available via Brightspace during the course.	
Assessment	The final grade of the course consists of the following components: - Project presentation + plenary discussion of projects results (100%) Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Project presentation + discussion: new presentation + discussion Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Recommended courses

AP3452	Quantum Error Correction	4
Responsible Instructor	Prof.dr. B.M. Terhal	
Contact Hours / Week x/x/x/x	0/0/0/3	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Fundamentals of Quantum Information (QIST4310) or a similar course which covers the basic model of quantum computing (qubits, quantum circuits). Optional: Quantum Hardware I (AP3442) or a course which goes into the physical description of quantum computing.	
Course Contents	We discuss the ideas behind quantum error correction and fault-tolerance such as the threshold theorem, code concatenation, stabiliser and topological codes, code constructions (e.g. color codes, hypergraph product construction), fault-tolerant gate constructions (code deformation), decoding and bosonic codes. There will some hands-on Python exercises of decoding.	
Study Goals	Obtain deep and clear understanding of the notion of quantum error correction and the tools and concepts that have been developed. Understand how quantum error correction applies to real noise of various sources. Understand the idea behind fault-tolerance.	
Education Method	Weekly Lecture + Weekly Exercise/Study Session. Exercises will be provided and discussed weekly (no grading).	
Literature and Study Materials	Lecture notes and slides.	
Assessment	Final oral exam (list of topics will be provided) (80%) + Hand-in of numerical work (20%).	

CS4090	Quantum Communication and Cryptography	5
Responsible Instructor	Dr. S.D.C. Wehner	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Linear Algebra, Probability & Statistics, Q101 (AP3421 Fundamentals of quantum information)	
Course Contents	This class will introduce you to the fascinating field of quantum communication! We will look at the state of the art of quantum networks, and explore techniques for building quantum repeaters that promise to deliver qubits over long distances. We also briefly look at one of the most famous application of quantum cryptography, quantum key distribution. Caution: 1. This class requires you to take "Fundamentals of Quantum Information" in Quarter 1 2. The focus of this class is presently on quantum communication, and we will only briefly look at quantum cryptography. As such, this class is not held in flipped classroom style in conjunction with edX QuCryptoX as in previous years.	
Study Goals	The student will acquire: - A good understanding of the fundamental concepts of quantum communication - Insight into the differences between classical and quantum communication and cryptography - Skill set required to follow the remainder of the quantum curriculum (Q301 Quantum hardware and Q401 Quantum electronics)	
Education Method	Lectures and tutorials. If remote classes continue this fall, then recorded lectures and live discussion session	
Literature and Study Materials	Primary: - Slides - Review Articles Auxilliary: - Nielsen and Chuang Quantum computation and information, Cambridge University Press.	
Assessment	The final grade of the course consists of the following components: - Homework (weighting 70%) - Final Project (weighting 30%) Final grade calculation: $0.7 * \text{Homework} + 0.3 * \text{Final Project}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance to TER Implementation Regulations Art 5, sub 5., for: - Homework: re-submit deliverable - Final Project: re-submit deliverable	
Tags	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). - Abstract - Adventurous - Algorithmics - Challenging - Group Dynamics/Project Organisation - Information & Communication - Integrated - Intensive - Involved - Lineair Algebra - Mathematics - Physics - Quantum - Signals - Technology - Telecommunication	

QIST4300	Qubit Dynamics and Quantum Control	4
Responsible Instructor	V.V. Dobrovitski	
Responsible Instructor	Dr.ir. M. Veldhorst	
Contact Hours / Week x/x/x/x	0/8/0/0/	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The students will learn fundamental methods and approaches to manipulating and controlling qubits (in the absence and in the presence of noise), which are used in virtually all existing architectures for quantum information processing. The students will study	
	* fundamental notions used for description and analysis of the qubit dynamics	
	* typical approaches to manipulation and control of most qubits, from superconducting qubits to trapped ions and atoms;	
	* basic concepts, such as rotating-frame description of qubits, Rabi driving, decoherence by noise, and decoherence suppression via dynamical decoupling.	
	* basic regimes of qubit dynamics and their theoretical description, basic approaches to description and analysis of the qubit decoherence by noise and by environment, as well as basic approaches to qubit manipulation and decoherence suppression.	
Study Goals	- Recognize, explain, and illustrate the basic regimes of qubit dynamics and the basic sources of noise and decoherence relevant for different qubits;	
	- Identify typical scenarios of free and driven qubit dynamics, assess the relevant sources of decoherence and noise, and determine relevant models for description and analysis of dynamics of single- and multi-qubit systems;	
	- Apply these relevant models and evaluate the possible influence of noise and decoherence on the qubit dynamics in different typical regimes;	
	- Choose appropriate models for describing different types of qubits, determine corresponding noise and decoherence models, and calculate the parameters of the qubit dynamics in the presence of noise and coupling to the environment;	
	- Explain the basics of the dynamical decoupling protocols and the physical processes underlying this method of decoherence suppression;	
	- Apply the knowledge about qubit control and decoherence suppression to diverse qubits, such as impurity centers, quantum dots, (partially) superconducting qubits, and other similar systems;	
	- Choose and identify the dynamical decoupling protocols, and evaluate efficiency of different dynamical decoupling protocols under different conditions.	
Education Method	Lectures followed by seminars/tutorials, including supervised problem solving, and independent reading and problem solving.	
	Final project to be performed during the course, and presented at the end of the course	
Assessment	Assessment will consist of the four tests given during the course, and the final project to be performed during the course and presented at the end of the course. Besides, optional (but graded) homework assignments will be given after each lecture.	
	The final grade will be determined as $0.75 \cdot [\text{average grade for the tests}] + 0.25 \cdot [\text{final project grade}]$. The average grade for the optional homework assignments will be taken into account only if it improves the final grade; in that case, the final grade will be determined as $0.15 \cdot [\text{average homework grade}] + 0.6 \cdot [\text{average test grade}] + 0.25 \cdot [\text{final project grade}]$.	
	If the student wants to improve the final grade then they can take a final exam over the content of the whole course, and the final grade will be a maximum of the two grades, the average grade over the tests/homework/final project, and the grade for the final exam.	

QIST4320	Quantum Computation and Quantum Programming	4
Responsible Instructor	Dr. M. Möller	
Contact Hours / Week x/x/x/x	0/4/0/0/	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Quantum computing has been a theoretical concept for decades that has brought a zoo of quantum algorithms in theory. However, with the advent of real-world quantum computers, it is possible to explore quantum computing in practice. This course will teach you - the theoretical background of quantum computation, - to implement certain quantum algorithms with state-of-the-art tools, - to execute them on simulators and real-world quantum computers and, - to interpret the outcome. During this course you will get hands-on experience with different quantum algorithms and the challenges involved in their application to real-world problems.	
Study Goals	At the end of the course, the student can: 1. name and explain basic quantum computing concepts and can differentiate them from their classical counterparts; 2. name and explain the different complexity classes, illustrate them by giving concrete examples, and compare the performance of classical and quantum algorithms based on their complexity; 3. outline properties of main quantum algorithms by clearly defining their input, output and computational complexity in contrast to corresponding classical complexity and by exemplifying some of their main applications 4. implement basic quantum and hybrid quantum-classical algorithms in at least one of the mainstream quantum programming languages and assess/interpret the results (with and w/o noise) 5. describe the limitations of real-world quantum computing	
Education Method	This course will include the following educational activities/methods: Read preparatory notes & watch video lecture: Students will engage in reading assigned notes and watching video lectures related to the course topics. This will provide a foundation of knowledge before the interactive sessions. Interactive session with lecturer: In these sessions, students will actively engage with the lecturer to discuss and further explore the topics that are introduced in the video lectures . Quiz with lecturer: Students will have quizzes with the lecturer to assess their understanding of the course material and reinforce key concepts. These quizzes provide opportunities to identify areas that require further clarification. WorkGroup Session: Students will participate in guided workgroup sessions, where they can work on their projects under supervision. During this course, students will gain knowledge and experience in the following academic skill(s): - Active Learning and Self-Study - Critical Thinking and Analysis - Communication and Collaboration	
Assessment	The assessment consists of two take-home exams to be made and handed in individually per student and two complementary exams in multiple-choice format in the weeks that the take-home exams are due and on the same topics. No exam registration is needed. Additionally, there is a presentation on the group project, which includes a question-and-answer session. The final grade is calculated as follows: 20% take-home exam 1 + 20% take-home exam 2 + 10% multiple-choice exam 1 + 10% multiple-choice exam 2 + 40% group project An oral exam is offered as "generic resit". Note that this option is not meant to improve the grade but to compensate for one or more failed assessment components.	

WI4046	Spectral Theory of Linear Operators	6
Responsible Instructor	Dr. M.P.T. Caspers	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Elements of functional analysis: Banach and Hilbert spaces, bounded linear operators, compact operators, elementary properties of the spectrum of bounded operators, spectral theorem for compact self-adjoint operators on a Hilbert space. If you have followed Applied Functional Analysis, then you have the right prerequisites. A student will benefit from a course in topology, but we do not expect this as a prerequisite. We shall use the notions: compactness, Hausdorff and topological base. We will briefly discuss this and provide material for the basics that we shall need as self-study.	
Course Contents	We will certainly treat: Single operator theory and functional calculi -Holomorphic functional calculus (also known as Dunford calculus). -Gelfand-Naimark theorem for commutative unital C*-algebras. -Spectral theorem for bounded operators. -Unbounded operators and notions as: symmetric, self-adjoint, closed, domain, dense domain, deficiency indices, self-adjoint extensions. -Spectral theorem for unbounded self-adjoint operators. -Stone's theorem on 1-parameter groups of unitary transformations. -Fredholm operators and Fredholm index. Operator algebras -C*-algebras -Positive operators in a C*-algebra -States on a C*-algebra, and the GNS-construction -Elements of von Neumann algebra theory: double commutant theorem and classification of abelian von Neumann algebras	
Study Goals	-The student is able to explain the main notions that occur in all the topics of the course content. This concerns: holomorphic functional calculus, continuous functional calculus, the Gelfand-Naimark theorem, GNS-representations, spectral measures, the spectral theorem for bounded and unbounded operators, unbounded operators and when they are densely defined/symmetric/self-adjoint, Stone's theorem, Fredholm operators and index. -From the most important theorems the student is able to analyse the proofs. This concerns: the Gelfand-Naimark theorem and continuous functional calculus, the spectral theorem both for bounded and unbounded operators, the correspondence between deficiency indices and symmetric extensions of operators, Stone's theorem, the GNS-construction, the double commutant theorem and the characterization of abelian von Neumann algebras. -The student is able to apply this knowledge to solve problems in spectral theory, C*-algebras, von Neumann algebras and examples from physics such as the harmonic oscillator.	
Education Method	Classroom lectures + homework exercises.	
Literature and Study Materials	We will extensively make use of the book: John B. Conway - A course in functional analysis (Springer). You should find yourself a copy of this book to follow the course.	
Assessment	The final grade of the course consists of the following components: - take-home mid-term exam - Oral exam Final grade calculation: $\text{Max}(E, 0.15 \cdot M + 0.85 \cdot E)$. If the estimated number of students that takes the exam is above 15 then the exam may be written. This will be decided and announced during the first quarter of the course. In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5 Oral exam: oral resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Elective	Yes	
Tags	Analysis	

WI4207	Continuous Optimization	6
Responsible Instructor	D. de Laat	
Contact Hours / Week x/x/x/x	1e semester Utrecht maandag	
Education Period	1 2	
Start Education	1	
Exam Period	2	
Course Language	English	
Expected prior knowledge	The student should have a solid knowledge of linear algebra and multivariable calculus. The student should also have knowledge of linear programming (including linear programming duality) and convex analysis to the level of being able to follow the text and do the exercises from: Chapters 1 and 2 (including the exercises) from the book 'Linear Programming, A Concise Introduction, Thomas S. Ferguson' https://www.math.ucla.edu/~tom/LP.pdf Exercises 2.1, 2.2, 2.12, 3.1, 3.3, 3.5, and 3.7 from the book 'Convex Optimization, Stephen Boyd and Lieven Vandenbergh' http://stanford.edu/~boyd/cvxbook	
Course Contents	Continuous optimization is the branch of optimization where we optimize a (differentiable) function over continuous (as opposed to discrete) variables. Here the variables can be subject to equality and inequality constraints. Optimization problems of this form are common in science and engineering, machine learning, and as relaxations of discrete optimization problems. One can use linear algebra, multivariable calculus, and convex geometry to study the properties of continuous optimization problems and to design and analyze efficient algorithms. In this course, we study theory, algorithms, and some applications of continuous optimization. In the theory part we discuss convexity, Lagrangian duality, optimality conditions, and conic programming. In the algorithmic part we discuss derivative-free methods, first-order optimization methods, neural networks/supervised learning, second-order optimization methods, interior-point methods, and support vector machines. For some of the algorithms, we analyze the convergence.	
Study Goals	See the Mastermath page.	
Education Method	There will be 14 lectures.	
Literature and Study Materials	- Convex Optimization by Stephen Boyd and Lieven Vandenbergh (http://stanford.edu/~boyd/cvxbook) - Additional notes distributed by the lecturer	
Assessment	The course will have a written exam that counts for 100% of the grade. The resit will also be a written exam. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	This is a LNMB Mastermath course taught by David de Laat.	

WI4211	Advanced Topics in Analysis	6
Responsible Instructor	Prof.dr. J.M.A.M. van Neerven	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Measure theory, functional analysis, Fourier analysis (Fourier series, Fourier transform, Plancherel's theorem). (last modified: 20/2/2023)	
Course Contents	The scope of the course is to introduce the students to the applications of group representation theory in signal analysis. The course will cover the following subjects: - Locally compact groups: Haar measures, modular functions. - Representation theory of locally compact groups: irreducible representations, Schurs lemma, voice transforms, square integrable representations, Duflo-Moore operator, reproducing formulae. - Voice transforms in signal analysis: wavelet transform, short time Fourier transform, shearlet transform. (last modified: 20/2/2023)	
Study Goals	After completing this course, the student is expected to: - Reproduce the definitions and the statements of the theorems covered in the course. - Reproduce the proofs of the results covered in the course in a clear and rigorous way. - Apply theoretical results and proof techniques in solving new exercises. - Name and describe some applications of group representation theory in signal analysis. (last modified: 20/2/2023)	
Education Method	Lectures and assigned exercises. (last modified: 20/2/2023)	
Literature and Study Materials	The course is based on my lecture notes Representation Theory in Signal Analysis (which will be made available via Brightspace). In the background, the following books are relevant: - G. B. Folland. A course in abstract harmonic analysis. Textbooks in Mathematics. CRC Press, second edition, 2016. - S. Dahlke, F. De Mari, P. Grohs, and D. Labate. From group representations to signal analysis. In: Harmonic and Applied Analysis, Applied Numerical Harmonic Analysis, vol 68. Birkhäuser, 2015. - H. Führ. Abstract harmonic analysis of continuous wavelet transforms. Springer, 2005. - K. Gröchenig. Foundations of time-frequency analysis. Springer Science & Business Media, 2001. - V. S. Varadarajan. Geometry of quantum theory. Springer-Verlag, 2nd edition, 1985. - G. Kutyniok and D. Labate. Shearlets. Applied Numerical Harmonic Analysis, Birkhäuser/Springer, 2012. (last modified: 20/2/2023)	
Assessment	The final grade of the course consists of the following components: - Oral exam (100%) Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Oral exam: oral resit	
Special Information	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). The instructor for this course will be Francesca Bartolucci. (last modified: 20/2/2023)	
Co-Instructor	S.E. Zegers	

WI4635	Linear Algebra and Optimization for Machine Learning	6
Responsible Instructor	Dr. A. Heinlein	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	The course material and literature are mostly self-contained, but prior knowledge equivalent to contents of the courses AM2010 Linear Algebra 2 and AM2020 Optimization from the Bachelor of Applied Mathematics programme is strongly recommended. Programming experience in Python or Matlab is expected.	
Course Contents	State-of-the-art machine learning algorithms combine techniques from many different areas of mathematics. In this course, we will discuss algorithmic foundations from the areas of linear algebra and optimization. This also includes aspects of their theory and efficient implementation. Among others, we will study + matrix decomposition and factorization techniques, + regression and classification problems and regularization, + iterative and continuous optimization methods, + hyper parameter optimization, + graph-based algorithms and clustering, + basic introduction to neural networks and algorithmic aspects neural networks (e.g. computational graphs and back-propagation)	
Study Goals	Upon completion of the course, the student should be able to: + Categorize typical data sets based on their structure + Derive suitable mathematical formulations for common supervised and unsupervised machine learning tasks + Choose suitable algorithmic building blocks for solving mathematical formulations of common machine learning tasks + Analyze machine learning algorithms with respect to their properties, such as efficiency and robustness + Implement machine learning algorithms in a computationally efficient way	
Education Method	The course comprises 14 lectures and 14 exercise classes.	
Literature and Study Materials	Lecture notes, slides, and the main literature: + Charu Aggarwal. Linear algebra and optimization for machine learning. Springer, 2020. + Gilbert Strang. Linear algebra and learning from data. Wellesley-Cambridge Press Cambridge, 2019. + Jeremy Watt, Reza Borhani, and Aggelos K Katsaggelos. Machine learning refined: Foundations, algorithms, and applications. Cambridge University Press, 2020.	
Assessment	The final grade of the course consists of the following components: - Two group projects (0.4) - Written exam (0.6) Final grade calculation: $(0.4 * \text{group projects} + 0.6 * \text{written exam})$ Note: For the written exam, a score of at least 5.0 must be obtained. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Group projects: resubmit deliverable - Written exam: There will be a resit for the written exam at a later point in the same academic year Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	D. de Laat	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Non-Mathematics Electives AM 2023

Introduction 1

The non-mathematics elective space may be used for

non-mathematics courses that show the role of mathematics in applications and society;
homologation, i.e. courses at bachelors level required to obtain the necessary prior knowledge for a course at masters level (for students with a bachelor degree different from (applied) mathematics);
courses of the verdiepingsdeel of SEC;
the Interdisciplinary Advanced AI Project (IFEEMCS520200);
the Joint Interdisciplinary Project (TUD4040);
an internship of 15EC (WI5215).

A maximum of 3 credits may be spent on language courses, for example:

WM1135TU Advanced English for the University
TPM301B Spoken English for Academic Purposes - Intermediate 2EC
TPM302B Spoken English for Academic Purposes - Advanced 2EC
TPM303A Intermediate Writing in English for the University 2EC
TPM304A Advanced Writing in English for the University 2EC
TPM305A Writing a Master's Thesis in English 2EC

Please note:

Language courses may only be chosen if required. Placement tests showing the necessity to take one or more of these courses must be taken and submitted to the master coordinator.

IFEEMCS520200	Interdisciplinary Advanced Artificial Intelligence Project	15
Responsible Instructor	Dr. P.K. Murukannaiah	
Responsible Instructor	J. Zatarain Salazar	
Contact Hours / Week x/x/x/x	10/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	This course is aimed at students who have already obtained at least 20EC computer science/artificial intelligence background. Having a solid shared pool of knowledge on AI enables the students embarking on this course to work in a team on an AI challenge at a master level. Prospective students, for example, have completed an AI minor, or have a good amount of AI knowledge from their bachelor (e.g., Mechanical Engineering).	
Course Contents	Artificial Intelligence (AI) is an interdisciplinary challenge. This course aims to provide students with a unique experience to learn, develop and apply AI to a real-world challenge, working in an interdisciplinary project team. During the course, the project team executes AI research, and/or produces AI solutions, that address real-world problems. Students collaborate in a group of 4-6 students, ideally, of diverse backgrounds (including Computer Science). Mini-tutorials will be organized during the course to provide students with knowledge and skill-based input required to successfully complete their project. The topics of the mini-tutorials will be based on the initiative from the project supervisors and other AI researchers, e.g., from the TU Delft AI Labs. The course will include a kick-off, mini-tutorials, and lab sessions with project supervisors.	
Study Goals	Upon finishing this course, the student should be able to: 1. position an AI challenge in the broader literature, e.g., by performing a literature review on the AI challenge in a domain; 2. set-up a research strategy for an interdisciplinary AI project; 3. develop and deploy AI solutions according to a research strategy; 4. effectively collaborate with the team and supervisor; 5. evaluate and reflect on their research / developed AI solution; 6. assess the impact of deploying AI-based solutions and interventions on individuals, organizations, and society; and 7. report and present their research on a scientific level.	
Education Method	The students may work and discuss in groups. Each student is responsible for gathering, sharing and implementing information within the overall project. This allows for collaboration and communication as well as self-directed learning. At least once a week, the students meet with their supervisor. Peer feedback as well as feedback from the supervisor is organized on essential steps in the research project, which include: 1. Project proposal 2. Mid-term review, and 3. A scientific report, describing the problem, methods, results, and conclusion 4. A scientific presentation The mini-tutorials are organized as electives, i.e., students identify which tutorials contribute to their assigned task and field of expertise within the group project.	
Assessment	The scientific report and the project presentation are assessed by the project supervisor and an external examiner. The assessment based on the clarity of presentation, the soundness of the methods used, and the conclusions drawn. Resit next quarter: A repair option is allowed only for students that received a fail mark (<5.8). Overall, the resit mark will be capped to 5.8.	

TPM301B	Spoken English for Academic Purposes Intermediate	2
Module Manager	Drs. M.A. Swennen	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Instructor	S.F. Johnson	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	0/3/0/3	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	This speaking course focuses on the language and skills students need to participate successfully at the university. Students will engage in a number of activities both inside and out of the classroom - to practise and improve their English. The vocabulary focus will be on general academic language, so that after the course students can interact with more confidence in an academic setting.	
Study Goals	The aim of the course is to develop language skills needed for the university environment. You will learn phrases and words for a range of skills needed here, for example participating in discussions; rephrasing and explaining; giving an opinion, (dis)agreeing, making suggestions; describing and presenting a concept.	
Education Method	Group and pair work, combined with assignments to do at home. Feedback on course components is given by the course lecturer and other students	
Literature and Study Materials	To be announced in Brightspace.	
Practical Guide	This is a course for both BSc and MSc students who would like to feel (more) confident when communicating with others at the university.	
Assessment	When you have taken our Placement Test you will be recommended a language level, either Intermediate or Advanced. With this recommendation you can sign up for both of the courses offered at that level. Class participation and weekly assignments. An 80% attendance rate is obligatory, which means you may only miss one session. To keep your place in the group you must attend the first session. The assignments must be submitted each week before the deadline set by the instructor. You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class after session 2.	
Enrolment / Application	Before enrolling in a group you must first take our Placement Test. This is not an actual test but a tool to ensure that you are taking the course that is the most suitable for your level of English. For more information on this 'test' please visit our website: www.itav.tudelft.nl Enrolment for the course (and the Placement Test) is via Brightspace. You do not need to register for this course in Osiris. Please note that you must also choose a group to reserve a place in the class. Type 'TPM301B' in Course search (Catalog). Please make sure you select the correct academic year and quarter. Register in the course AND in a group that suits your schedule.	
Remarks	You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class after session 2. The course will only be offered if there is a sufficient number of participants.	
Elective	Yes	
Tags	English proficiency Group work Small groups	
Category	BSc and MSc students	

TPM302B	Spoken English for Academic Purposes Advanced	2
Module Manager	Drs. M.A. Swennen	
Instructor	Drs. K.I. van der Linden	
Instructor	Drs. D.W. Laponder	
Instructor	S.F. Johnson	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	3/0/3/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	This speaking course focuses on expanding and improving the language students need to participate at the university. A particular focus point of the course is the vocabulary needed at a higher level of English, both in an academic and a more general context. A range of sources for language input is offered, such as podcasts, videos, websites, or (academic) articles. The sessions revolve around student interaction.	
Study Goals	The first aim of the course is to improve and expand the language skills needed at a higher level of English. These include paraphrasing and sharing information with others, using polite language, giving feedback, and presenting information. Another study course goal is to expand and improve the students' general vocabulary.	
Education Method	Group and pair work, combined with (individual) assignments to do at home. Feedback on course components is given by the course lecturer and other students.	
Literature and Study Materials	To be announced in Brightspace.	
Practical Guide	This is a course for both BSc and MSc students who would like to expand and improve their English language skills. When you have taken our Placement Test you will be recommended a language level. With this recommendation you can sign up for either of the courses offered at that level.	
Assessment	For this course you need a recommendation to take courses at the Advanced level. Class participation and weekly assignments. An 80% attendance rate is obligatory, which means you may only miss one session. To keep your place in the group you must attend the first session. The assignments must be submitted each week before the deadline set by the instructor.	
Enrolment / Application	You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class after session 2. Before enrolling in a group you must first take our Placement Test. This is not an actual test but a tool to ensure that you are taking the course that is the most suitable for your level of English. For more information on this 'test' please visit our website: www.itav.tudelft.nl Enrolment for the course (and the Placement Test) is via Brightspace. You do not need to register for this course in Osiris. Please note that you must also choose a group to reserve a place in the class. Type 'TPM302B' in Course search (Catalog). Please make sure you select the correct academic year and quarter. Register in the course AND in a group that suits your schedule.	
Remarks	The course will only be offered if there is a sufficient number of participants. Please note that it is not a presentation course.	
Elective	Yes	
Tags	English proficiency Group work Small groups	
Category	BSc and MSc level	

TPM303A	Intermediate Writing in English for the University	2
Module Manager	Drs. K.I. van der Linden	
Instructor	Drs. K.I. van der Linden	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. M.A. Swennen	
Instructor	S.F. Johnson	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	2/0/2/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	This writing course is a 7 week course with 1 lesson a week (2 hours) plus approx. 6 hours of homework assignments per week. The course focuses on the acquisition of necessary language skills in the areas of grammar and usage, as well as reading and writing at a level appropriate to university study. Students will develop skills that develop an idea with relevant support for that idea by structuring the text according to current writing conventions, using various transitions to link ideas and demonstrating a knowledge of grammar and punctuation. Students will edit their own texts and work with other students in peer-editing groups.	
Study Goals	The aim of the course is to develop language skills and writing skills needed for producing academic texts. This course will give the student an understanding of the conventions of academic writing in English and some of the grammatical structures common in academic texts.	
Education Method	Individual, pair and group work, combined with assignments to do at home. Feedback on course components is given by the course lecturer and other students.	
Practical Guide	This is a course for both BSc and MSc students who would like to expand and improve their English writing skills. When you have taken our Placement Test you will be recommended a language level. With this recommendation you can sign up for any of the courses offered at that level. For this course you need a recommendation to take courses at the Intermediate level.	
Books	Course material will be announced on Brightspace	
Assessment	Students will be given a final writing assignment. In borderline cases teachers will take the final decision, based on the student's class participation and homework assignments.	
Enrolment / Application	Before enrolling in a group you must first take our Placement Test. For more information on this test please visit our website: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit/placement-test For this course you need a recommendation to take courses at intermediate level. After receiving your Placement Test results you can enrol via Brightspace. Please note that you must also choose a group to reserve a place in the class. Once you have enrolled in the course, go to collaboration and select the group that suits your schedule. You do not need to register in Osiris for this course.	
Special Information	Note that you are required to spend an additional 6 hours (approx.) per week doing homework assignments. Please make sure that your schedule allows you this time.	
Remarks	An 80% attendance rate is obligatory, which means you may miss one session. To keep your place in the group you must attend the first session. You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class. The course is offered in Q1 and Q3. However, the course will only be offered if there are sufficient participants.	
Elective	Yes	
Tags	English proficiency	
Targetgroup	This course is available to BSc and MSc students	

TPM304A	Advanced Writing in English for the University	2
Module Manager	Drs. K.I. van der Linden	
Instructor	Drs. K.I. van der Linden	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. M.A. Swennen	
Instructor	S.F. Johnson	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	0/2/0/2	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	This writing course is a 7 week course with 1 lesson a week (2 hours) plus approx. 6 hours of homework assignments per week. The course focuses on writing effective, well-structured and coherent academic texts in English. It looks at the writing process, including planning and structuring the text. Students will then develop paragraphs into longer academic texts and work on text coherence and cohesion. The course also pays attention to advanced academic vocabulary, sentence structure, punctuation and grammar.	
Study Goals	The aim of the course is to develop language skills and writing skills needed for producing academic texts. This course will give the student an understanding of the conventions of academic writing in English and some of the grammatical structures common in academic texts.	
Education Method	Individual, pair and group work, combined with assignments to do at home. Feedback on course components is given by the course lecturer and other students.	
Practical Guide	This is a course for both BSc and MSc students who already have a good understanding of academic writing, but who wish to expand and improve their English writing skills. When you have taken our Placement Test you will be recommended a language level. With this recommendation you can sign up for any of the courses offered at that level. For this course you need a recommendation to take courses at Advanced level.	
Books	Course material will be announced on Brightspace	
Assessment	Students will be given a final writing assignment. In borderline cases teachers will take the final decision, based on the student's class participation and homework assignments.	
Enrolment / Application	Before enrolling in a group you must first take our Placement Test. For more information on this test please visit our website: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit/placement-test For this course you need a recommendation to take courses at Advanced level. After receiving your Placement Test results you can enroll via Brightspace. Please note that you must also choose a group to reserve a place in the class. Once you have enrolled in the course, go to collaboration and select the group that suits your schedule. You do not need to register in Osiris for this course.	
Special Information	Note that you are required to spend an additional 6 hours (approx.) per week doing homework assignments. Please make sure that your schedule allows you this time.	
Remarks	An 80% attendance rate is obligatory, which means you may miss one session. To keep your place in the group you must attend the first session. You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class. The course is offered in Q2 and Q4. However, the course will only be offered if there is a sufficient number of participants.	
Elective	Yes	
Tags	English proficiency	
Targetgroup	This course is available to BSc and MSc students	

TPM305A	Writing a Masters Thesis in English	2
Module Manager	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. M.A. Swennen	
Instructor	S.F. Johnson	
Responsible for assignments	Drs. D.W. Laponder	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This course is designed for students who have started their MSc thesis or are about to start on their graduation project. The main focus is on the writing process, which means that matters such as grammar, academic style and precision of formulation are considered at length. Particular attention is also paid to the importance of structure and organising ideas. Aspects of the writing process will be backed up by weekly reading and writing assignments.	
Study Goals	The course is primarily designed for students working on their Masters thesis. As the name of the course indicates, the focus throughout is on writing. Samples of your academic writing will be corrected in considerable detail and you will be taught how to structure an essay, report or thesis. Plenty of tips will be given on ways of perfecting your English and invigorating your writing. Your systematic mistakes will be pointed out to you so that you can become critical about your own writing and go on to do your own editing.	
Education Method	Learning by doing and learning from lectures, corrections, peer review and tutor feedback.	
Books	Academic Writing for Graduate Students, Essential Tasks and Skills ,3rd edition by John M. Swales & Christine B. Feak. ISBN: 978-0-472-03475-8	
Assessment	Continuous assessment. Participants are to submit a number of text analysis and writing assignments in which they demonstrate their writing skills and their understanding of text composition and academic English. You may not submit texts that were previously submitted for any other course. Students who submit their weekly assignments late, repeatedly miss deadlines or lack commitment may be asked to leave the course. An 85% attendance record is required. This means students may miss no more than one session (preferably none). Students will receive their course credits when their work shows sufficient progress and weekly deadlines and all other conditions are met.	
Special Information	Enrolment is through Brightspace.	
Remarks	The course will only be offered when there is sufficient demand. If there are not enough enrollments, the course may be cancelled.	
Elective	Yes	
Tags	English proficiency	
Category	MSc level	

TUD4040	Joint Interdisciplinary Project	15
Responsible Instructor	Prof.dr.ir. J. Hellendoorn	
Project Coordinator	Ir. B.J.E. de Bruin	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Summary	JIP consists of three sets of activities: 1. The project which takes place at a company. Students are responsible for the project work (collaboration, integration of different perspectives, the content quality of their work). They plan their activities via a scrum method, are challenged to realise field trips, consult with experts and realise the necessary research work. The team keeps a Scrum log, run a blog for interaction with the outside world and every team member keeps a personal development log. 2. Meeting Professionals lectures and workshops about specialised topics, by academic staff or senior professionals from the companies involved. 3. Plenary meetings and design reviews for all the teams, company coaches and academic staff. At the meetings the students present their intermediate resp. final outcomes and plans, and are provided with feedback.	
Course Contents	The aim of the Joint Interdisciplinary Project is to prepare students to contribute to solving impactful technological challenges. The projects not only demand good engineering working knowledge but also experience with interdisciplinary and systems theory, and both knowledge and mindsets of innovation and entrepreneurial behavior. The project brief is provided by renowned companies like Airbus, Arcadis, etc. Teams of interdisciplinary student teams guided by a company coach and offered academic and industry expertise, are invited to realise an innovative problem solution to a complex problem and contributing to the sustainable development goals.	
Study Goals	1. Cognitive abilities attributable to interdisciplinary learning Demonstrate the ability to engage in perspective-taking; Develop structural knowledge pertaining to the problem; Integrate knowledge and modes of thinking drawn from two or more disciplines; Produce an interdisciplinary understanding of complex problem or intellectual question. 2. Scientific and intellectual development Capable to analyze scientific and societal consequences (economic, social, cultural, environmental) of the innovation; 3. Research and design capabilities Demonstrate engineering skills: technical skills, interpreting results, creativity, usability for company/institute; Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; 4. Collaboration and communication in an interdisciplinary team Demonstrate behavioral competences and skills: taking initiative, responsibility, showing communication skills, independency, collaboration and the ability to respect different disciplines and adapt to different cultures); Show ability to write a technical report: structured/consistent, language proficient, with correct use of literature/references, use of figures/tables/equations, and has a concise format (30 pages); Present work performed in a structured way through an oral presentation to their peers and customer. 5. Self-adjustment and reflection capabilities Plan and control the project efficiently considering resources and methodology; Being able to reflect on personal functioning in an evaluation report: reflect on personal objectives, indicate personal strengths/weaknesses. Indicate future personal improvement, drawing conclusions for future career. Cognitive abilities attributable to interdisciplinary learning: The ability to integrate (scientific and practical technological) knowledge from different disciplines to solve complex problems Scientific and intellectual development The capacity to evaluate the ethical, scientific and societal consequences of the proposed innovation Research and design capabilities The ability to create reasonable and relevant research or design , according to the academic standards of the involved disciplines Collaboration and communication in an interdisciplinary team Demonstrate behavioural competences and skills relevant for teamwork and effective communication with different stakeholders Self-adjustment and reflection capabilities To carry out regular reflections on professional and personal development and being able to improve upon those reflections Understand contemporary and societal issues in their work.	
Education Method	Full-time project work in an interdisciplinary team of about five students. The project work is interspersed with some just in time seminars and workshops about specialized subjects, methods or practical situations that play an essential role in interdisciplinary project work.	
Assessment	The assessment criteria pertain to the process, product and presentation and are assessed at an individual and team level, during 3 review sessions at three levels of accomplishment (each review assesses at a different level): Interdisciplinary work Scientific reasoning and ethical mindset Innovation process Presentation and communication Academic staff, supported by the input from company coaches, grade the students during the Problem Statement, the Midterm and the Final Review. The final group mark for JIP is differentiated per person and is based on: The team presentations - 30% The Problem definition, progress report and final report 30% The individual contribution to the team 20% The final individual reflection report 5% The Blog 5%	
Remarks	Project Outcomes (product, model, system) are presented at: 1. Problem Statement Review 2. Midterm meeting 3. Final Review 4. Symposium presentation	

WI5215	Internship AM	15
Responsible Instructor	Prof.dr.ir. M.B. van Gijzen	
Contact Hours / Week x/x/x/x	self study	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	Students are required to gain experience as a mathematician in interdisciplinary work. This requirement can be satisfied in four ways: - Through an internship of 15 EC (WI5215) - Through a Joint Interdisciplinary Project (TUD4040) - Through an Interdisciplinary Advanced AI Project (IFEEMCS520200) - Through a graduation project performed in an interdisciplinary setting outside the Delft Institute of Applied Mathematics. Students that have been admitted through a bridging programme (i.e. with a bachelor degree different from (Applied) Mathematics) or that are following the SEC specialisation are exempt from this requirement, but are still allowed to include an internship in their IEP. When carrying out the internship at a company or institute, the following conditions apply: - The student takes part in the company or institute culture in general, including sharing the company-established balance between on-site and online work. - The student takes part in the day-to-day running of the company or institute similar to regular employees. This includes having regular consultation and interaction with colleagues and manager(s), and use of and access to the company-specific resources and tools applicable to the internship. Such resources and tools can be physical/on-site (e.g. measurement equipment, company machines and infrastructure) as well as virtual (online, software tools, databases, ...). You are only allowed to start if you have an approved IEP. Detailed information on projects, procedures and application is available on the Brightspace page Internship AM.	
Study Goals	After this course, you should be able to: - plan and carry out an internship project using mathematics for solving real life problems - report the results to the external supervisor - communicate and collaborate with an interdisciplinary team - provide a mathematical framework for the work carried out during the internship - take the societal and ethical context into account while considering technological and societal problems - reflect on your own role and experience during the internship	
Education Method	For the internship, students must do an interdisciplinary project at a company or at a non-mathematics department of a university. The main goal of the internship is to apply mathematical knowledge in a non-mathematical environment, not to gain mathematical knowledge. The duration of a project is three months. Before the project starts the student has to fill out an internship application form. Once the form is approved, the student can start with the project.	
Assessment	Assessment: - You need to write a report during the internship and send it to your company supervisor and the TU Delft coordinator on the final day of the internship. - You need to write your reflection note and testimonial and send it, together with the report, to the TU Delft coordinator on the final day of your internship. - Your company supervisor needs to send the evaluation within two weeks after the end of the internship to the TU Delft coordinator. - The TU Delft coordinator needs to send the TU assessment rubric and the company evaluation form to the responsible instructor, within two weeks after receiving the evaluation from the company supervisor. - When the responsible instructor has received all documents, including the two assessment forms, your grade will be processed within a week. The responsible instructor is responsible for the final grading. The internship will be graded with a pass or a fail (not numerically). Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Report: resubmit deliverable	
Remarks	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). AM students from cohort 2022-2023 and before can include WI5012 or WI5118 in their IEP. For information on WI5012 or WI5118, please consult the study guide of 2022-2023.	

WM0375TU	Ethics of Technological Risks	3
Module Manager	Prof.dr. S. Roeser	
Instructor	K. Maheshwari	
Instructor	Prof.dr. S. Roeser	
Responsible for assignments	Prof.dr. S. Roeser	
Co-responsible for assignments	Prof.dr.mr.ir. N. Doorn	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course investigates ethical aspects of technological risks. Debates about technological risks related to, for example, energy technologies, robotics and biotechnology frequently culminate in stalemates. This is due to the complexities and intricacies inherent to such debates as they involve scientific information and uncertainties, as well as ethical and emotional considerations. Conventional, quantitative approaches focus primarily on statistical information to risk. They do not explicitly incorporate important ethical considerations such as justice, fairness and autonomy. Emotions such as compassion, care and feelings of responsibility can draw attention to such ethical aspects of risky technologies. Taking emotions and ethical considerations seriously can lead to more fruitful deliberations between different stakeholders in which relevant concerns are taken seriously and are explicitly reflected upon. This course will study how approaches to ethical aspects of risk can lead to more morally responsible decision making and design of technological innovations.	
Study Goals	This course will provide students with an understanding of approaches to risk ethics and students will get experience with discussing and evaluating ethical aspects of risk	
Education Method	Self-study, lectures and tutorials (in smaller groups, ca 25-35 students, depending on the total number of students). All students will hold one group presentation (ca 5 students) at one of the lectures. Attending lectures and tutorials and giving a presentation is obligatory.	
Assessment	On campus written exam (individual assignment), this will count for 100% of the grade for this course. Students need to have a pass (5.75 or higher) in order to pass the course. There is a resit of the written exam in the 4th quarter (date to be announced). In case of unforeseen circumstances, eg related to COVID-19 measures, the on campus exam may be replaced with an online (eg take home) exam. [There is also a 5 ECTS version of this course, i.e. WM0376TU; the only difference is that for the 5 ECTS version students also write an essay in addition to the exam.]	
Special Information	Contact person for additional questions: S.Roeser@tudelft.nl	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Tags	Ethics Philosophy	
Category	MSc level	

WM1135TU	Advanced English for the University	3
Module Manager	Drs. K.I. van der Linden	
Instructor	Drs. K.I. van der Linden	
Instructor	Drs. M.A. Swennen	
Instructor	S.F. Johnson	
Responsible for assignments	Drs. K.I. van der Linden	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	2/2/2/2	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	This course is given at C1 CEFR (Common European Framework of Reference) level. Advanced English for the University is the highest level integrated skills course offered by the English Unit within the ITAV institute at TPM. It is designed for those who already have a thorough working knowledge of English grammar and vocabulary but who still wish to further perfect their proficiency in order to become more adept in both writing and speaking. Time will be devoted to: Academic texts and writing conventions Academic vocabulary and collocations Grammar and punctuation Presentations and discussions Language learning strategies Advanced English for the University is a 14-week course with 1 lesson a week (2 hours) plus approx. 4 hours of homework assignments per week. The groups tend to be small (max. 18 participants) and the sessions are highly interactive. Attendance is mandatory. Students can miss a maximum of two lessons. Courses begin in the first week of September and in the first or second week of February . Please see the website for Placement Test dates and for any further course details: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit	
Study Goals	By the end of the course students should be familiar with academic writing conventions academic vocabulary with a specific focus on collocations common grammatical errors punctuation conventions structural devices to improve coherence various language resources such as AI tools	
Education Method	Students will do in-class exercises completed either in small groups or in pairs. They will also do a group PowerPoint presentation and a discussion led in pairs. The feedback on course components will be given both by the lecturer and by fellow course participants.	
Literature and Study Materials	Course book: Cambridge Academic English - An integrated skills course for EAP (Student's Book; C1 Advanced). Authors: Martin Hewings and Craig Thaine Publisher: Cambridge University Press, 2012 ISBN: 978-0-521-16521-1	
Prerequisites	To qualify for the course students need to have received an Advanced English for the University recommendation following the completion of the Placement Test. Please see in this connection: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit/placement-test	
Assessment	This is a pass/fail course. There is no final exam. Assessment is based on the six written assignments and on the in-class discussions and presentations.	
Enrolment / Application	After receiving their Placement Test results students can enrol via Brightspace by first registering for the course and then selecting the group of their choice.	
Remarks	More information can be found on Brightspace WM1135TU	
Elective	Yes	
Tags	English proficiency	
Targetgroup	BSc, MSc and PhD students are welcome	
Category	BSc and MSc level	

WM1301TU	Ethics of Transportation	3
Module Manager	Dr. F. Santoni De Sio	
Co-responsible for assignments	Dr. J.B. van Grunsven	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	In this course you will explore some ethical and societal aspects and problems related to the design, development, management and use of transportation systems, with a particular focus on their digitalisation and robotisation. You will be introduced to a range of relevant aspects and concepts, including Responsible Innovation, moral dilemmas with self-driving cars, philosophical theories of freedom and justice, moral and legal responsibility, privacy, and the future of human employment. You will analyse ethical, legal and social problems of transportation in relation to case studies, and you will explore possibilities for resolving, diminishing or preventing these problems.	
Study Goals	After having completed the course you: - can better recognise and analyse ethical and social aspects and problems inherent in (automated) transportation and in the work of professionals and managers active in the design, development, management and control of (automated) transportation systems - can see these problems within the broader background of moral, legal and political philosophy - have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds - are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to (automated) transportation and the work of professionals and managers in this domain. - are better prepared to perform your future work as a professional or manager in the design, development, production and control of (transportation) technology in an ethical and socially responsible way	
Education Method	A series of 6 lectures and some working group sessions, focused on the discussion and application of the theories presented in the course	
Assessment	Written exam and assessment participation in working groups	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Category	MSc level	

WM1401TU	Ethics of Healthcare Technologies	3
Module Manager	Dr. S.M. Copeland	
Co-responsible for assignments	Prof.dr. S. Roeser	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Introduction Ethics and Healthcare Technologies introduces students to the range of ethical issues that tend to arise during the development, design and application of devices used in healthcare or for medical purposes. The class will gain insight into the different aspects of healthcare ethics, such as the distinction between medical research and practice, the ethics of using human or animal research participants, and the ethics of technology, as well as value-sensitive design and responsible innovation approaches within the context of healthcare technologies. The course will provide students with skills in critical reflection, sensitivity to ethical concerns, and the ability to take up considerations of value to society as well as governance in the development and applications of technology. Assignments and cases examined in class will address contemporary developments in these technologies, the reasons behind new regulations that pertain, as well as giving a historical perspective on the ethical principles that underlie current approaches. Thus, the course should prepare students for common problems they will face in their current studies and prepare them for addressing issues that will arise in their future careers. Topics include: Introduction to ethics in medicine and healthcare: The research-practice and normative-descriptive distinctions, Bioethics, and the ethics of clinical research (historical and present), Medical devices and some relevant regulations & codes. Applying ethical frameworks e.g., utilitarianism, principlism, care ethics; Scientific integrity and the social value of research. Introduction to ethics of technology: Technology as a socio-technical system, Innovation in healthcare, Value-sensitive design, responsible innovation. Specific issues in developing healthcare technologies, such as: wearable and other health tracking devices; transhumanism and augmentation; human and animal research ethics; genetic modification; 3D tissue/organ printing; datamining, imagery and related privacy issues; the role of AI, machine learning and robotics.	
Study Goals	Regarding healthcare ethics & ethics of innovation in healthcare, students can: identify and describe various canonical forms of debate and ethical approaches within healthcare and biomedical ethics, compare diverging viewpoints in the field and assess the strength of an argument made in the field, evaluate different ethical issues as they arise at different points along the development and application trajectory. Regarding ethics and technology, students can: identify and describe normative aspects of technology; analyse and evaluate the ethical and social aspects of a case in contemporary healthcare technology development and application, and make recommendations for value sensitive design.	
Education Method	Contact hours: Plenary sessions that include blended learning activities, 2x Tutorials, Online discussion forums. Study materials: Reader (online) - becomes available during the course; Lecture slides, available on Brightspace after the lecture; Online materials to prepare for in-class activities, including self-directed research.	
Assessment	Grade calculation WM1401 (3ECs): Mandatory attendance in tutorials and short formative assignments, Exam (multiple choice and/or short open questions).	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Category	MSc level	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Thesis project 2023

Introduction 1

The thesis project is performed under supervision of (at least) one of the research groups at DIAM.

In case an external thesis project is mandatory (for students that do not have the internship, IAAIP or JIP as part of their IEP), the course code is WI5010EXT. The code for students that are free to choose between an external or internal project is WI5010. Students that are exempt from the requirement to gain experience as a mathematician in interdisciplinary work should use the code WI5010 for their thesis project in any case.

WI5010	Thesis Project	45
Responsible Instructor	Prof.dr.ir. M.B. van Gijzen	
Contact Hours / Week x/x/x/x	N/A	
Education Period	None (Self Study)	
Start Education	2	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	The master thesis project is the last study unit of the MSc AM programme. It is an individual in-depth research with sufficient academic content in the specific field of expertise chosen by the student and needs approval beforehand from the thesis supervisor. The thesis project may be executed within a research program at TU Delft, or in a suitable research institute or company. The nominal duration of a thesis project is 8 to 9 months (45 ECTS). The student carries out the thesis project independently, with the guidance of one or more thesis supervisors, at least one of them from the scientific staff of Delft University of Technology, and under the responsibility of the full or associate professor who approved the IEP. The thesis project is governed by the EEMCS Graduation Policy: https://www.tudelft.nl/studenten/ewi-studentenportal/onderwijs/afstuderen-msc	
Study Goals	1. The student is able to design a research project: - The student is able to use, explain and justify adequate research and design methodologies; - The student is able to apply theory to the performed project; - The student is able to use techniques for interpretation and verification and bases his/her conclusions on results; - The student is able to do reliable work with scientific significance. 2. The student is able to execute a research project: - The student has a critical attitude towards his/her own results, literature and specialists; - The student makes an original contribution to the project; - The student takes initiative (together with the supervisor) to give his/her own input within the research project; - The student interacts sufficiently with peers and superiors; - The student is able to make and execute a project plan. 3. The student is able to write a research report: - The student is able to write a research report that shows sufficient coherence of content; - The student is able to structure the research report and sufficiently present the content (text and figures); - The student expresses argumentation using correct spelling and grammar; 4. The student is able to present and defend the research project: - The student is able to present the content using sufficient detail to support conclusions; - The student is able to logical structure the presentation and use visual aids; - The student is able to adequately formulate and express himself/herself as well as sufficiently address the audience; - The student is able to argument and answer the questions asked by the committee.	
Education Method	Project	
Prerequisites	Before starting the project, students must have completed at least 60 EC of the Master's degree programme and be in possession of a Declaration of Credits Obtained (DCO). To be able to get a DCO, the individual exam program (IEP) should have been approved by the Board of Examiners.	
Assessment	The thesis committee assesses the thesis and the defense on the following criteria: - quality of work: novelty, volume, grasp, methodology, publishable; - personal performance: autonomy, planning, creativity, attitude; - quality of thesis report: clarity, organisation, argumentation; - oral presentation and defense: clarity, focus, relevance, discussion. More information on thesis grading: https://www.tudelft.nl/studenten/ewi-studentenportal/onderwijs/afstuderen-msc/assessment Weighting of assessment: - Quality of the work (research) (40% weight of the final grade), - Performance (30%), - Report (20%) - Presentation & defence (10%). Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Report: resubmit deliverable - Defense: resit presentation and defense Depending on the planning of the student it might happen that it is not possible to repair the report or the presentation within the same academic year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI5010EXT	Thesis Project (external)	45
Responsible Instructor	Prof.dr.ir. M.B. van Gijzen	
Contact Hours / Week x/x/x/x	N/A	
Education Period	None (Self Study)	
Start Education	2	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	The master thesis project is the last study unit of the MSc AM programs. It is an individual in-depth research with sufficient academic content in the specific field of expertise chosen by the student and needs approval beforehand from the thesis supervisor. A thesis project with the code WI5005EXT must be executed outside DIAM, in a suitable research institute or company. The nominal duration of a thesis project is 8 to 9 months (45 ECTS). The student carries out the thesis project independently, with the guidance of one or more thesis supervisors, at least one of them from the scientific staff of Delft University of Technology, and under the responsibility of the full or associate professor who approved the IEP. The thesis project is governed by the EEMCS Graduation Policy: https://www.tudelft.nl/studenten/ewi-studentenportal/onderwijs/afstuderen-msc	
Study Goals	1. The student is able to design a research project: - The student is able to use, explain and justify adequate research and design methodologies; - The student is able to apply theory to the performed project; - The student is able to use techniques for interpretation and verification and bases his/her conclusions on results; - The student is able to do reliable work with scientific significance. 2. The student is able to execute a research project: - The student has a critical attitude towards his/her own results, literature and specialists; - The student makes an original contribution to the project; - The student takes initiative (together with the supervisor) to give his/her own input within the research project; - The student interacts sufficiently with peers and superiors; - The student is able to make and execute a project plan. 3. The student is able to write a research report: - The student is able to write a research report that shows sufficient coherence of content; - The student is able to structure the research report and sufficiently present the content (text and figures); - The student expresses argumentation using correct spelling and grammar; 4. The student is able to present and defend the research project: - The student is able to present the content using sufficient detail to support conclusions; - The student is able to logical structure the presentation and use visual aids; - The student is able to adequately formulate and express himself/herself as well as sufficiently address the audience; - The student is able to argument and answer the questions asked by the committee.	
Education Method	Project	
Prerequisites	Before starting the project, students must have completed at least 60 EC of the Master's degree programme and be in possession of a Declaration of Credits Obtained (DCO). To be able to get a DCO, the individual exam program (IEP) should have been approved by the Board of Examiners.	
Assessment	The thesis committee assesses the thesis and the defense on the following criteria: - quality of work: novelty, volume, grasp, methodology, publishable; - personal performance: autonomy, planning, creativity, attitude; - quality of thesis report: clarity, organisation, argumentation; - oral presentation and defense: clarity, focus, relevance, discussion. More information on thesis grading: https://www.tudelft.nl/studenten/ewi-studentenportal/onderwijs/afstuderen-msc/assessment Weighting of assessment: - Quality of the work (research) (40% weight of the final grade), - Performance (30%), - Report (20%) - Presentation & defence (10%). Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Report: resubmit deliverable - Defense: resit presentation and defense Depending on the planning of the student it might happen that it is not possible to repair the report or the presentation within the same academic year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

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